NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Week 1 – The Business of Shipping





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13 August 2026

Learning Objectives

1. Understand the function of maritime transport.
2. Discuss the characteristics of maritime transport.
3. Understand the role of maritime transport in supply chains.
4. Describe the different stakeholders involve in the maritime eco-system.

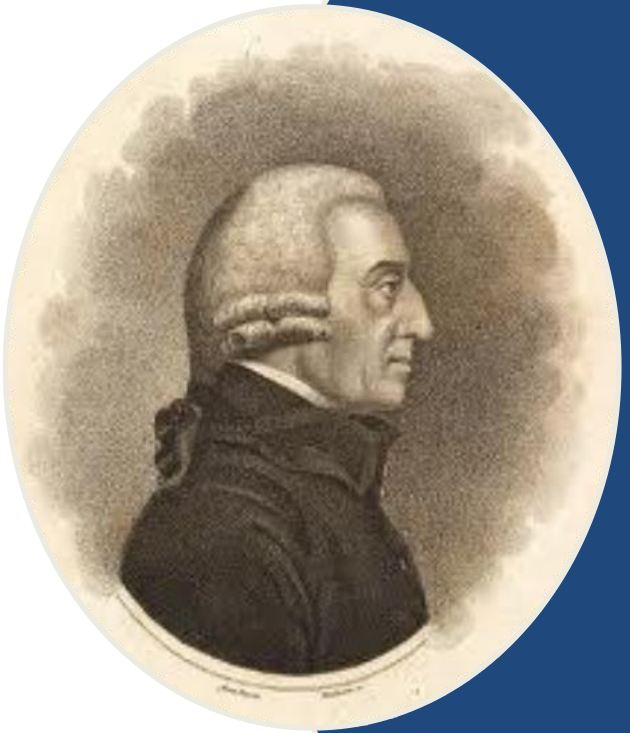
Learning Objectives

5. Describe the maritime organisations (both regulatory and industry) involved in shipping at the international and national level.
6. Discuss the two main types of shipping services.
7. Understand the ton-miles concept.
8. Understand the different maritime trading networks.

LO1

1. Function of Maritime Transport

1.1 Function of Maritime Transport



Adam Smith (1776) – shipping as a source of **low-cost transport** that could open up market:

…"as by means of water carriage a more extensive market is **opened to every sort of industries**… it is upon the sea-coast that industry of every kind naturally begins to **subdivide and improve itself**" …..

1.1 Function of Maritime Transport

- Maritime transport is the backbone for the development of international trade and a key engine driving globalization.
- Around ___% of global trade by volume and over ___% by value is carried by sea and handled by ports worldwide.
- Ships are sophisticated large valuable assets, costing over USD100 million to build.

1.1 Function of Maritime Transport

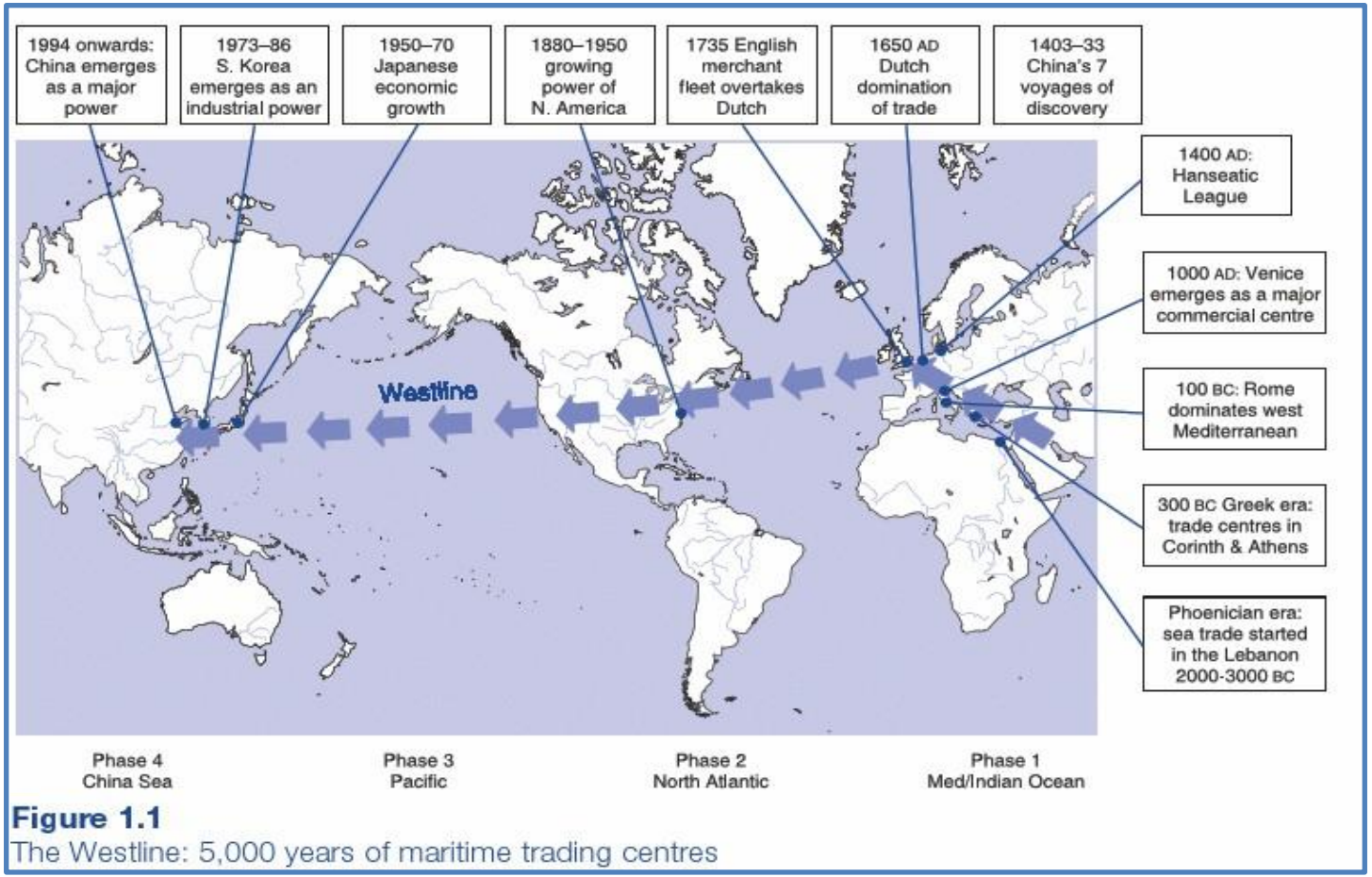


Figure 1.1
The Westline: 5,000 years of maritime trading centres

LO2

2. Characteristics of Maritime Transport

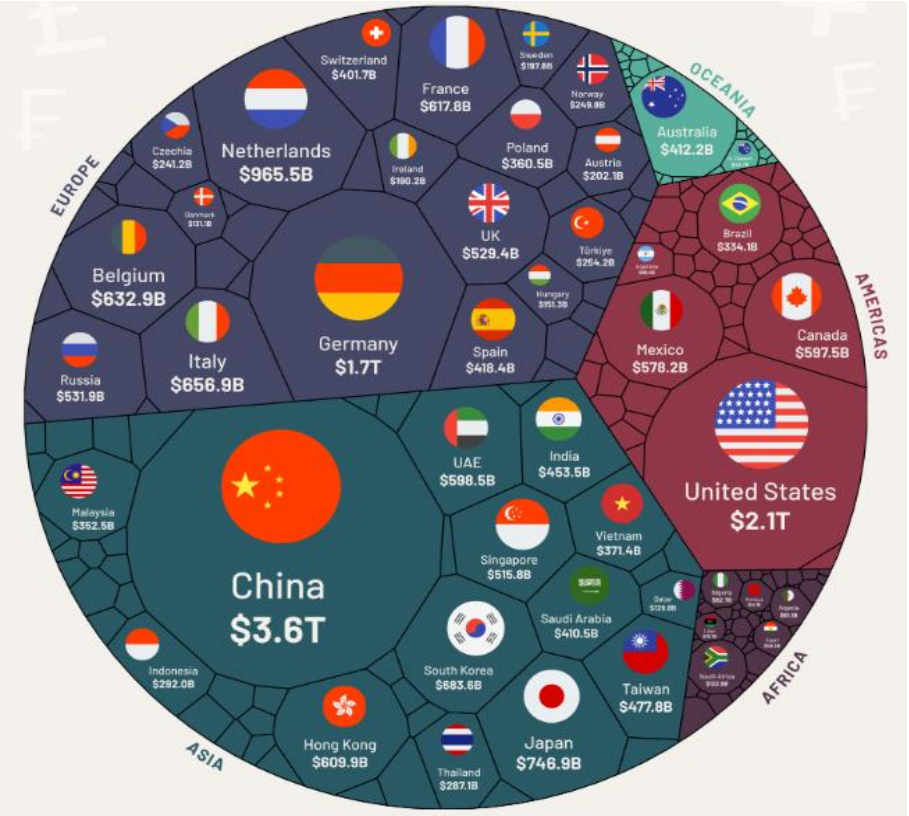
2.1 Derived Demand

- Occurs when there is a demand for a good or factor of production resulting from demand for a primary, intermediate or finished good or service.
- Demand for maritime transport service results from the demand for goods it transport.



2.2 High Levels of Internationalisation

- Maritime transport operations span across countries and/or regions
- Many shipping companies have local offices around the world.



2.3 Openness and Freedom

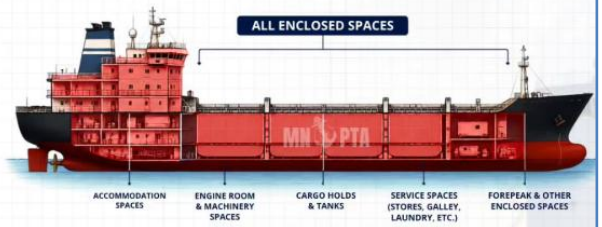
- Registered ships are granted freedom of navigation.
- Freedom of navigation (FON) is a principle of customary international law.
  - ships flying flag of any sovereign states shall not suffer interference from other states (apart from the exceptions provided for in international law).
- This right codified as Article 87(1)a of the 1982 United Nations Convention on the Law of the Sea (UNCLOS).

2.4 Economies of Scale

- Maritime transport often associated with gigantism.
- Economies of scale is the "*cost advantage experienced by a firm when it increases its level of output*".
- In maritime transport context, output refers to the ship capacities or ship size (*see next slide*).

2.4.1 Ship Capacities

GROSS TONNAGE



What Does It Mean?

- Gross Tonnage (GT) is a measure of the **total internal volume** of all enclosed spaces in a ship.

Included in GT	Excluded from GT
Cargo holds	Open decks
Engine room	Cargo hatches/openings
Crew accommodation	External superstructure
Bridge and control rooms	Spaces not completely enclosed
Stores, galley and service areas	–
Other fully enclosed spaces inside the ship	–

- ***GT is mainly used for :*** *ship registration, port dues, safety rules and classification purposes.*

NET TONNAGE



What Does It Mean?

- Net Tonnage (NT) is a measure of the **cargo- or passenger-earning enclosed volume of a ship.** It represents the spaces available for commercial use and excludes spaces required for operating the vessel.

Included in NT	Excluded from NT
Cargo holds	Engine room & machinery spaces
Passenger accommodation	Crew accommodation
Ro-Ro vehicle decks	Bridge & navigation spaces
Container/cargo decks used for earning revenue	Stores, galley & service spaces
Enclosed spaces used for carrying cargo or passengers	Forepeak, aftpeak & other technical spaces
Other revenue-earning enclosed spaces	Ballast tanks, fuel tanks & non-cargo spaces

- ***NT is mainly used for :*** *port dues, canal dues, harbour charges and commercial registration.*

DEADWEIGHT TONNAGE



What Does It Mean?

- Deadweight Tonnage (DWT) is the **maximum weight a ship can safely carry,** excluding the ship's own weight (Lightship).

DWT Includes

Included in DWT	Description
Cargo	All goods being transported
Fuel	Fuel oil and diesel for propulsion
Fresh water	Drinking and operational water
Ballast water	Water carried for stability
Crew & Passengers	People on board
Stores & Provisions	Food, spare parts and supplies
Lubricants & Consumables	Oils and other operational consumables

2.4.2 Container Ship Size



TOP 10 CONTAINER SHIPPING LINES IN 2025

BY TOTAL VESSELS OPERATED

Rank	Company	Total Vessels
1	MSC	815
2	Maersk	705
3	CMA CGM	625
4	COSCO Shipping Lines	488
5	Hapag-Lloyd	290
6	ONE (Ocean Network Express)	267
7	Evergreen Marine	214
8	ZIM	146
9	Yang Ming	92
10	HMM	78

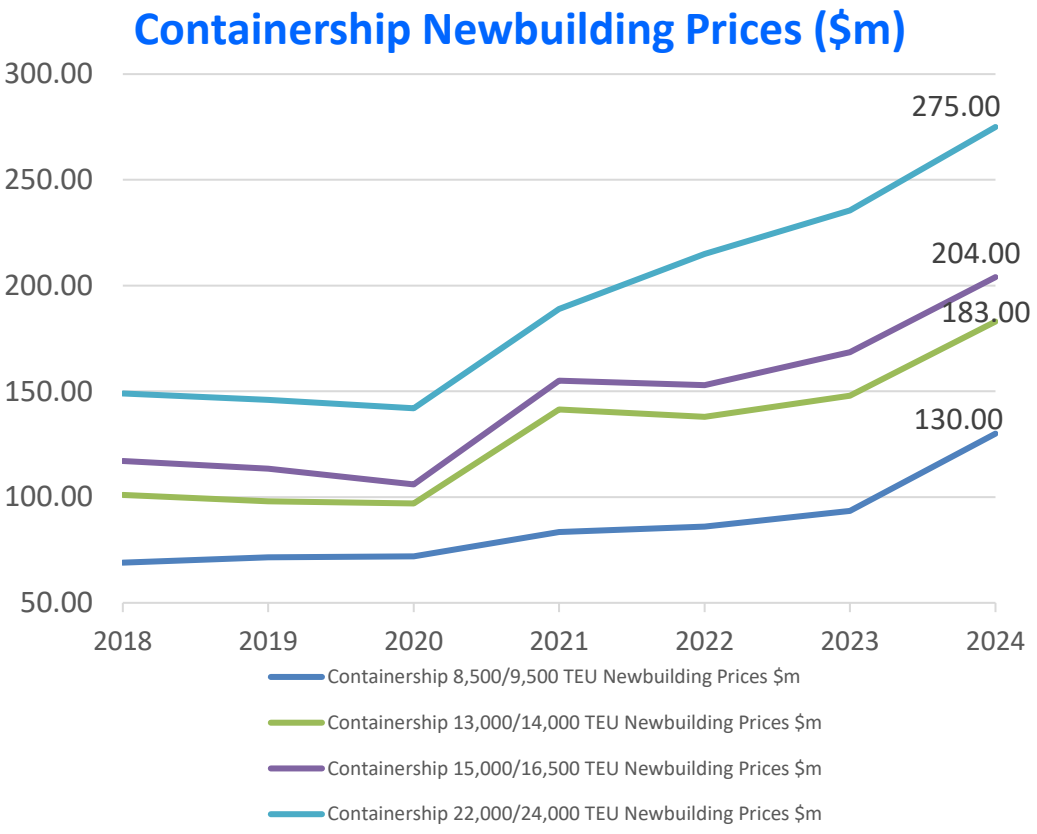
2.4.2 Ship Size



- ____ = **T**wenty-foot **E**quivalent **U**nits
- ____ = **F**orty-foot **E**quivalent **U**nits



What is the capital slot cost for the different new containerships sizes?



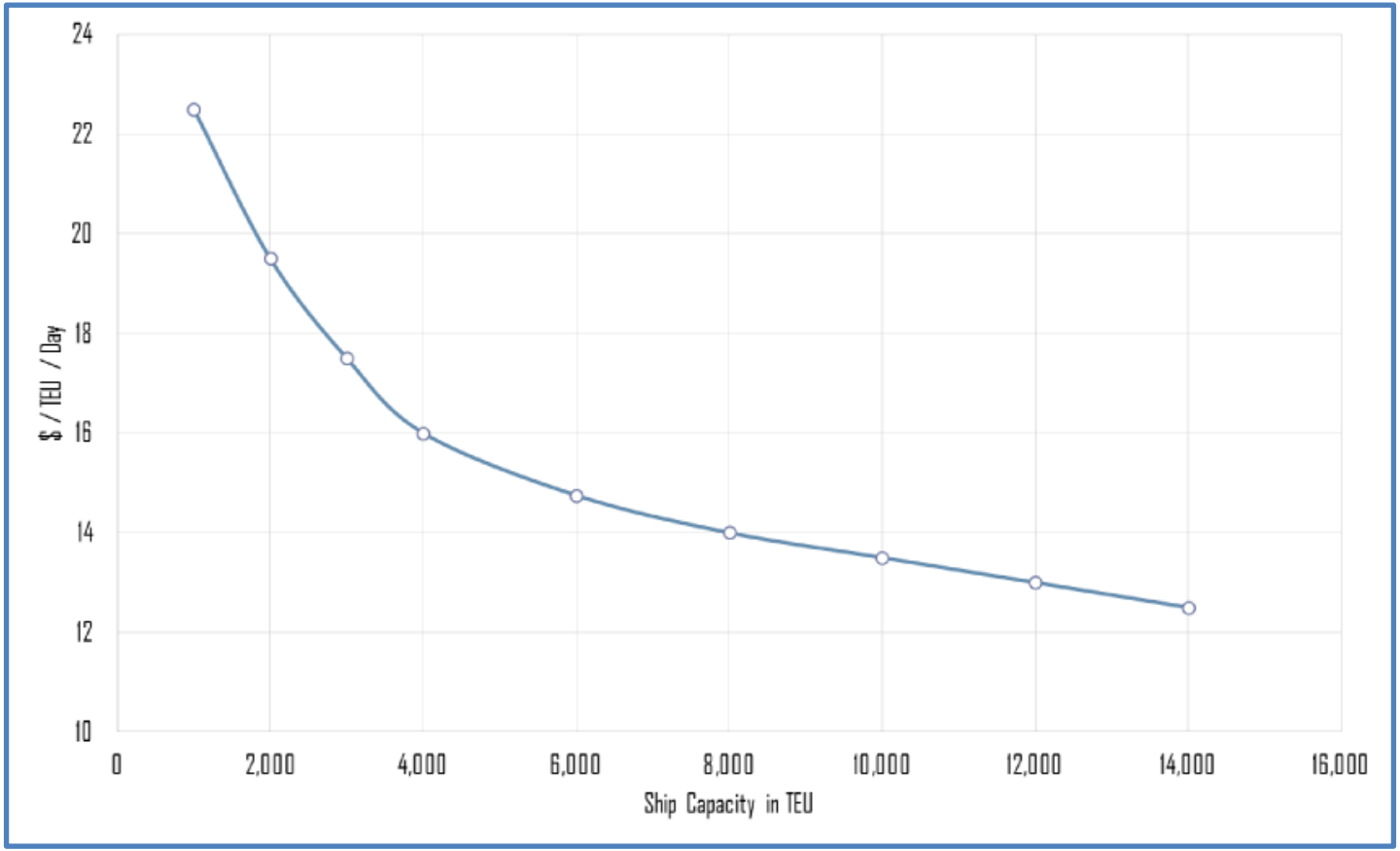
9,000 TEU =

13,500 TEU =

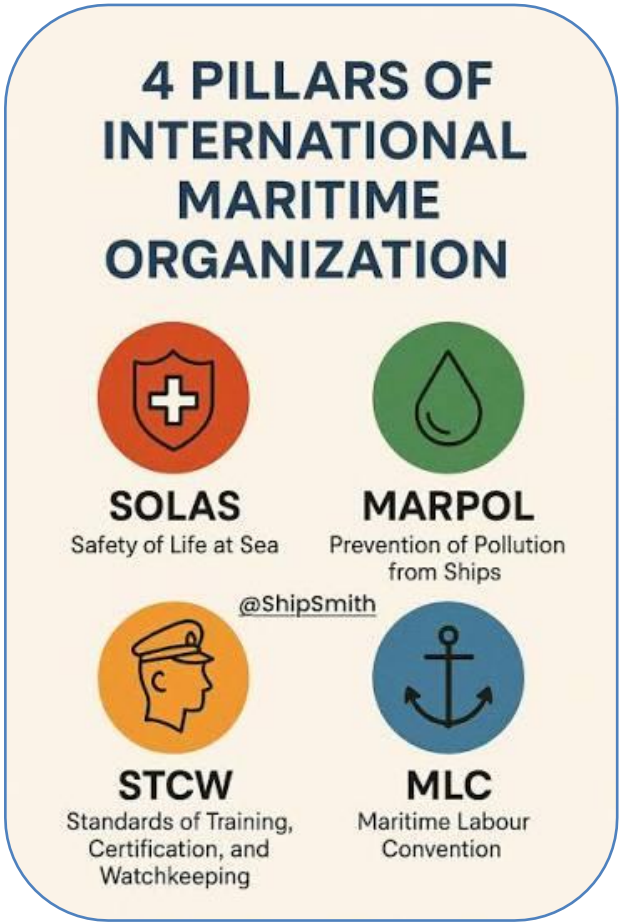
15,750 TEU =

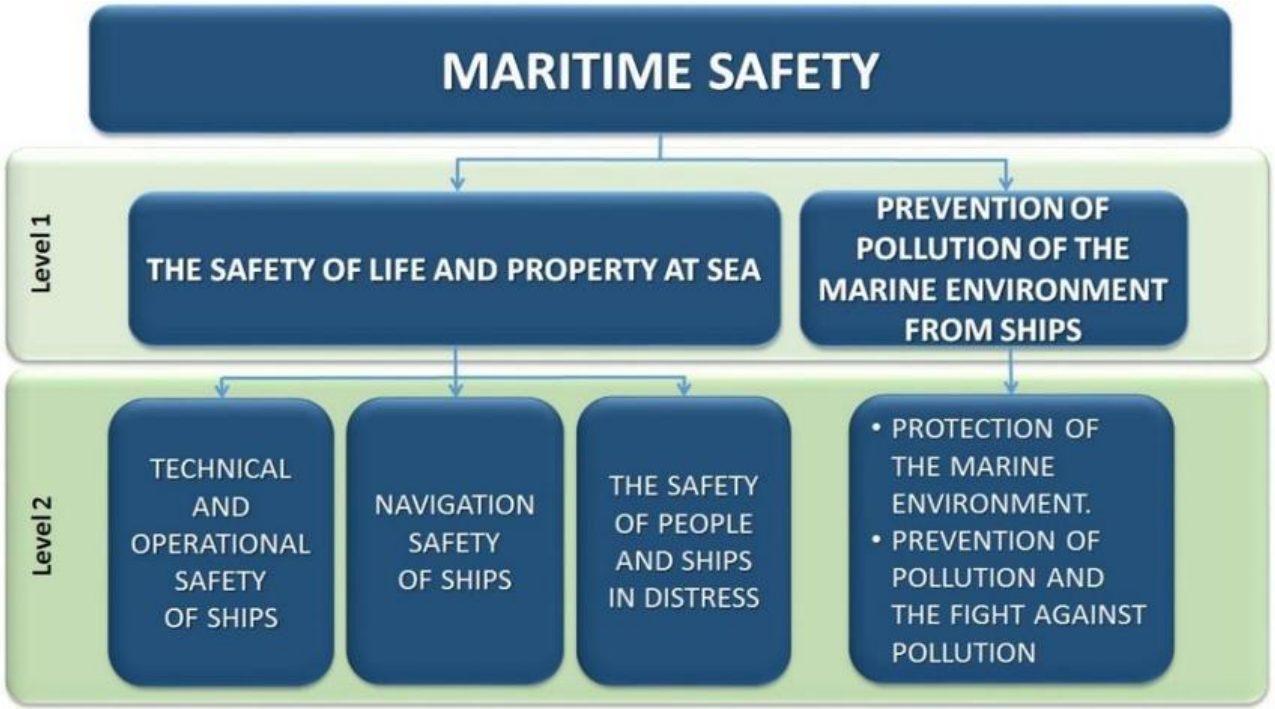
23,000 TEU =

2.4 Economies of Scale

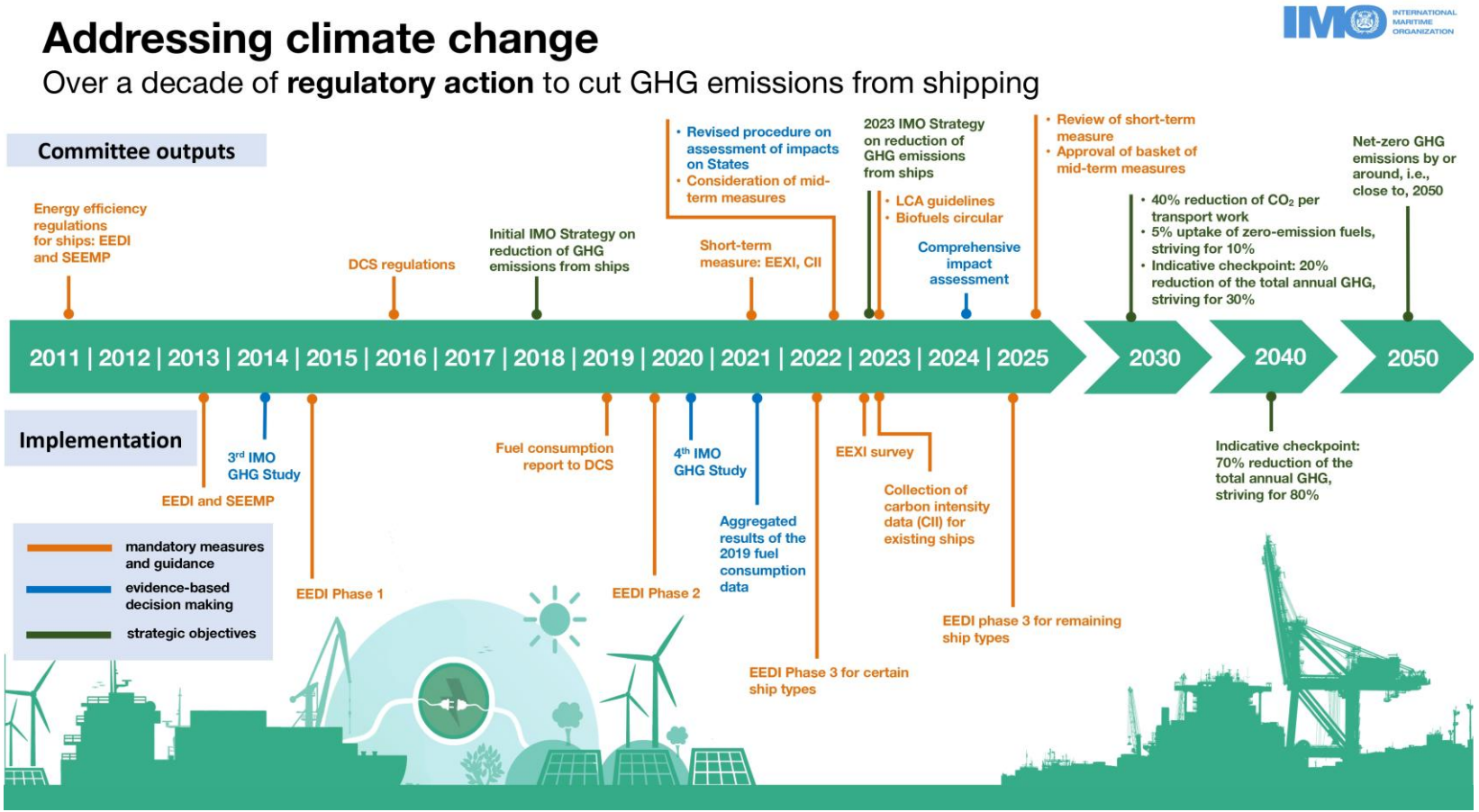


2.5 Regulations





2.5 Regulations



Review Questions

1. **What percentage of the world's total trade value is carried by ship and handled by ports worldwide?**

a. 55%

b. 60%

c. 65%

d. 70%

Answer:

Review Questions

2. Which of the following is NOT a characteristic of maritime transport?

a. Derived demand

b. Internationalisation

c. High transport cost

d. Economies of scale

Answer:

Review Questions

3. Which of the following is NOT a factor contributing to economies of scale in maritime transport?

a) Savings in capital costs

b) Savings in operating costs

c) Savings in bunker costs

d) Berth and draft limitations

Answer:

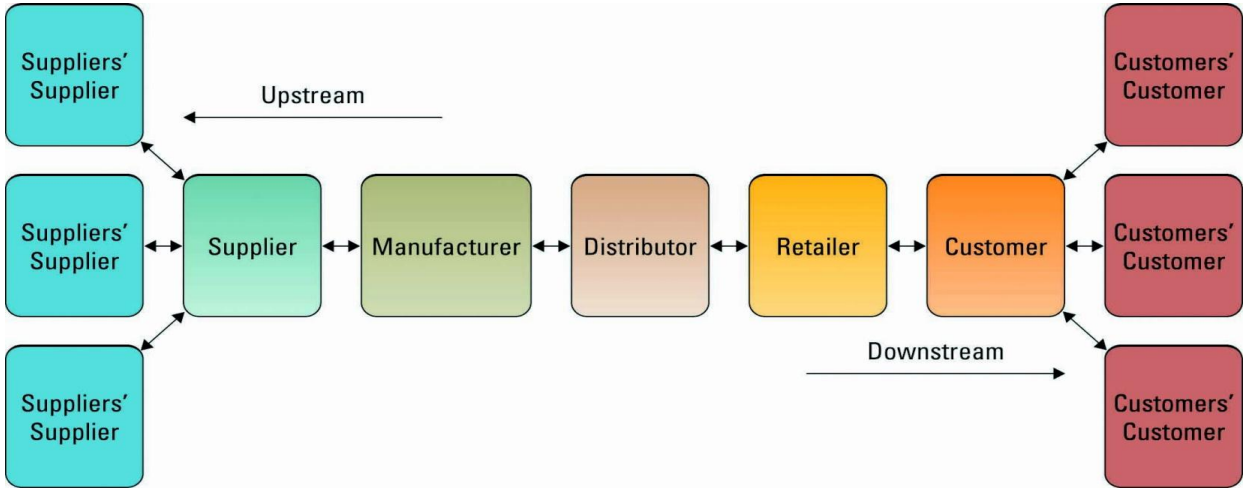
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3. Role of Maritime Transport in Supply Chains

3. Role of Maritime Transport in Supply Chains

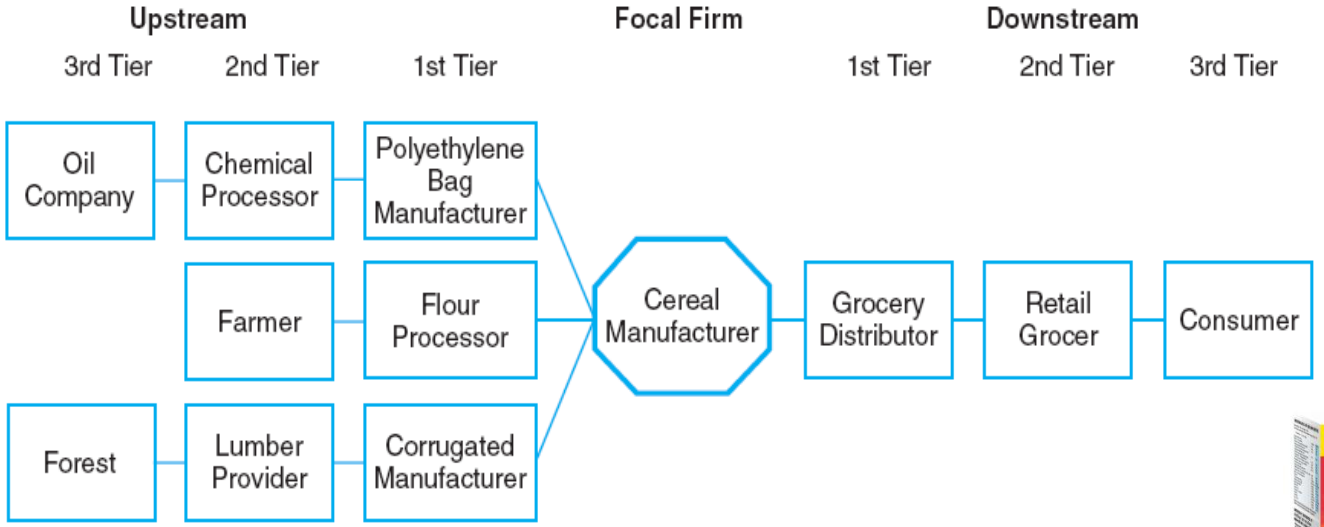
What is a supply chain?

- All **stages** involved, directly or indirectly, in fulfilling a customer's request.
  - includes manufacturers, suppliers, transport providers, warehouses, retailers and customers.



3. Role of Maritime Transport in Supply Chains







3. Role of Maritime Transport in Supply Chains

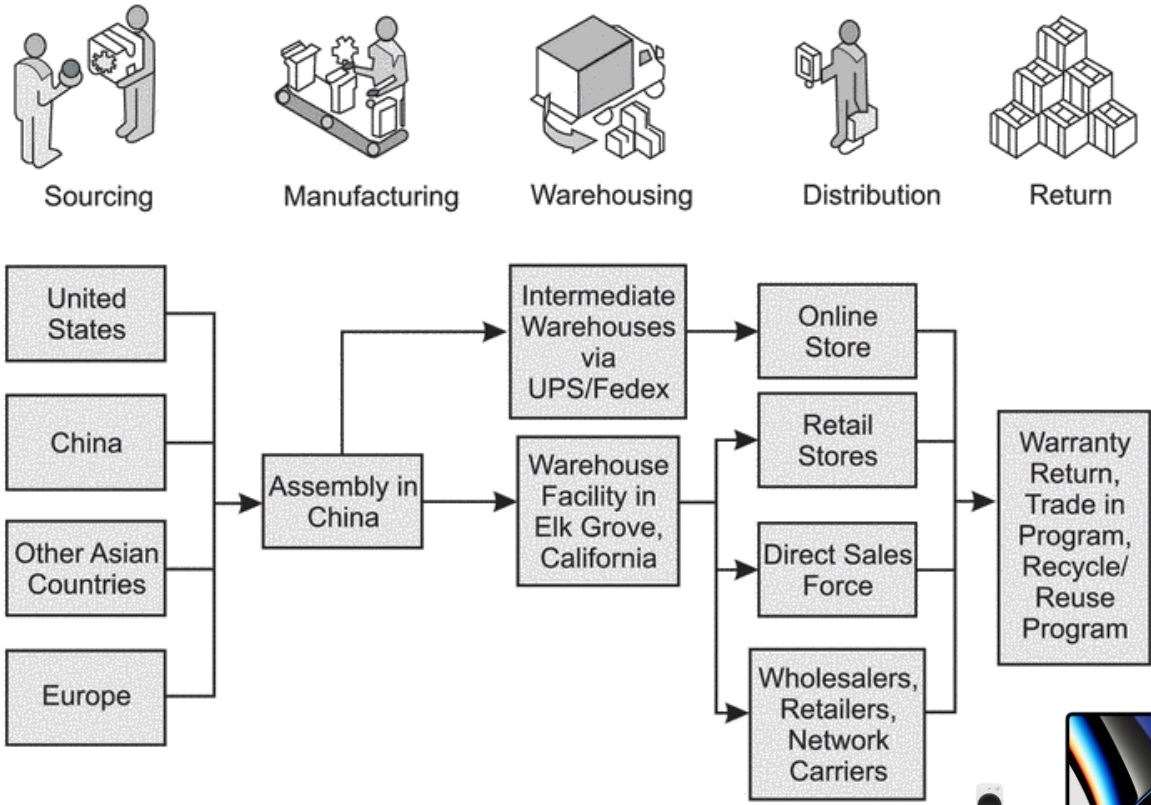
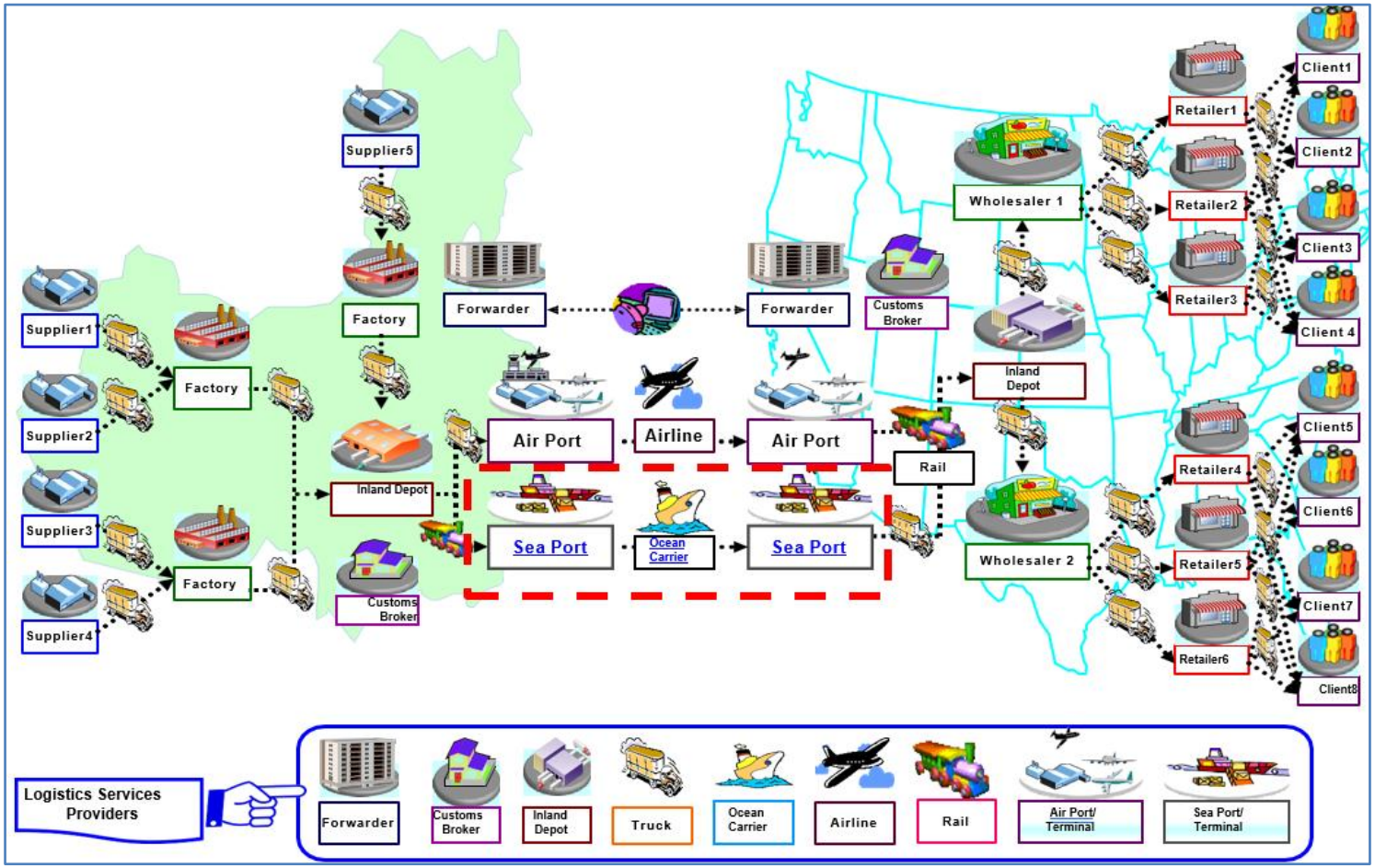
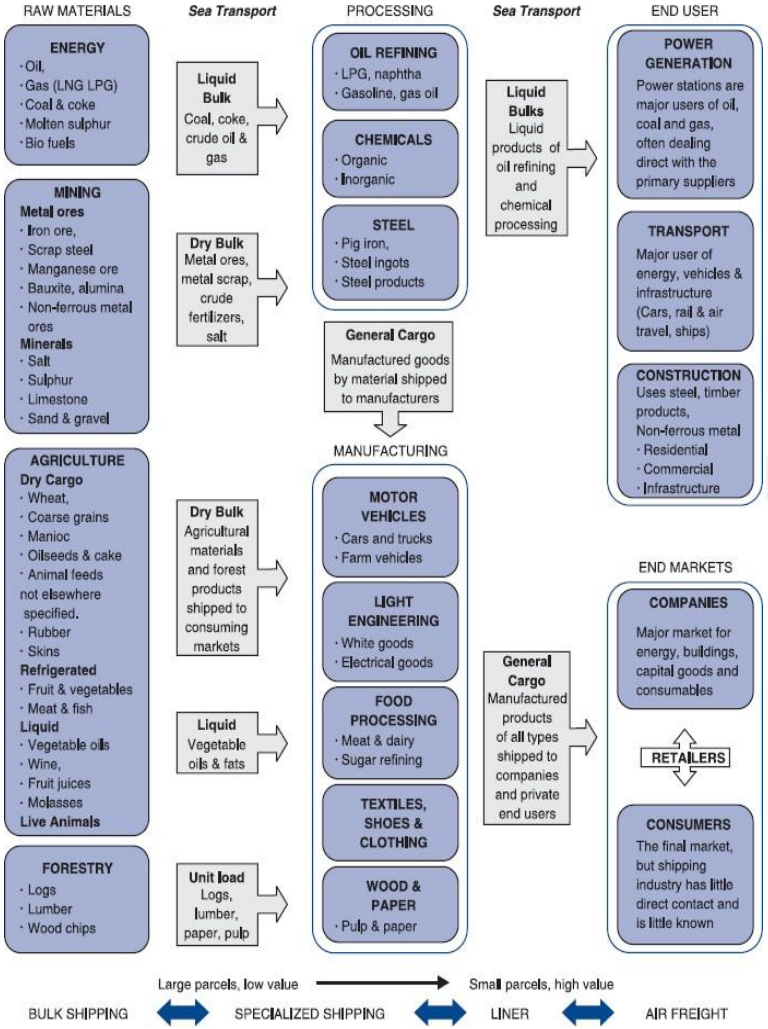


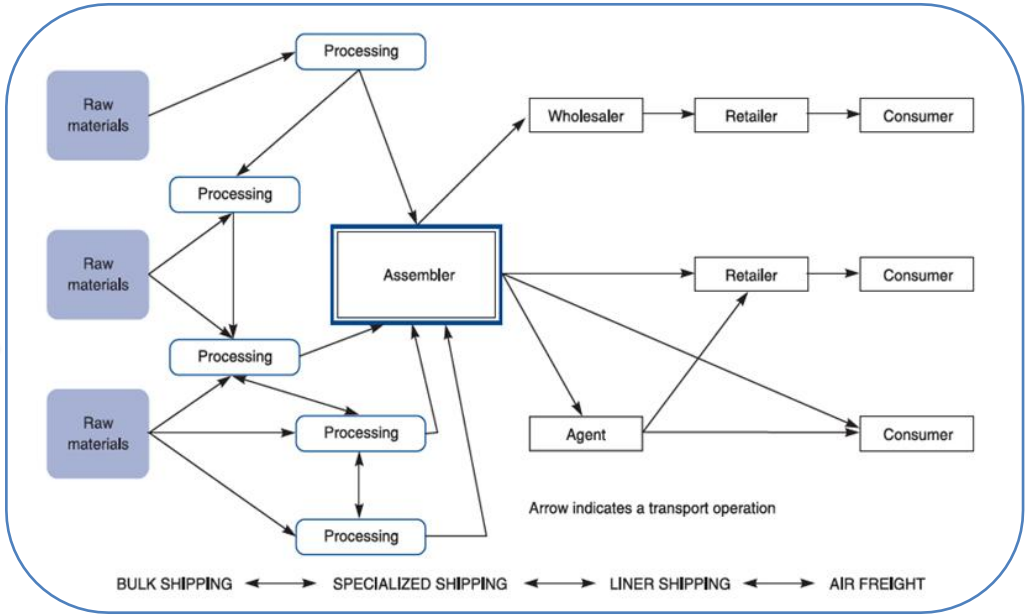
Figure 11.1: Apple's supply chain worldwide



3. Role of Maritime Transport in Supply Chains

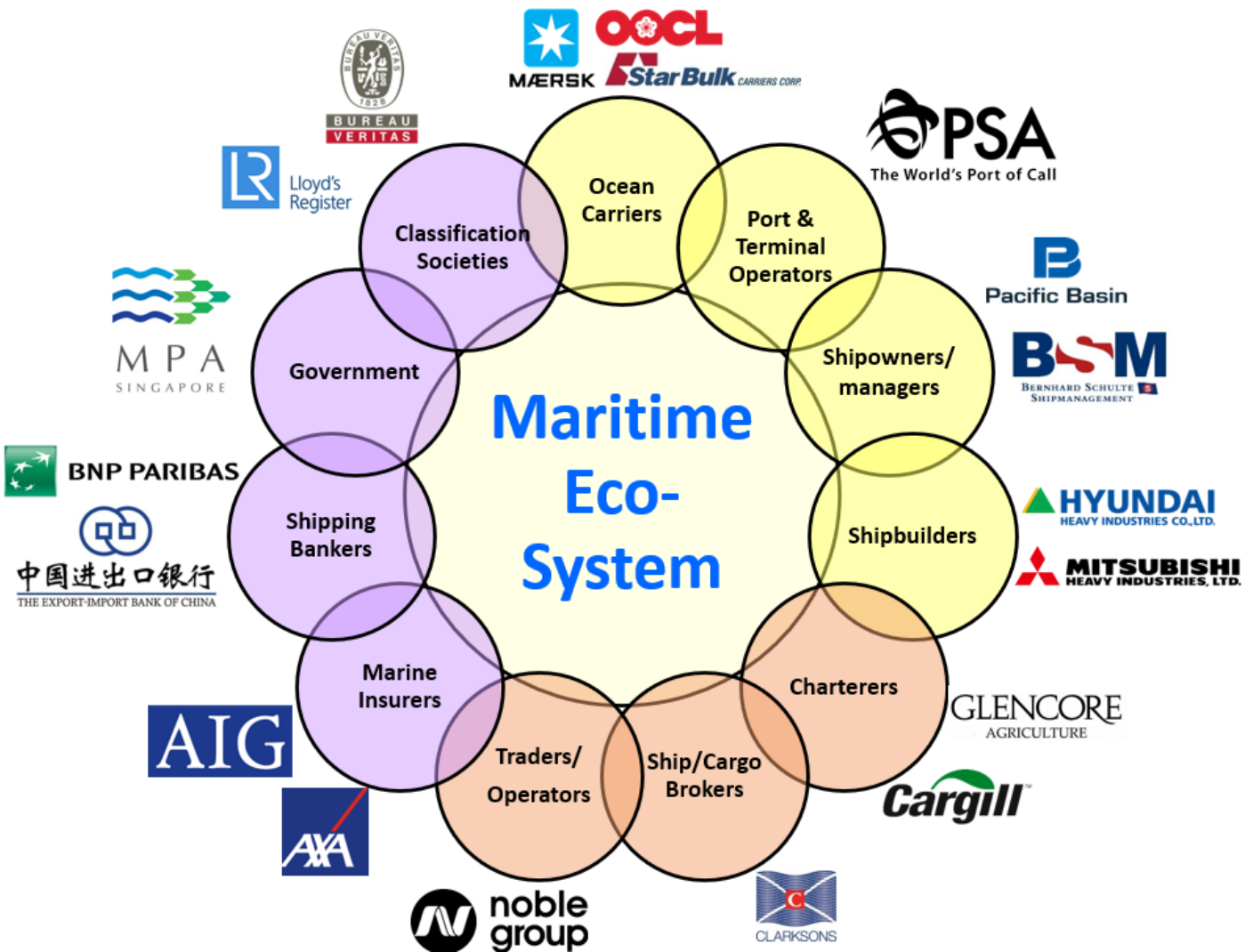



RAW MATERIALS
Sea Transport
PROCESSING
Sea Transport
END USER
ENERGY
· Oil,
· Gas (LNG LPG)
· Coal & coke
· Molten sulphur
· Bio fuels
MINING
Metal ores
· Iron ore,
· Scrap steel
· Manganese ore
· Bauxite, alumina
· Non-ferrous metal ores
Minerals
· Salt
· Sulphur
· Limestone
· Sand & gravel
AGRICULTURE
Dry Cargo
· Wheat,
· Coarse grains
· Manioc
· Oilseeds & cake
· Animal feeds not elsewhere specified.
· Rubber
· Skins
Refrigerated
· Fruit & vegetables
· Meat & fish
Liquid
· Vegetable oils
· Wine,
· Fruit juices
· Molasses
Live Animals
FORESTRY
· Logs
· Lumber
· Wood chips
Liquid Bulk
Coal, coke, crude oil & gas
Dry Bulk
Metal ores, metal scrap, crude fertilizers, salt
Dry Bulk
Agricultural materials and forest products shipped to consuming markets
Liquid
Vegetable oils & fats
Unit load
Logs, lumber, paper, pulp
OIL REFINING
· LPG, naphtha
· Gasoline, gas oil
CHEMICALS
· Organic
· Inorganic
STEEL
· Pig iron,
· Steel ingots
· Steel products
General Cargo
Manufactured goods by material shipped to manufacturers
MANUFACTURING
MOTOR VEHICLES
· Cars and trucks
· Farm vehicles
LIGHT ENGINEERING
· White goods
· Electrical goods
FOOD PROCESSING
· Meat & dairy
· Sugar refining
TEXTILES, SHOES & CLOTHING
WOOD & PAPER
· Pulp & paper
Liquid Bulks
Liquid products of oil refining and chemical processing
General Cargo
Manufactured products of all types shipped to companies and private end users
POWER GENERATION
Power stations are major users of oil, coal and gas, often dealing direct with the primary suppliers
TRANSPORT
Major user of energy, vehicles & infrastructure (Cars, rail & air travel, ships)
CONSTRUCTION
Uses steel, timber products, Non-ferrous metal
· Residential
· Commercial
· Infrastructure
END MARKETS
COMPANIES
Major market for energy, buildings, capital goods and consumables
RETAILERS
CONSUMERS
The final market, but shipping industry has little direct contact and is little known
Large parcels, low value
Small parcels, high value
BULK SHIPPING
SPECIALIZED SHIPPING
LINER
AIR FREIGHT


Raw materials
Raw materials
Raw materials
Processing
Processing
Processing
Processing
Processing
Assembler
Wholesaler
Retailer
Consumer
Retailer
Consumer
Agent
Consumer
Arrow indicates a transport operation
BULK SHIPPING
SPECIALIZED SHIPPING
LINER SHIPPING
AIR FREIGHT

LO4

4. Stakeholders in Maritime Eco-System


Maritime Eco-System
Ocean Carriers
Port & Terminal Operators
Shipowners/ managers
Shipbuilders
Charterers
Ship/Cargo Brokers
Traders/ Operators
Marine Insurers
Shipping Bankers
Government
Classification Societies
MÆRSK
OOCL
StarBulk CARRIERS CORP.
PSA
The World's Port of Call
Pacific Basin
BSM
Bernhard Schulte Shipmanagement
HYUNDAI
HEAVY INDUSTRIES CO.,LTD.
MITSUBISHI
HEAVY INDUSTRIES, LTD.
GLENCORE
AGRICULTURE
Cargill
CLARKSONS
noble group
AIG
AXA
BNP PARIBAS
中国进出口银行
THE EXPORT-IMPORT BANK OF CHINA
MPA
SINGAPORE
Lloyd's Register
BUREAU VERITAS

4.1 Shipowners

- An individual or company who, _____ and _____ a ship, to delivery cargo at a certain freight rate.

World fleet ownership by capacity and flag of registration, 1 January 2025

	Country or territory of ownerhip	Vessels (number)			Carrying capacity (dead weight tons)				
		National flag	Foreign flag	Total	National flag	Foreign flag	Total	Foreign flag share (percentage of world total)	Capacity share (percentage of world total)
1	Greece	562	4 555	5 124	47 554 402	350 065 296	397 649 662	88	16.4
2	China	7 103	3 282	10 440	134 110 270	210 763 710	347 215 014	61	14.4
3	Japan	958	3 123	4 083	39 524 134	201 146 629	240 678 389	84	9.9
4	Singapore	1 358	1 527	2 922	71 301 599	81 844 936	153 428 741	53	6.3
5	Hong Kong, China	919	1 122	2 081	82 621 846	56 608 620	139 502 591	41	5.8

4.2 Shipbuilders

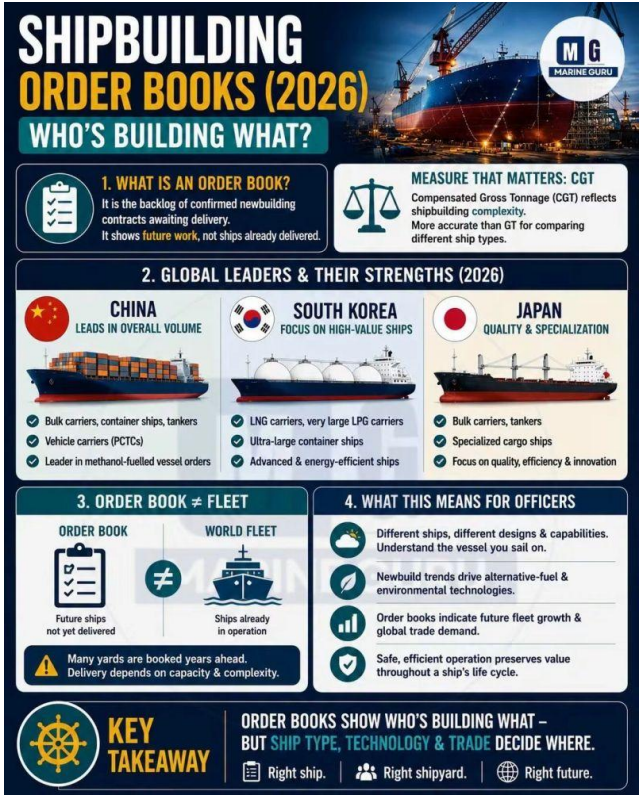
YANGZIJIANG SHIPBUILDING GROUP

SAMSUNG HEAVY INDUSTRIES

- Receives orders from shipowners to build/construct ships based on ships' specifications.

Deliveries of different types of newbuilt vessels, 2024

	China (thousands of gross tons)	Republic of Korea (thousands of gross tons)	Japan (thousands of gross tons)	Viet Nam (thousands of gross tons)	Philippines (thousands of gross tons)	Europe (thousands of gross tons)	Rest of the world (thousands of gross tons)	Total (thousands of gross tons)	Share (Percentage)
Bulk carriers	12 510 129	NA	5 729 387	148 469	666 846	NA	17 654	19 072 485	26.6
Container ships	16 577 611	11 325 793	1 735 128	NA	NA	NA	NA	29 638 532	41.3
Offshore supply	1 674 069	257 004	7 795	25 969	NA	358 042	83 236	2 406 115	3.4
Oil tankers	1 498 699	1 310 389	205 573	519 656	1 361	40 723	53 561	3 629 962	5.1
Gas garriers	1 710 487	6 817 843	220 425	NA	NA	NA	NA	8 748 755	12.2
General gargo	1 132 642	229 469	304 773	14 068	NA	141 729	142 995	1 965 676	2.7
Ferries and passenger ships	247 937	NA	19 773	3 170	NA	39 194	1 123 739	1 433 813	2
Chemical tankers	860 205	70 206	234 581	8 593	NA	18 842	9 678	1 202 105	1.7
Other	2 906 579	80 168	545 007	959	NA	26 142	34 685	3 593 540	5
Total	**39 118 358**	**20 090 872**	**9 002 442**	**720 884**	**668 207**	**624 672**	**1 465 548**	**71 690 983**	**100**
Share (percentage)	**54.6**	**28**	**12.6**	**1**	**0.9**	**0.9**	**2**	**100**	**0**



01



CONTAINER SHIPS

Move TEU and FEU boxes on global liner routes.

02



FEEDER VESSELS

Connect smaller ports with major hub terminals.

03



BULK CARRIERS

Carry dry bulk cargo such as grain, coal and ore.

04



TANKERS

Transport crude oil, products and liquid fuels.

05



CHEMICAL TANKERS

Move parcel chemicals with segregated cargo tanks.

06



GAS CARRIERS

Carry LNG, LPG and energy-transition gases.

07



RO-RO / PCTC SHIPS

Carry vehicles, EVs and rolling cargo.

08



REEFER SHIPS

Move chilled food, pharma and cold-chain cargo.

09



GENERAL CARGO SHIPS

Handle breakbulk and non-containerized freight.

10



MULTIPURPOSE VESSELS

Carry containers, bulk and project cargo.

11



HEAVY-LIFT VESSELS

Move oversized modules, turbines and machinery.

12



LASH / BARGE CARRIERS

Carry barges for inland and shallow-water routes.

4.3 Ship Breakers (Demolition/Recycling)

- Receives orders from shipowners to _____ _____ ships or ships who have reached the end of their life cycle.



Ship tonnage sold for scrapping, 2024

	Bangladesh (thousands of gross tons)	India (thousands of gross tons)	Türkiye (thousands of gross tons)	Pakistan (thousands of gross tons)	Denmark (thousands of gross tons)	Rest of the world (thousands of gross tons)	World total (thousands of gross tons)	Share (Percentage)
Bulk carriers	1 375.9	202.9	NA	400.6	NA	75.3	2 054.7	32.4
Container ships	203.5	584.9	113.6	14.5	NA	49.7	966.1	15.2
Offshore supply	187.2	289.6	14.9	1.1	310.7	9.1	812.5	12.8
Oil tankers	324.1	437.5	2.7	NA	NA	3.7	768	12.1
Liquefied gas carriers	400.6	289.9	NA	NA	NA	4.9	695.3	11
General cargo ships	214.2	128.7	135.2	38.5	5.5	43.8	565.9	8.9
Ferries and passenger ships	18.6	56.8	180.1	NA	0.8	10.7	267.1	4.2
Chemical tankers	2.1	46.6	NA	NA	NA	3.3	52	0.8
Other/n.a.	11.5	110.1	23	0.2	0.6	14.2	159.6	2.5
Total (thousands of gross tons)	**2 737.7**	**2 146.8**	**469.6**	**454.8**	**317.7**	**214.6**	**6 341.1**	**100**
Share (percentage)	**43.2**	**33.9**	**7.4**	**7.2**	**5.0**	**3.4**		

4.4 Port Terminal Operators

PSA The World's Port of Call

Lifting Global Trade. APM TERMINALS

- Operated by government (public) or private companies.
- Private companies purchases a contract with the local government to move cargo through a port at contracted minimum level of productivity.
- Measured in terms of how much cargo it moves in a given amount of time (e.g., *how many containers moved in an hour*).

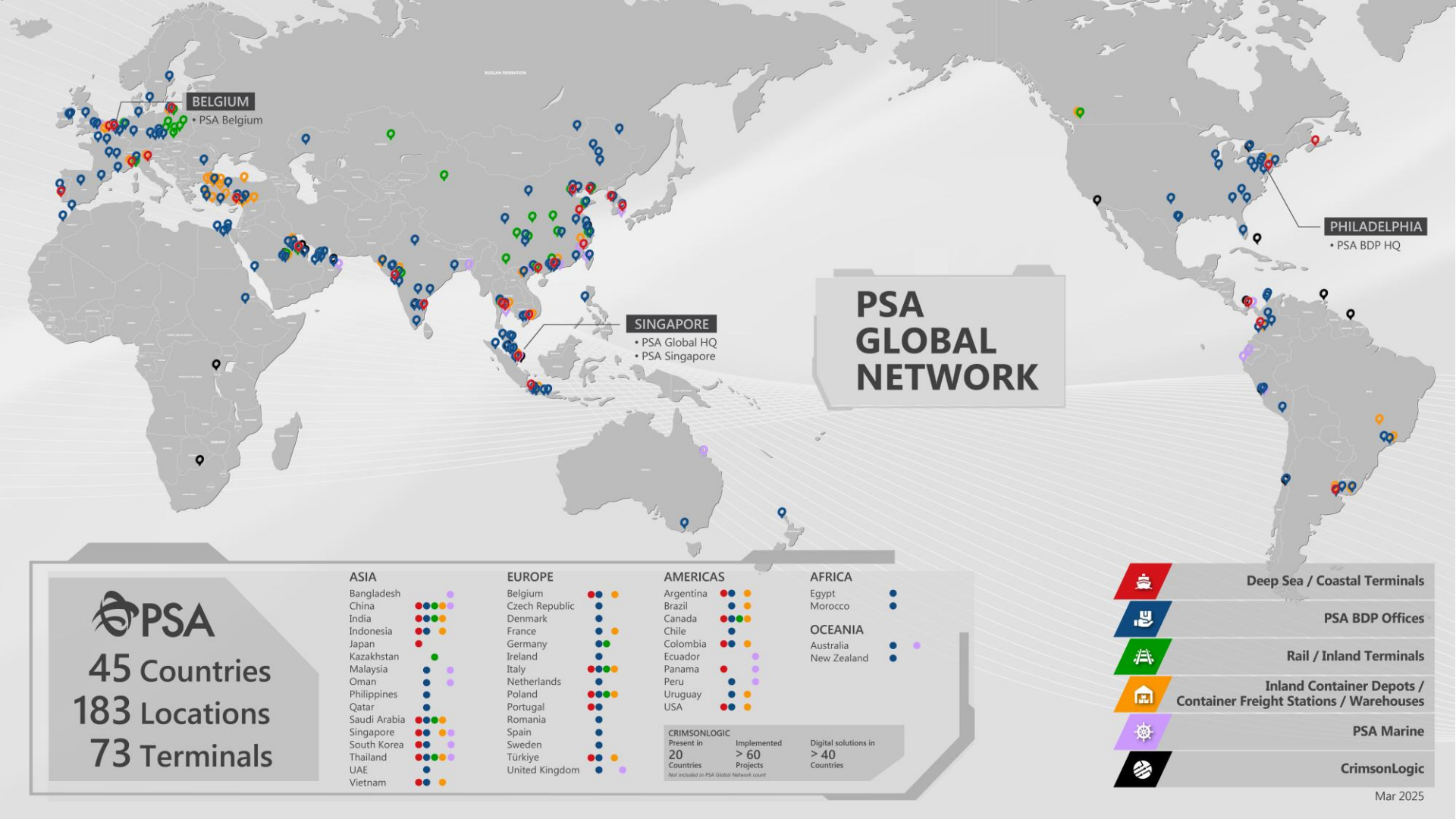
4.4 Port Terminal Operators

Global terminal operators' equity-based throughput league table, 2024

Ranking 2024	Operator	2024 Equity-adjusted throughput, 2024 (mteu)	2024 Volume growth/decline vs. 2023 (%)	2024 Share of global port throughput, 2024 (%)
1	PSA International	67.3	7.3%	7.2%
2	China Merchants Port	61.2	8.4%	6.6%
3	China COSCO SHIPPING	54.2	0.9%	5.8%
4	APM Terminals	53.2	8.9%	5.7%
5	Hutchison Ports	46.2	7.0%	5.0%
6	MSC Group	44.4	5.2%	4.8%
7	DP World	44.3	0.0%	4.8%
	subtotal	**370.9**	**5.5%**	**40.0%**
	Other GTOs	85.5	18.6%	9.2%
Global/international operators total		**456.4**	**7.7%**	**49.2%**

Source: Drewry's Global Container Terminal Operators Annual Review and Forecast 2025/26





PSA

45 Countries
183 Locations
73 Terminals

ASIA
Bangladesh
China
India
Indonesia
Japan
Kazakhstan
Malaysia
Oman
Philippines
Qatar
Saudi Arabia
Singapore
South Korea
Thailand
UAE
Vietnam

EUROPE
Belgium
Czech Republic
Denmark
France
Germany
Ireland
Italy
Netherlands
Poland
Portugal
Romania
Spain
Sweden
Türkiye
United Kingdom

AMERICAS
Argentina
Brazil
Canada
Chile
Colombia
Ecuador
Panama
Peru
Uruguay
USA

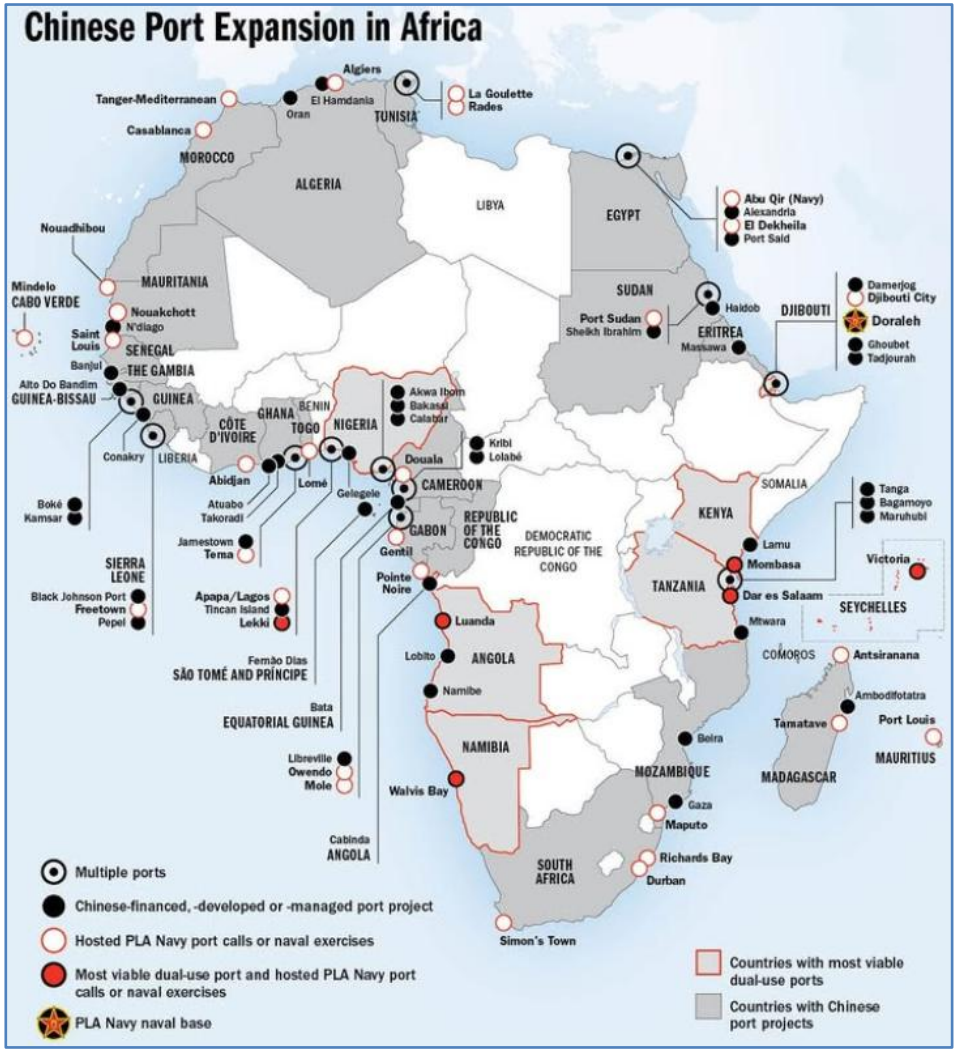
AFRICA
Egypt
Morocco

OCEANIA
Australia
New Zealand

CRIMSONLOGIC
Present in 20 Countries
Implemented > 60 Projects
Digital solutions in > 40 Countries
Not included in PSA Global Network count

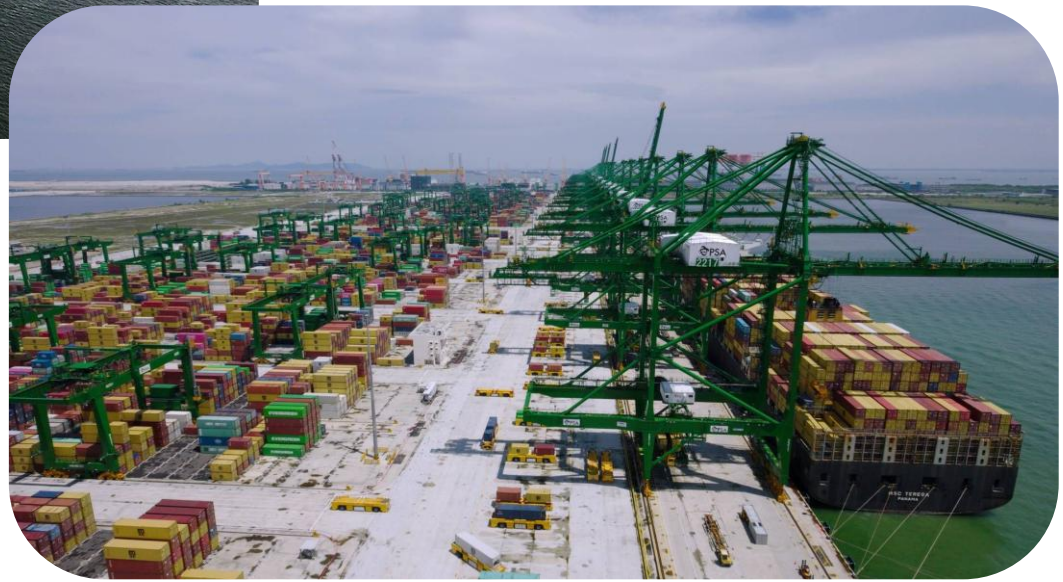
- Deep Sea / Coastal Terminals
- PSA BDP Offices
- Rail / Inland Terminals
- Inland Container Depots / Container Freight Stations / Warehouses
- PSA Marine
- CrimsonLogic

Mar 2025


Chinese Port Expansion in Africa
Algiers
El Hamdania
Oran
La Goulette
Rades
TUNISIA
Tanger-Mediterranean
Casablanca
MOROCCO
ALGERIA
LIBYA
EGYPT
Abu Qir (Navy)
Alexandria
El Dekheila
Port Said
Nouadhibou
MAURITANIA
Mindelo
CABO VERDE
Nouakchott
N'diago
Saint Louis
SENEGAL
Banjul
THE GAMBIA
Alto Do Bandim
GUINEA-BISSAU
GUINEA
Conakry
LIBERIA
BENIN
GHANA
TOGO
CÔTE D'IVOIRE
NIGERIA
Akwa Ibom
Bakassi
Calabar
Kribi
Lolabé
Douala
CAMEROON
Abidjan
Lomé
Gelegele
Atuabo
Takoradi
Jamestown
Tema
Boké
Kamsar
SIERRA LEONE
Black Johnson Port
Freetown
Pepel
Apapa/Lagos
Tincan Island
Lekki
Gentil
GABON
REPUBLIC OF THE CONGO
DEMOCRATIC REPUBLIC OF THE CONGO
Pointe Noire
Luanda
Lobito
ANGOLA
Namibe
Femão Dias
SÃO TOMÉ AND PRÍNCIPE
Bata
EQUATORIAL GUINEA
Libreville
Owendo
Mole
NAMIBIA
Walvis Bay
Cabinda
ANGOLA
SOUTH AFRICA
Simon's Town
Richards Bay
Durban
Maputo
Gaza
MOZAMBIQUE
Beira
SUDAN
Port Sudan
Sheikh Ibrahim
Haidob
ERITREA
Massawa
DJIBOUTI
Damerjog
Djibouti City
Doraleh
Ghoubet
Tadjourah
SOMALIA
KENYA
Lamu
Mombasa
TANZANIA
Dar es Salaam
Mtwara
Tanga
Bagamoyo
Maruhubi
Victoria
SEYCHELLES
COMOROS
Antsiranana
Ambodifotatra
Tamatave
Port Louis
MAURITIUS
MADAGASCAR
Multiple ports
Chinese-financed, -developed or -managed port project
Hosted PLA Navy port calls or naval exercises
Most viable dual-use port and hosted PLA Navy port calls or naval exercises
PLA Navy naval base
Countries with most viable dual-use ports
Countries with Chinese port projects



World’s largest cement terminal – Jurong Port



PSA Tuas Port

4.5 Charterers

a. may be a party _____ (e.g., *100,000 tons of coal*) and employ a shipbroker to find a ship to deliver the cargo.

b. may be a party, _____, who takes a vessel on charter (hire) for a specified period (e.g., *1/3/5/10 years*) from the shipowner and trades the ship to carry cargoes, hopefully at a profit, above the hire rate + other expenses.

4.6 Shipbrokers

- Specialist _____ between shipowners and charterers.
- Sale & purchase (S&P) brokers handle buying and selling of existing vessels in secondary market or newbuildings.
- Demolition brokers handle sale of vessels for scrapping.

4.7 Ship Managers

- Two main categories:

i. Manages its own ships and offers the same service to other ship owners.

ii. No ships of their own and solely providing ship management services to ship owners.

4.7.1 Ship Management Services

Crewing	**Where to employ workers from?**
Storing	**Crew and Technical (deck and engine room).**
Technical	**Marine and engineering superintendent.**
Insurance	**Hull & Machinery and 3rd Party.**
Operations	**Runs the ship. Close links with other departments.**

4.8 Marine Insurers





- A shipowner/cargo owner has to make the right decisions in choosing the correct combination of credible insurance cover at a reasonable cost.
- Marine insurers are specialized corporate entities, syndicates, or mutual associations that underwrite risks and provide financial protection for ships, cargoes, terminals and maritime liabilities.

4.8 Marine Insurers

- Key providers include:

  - **Commercial insurance companies**: Allianz Commercial and QBE Singapore that offers hull, machinery and cargo policies.

  - **P&I Clubs (Mutual Insurers)**: Non-profit associations owned by shipowners that mutually insure third-party liabilities and operational risks (more in Week 2/3).

  - **Lloyd's Syndicates**: specialized underwriting groups operating within the Lloyds market framework to cover complex maritime and war risks.

4.8 Marine Insurers

Ship War Risk Insurance Canceled After Iran Strikes

Reuters

Total Views: 6616

March 2, 2026

By Emily Chow and Jeslyn Lerh

REUTERS

SINGAPORE (Reuters) – Marine insurers are cancelling war risk coverage for vessels and oil shipping rates are set to surge further after the widening Iran conflict left at least three tankers damaged, a seafarer killed and 150 ships stranded around the Strait of Hormuz.

Iran has responded to U.S. and Israeli strikes that began on Saturday with retaliatory attacks that have sharply increased risks to commercial shipping in the past 24 hours.

4.9 Shipping Banks

CRÉDIT AGRICOLE
CORPORATE & INVESTMENT BANK

- Provides shipowners with a range of financial products and lending solutions.
- Services included: ship mortgages, sale and leaseback, initial public offerings (IPOs), green financing.
- Most banks have dedicated research teams to analyse risks and trends in the shipping market.

Seacon signs sale-and-leaseback deal with Bank of China for handysize bulker

The capital unlocked from the transaction will support Seacon's strategy of renewing and expanding its fleet

03 Jun 2024 | **NEWS**

Singapore tanker broker Vantage to raise $13m in US IPO

Vantage Corp is set to begin trading on the NYSE American today, using the proceeds to drive global expansion and digital innovation

12 Jun 2025 | **NEWS**

Hapag-Lloyd ties up green ship finance for latest newbuildings

- *Finance concluded for 12 ultra-large containerships and 12 mid-sized boxships ordered in China in October 2024*
- *Financing will be undertaken using Hapag-Lloyd's 'Green Financing Framework'*
- *Funding comprises a mix of internal funds, mortgage loans, leasing and syndicated credit facility*

04 Feb 2025 | **NEWS**

The financing arrangements for the 24 newbuildings comprises four components. They include some $900m of Hapag-Lloyd's own funds while $500m will be provided by two, undisclosed, banks in the form of bilateral mortgage loans.

The third component, of $1.8bn, will be provided via three leasing structures with the final $1.1bn financed via a syndicated credit facility backed by the China Export & Credit Insurance Corporation (Sinosure).



4.10 Ship Agents

- "The party representing the ship's owner and/or charterer (**the Principal**) in port…responsible for arranging the *port, berth, all relevant port and husbandry services, tending to the requirements of the Master and crew, clearing the ship with the port and other authorities along with releasing or receiving cargo* on behalf of the Principal."
- Commonly referred to as ____ *agents or* _____ *agents*.

4.11 Classification Societies



4.11 Classification Societies



- Makes unique contribution to maritime safety and regulation through technical support, compliance verification and R&D.
- Consists of __ member societies.
- Covers more than 90% of world's carrying tonnage.

4.11 Classification Societies

- Tasks include:
  1. Classification on behalf of shipowner – certify ships for compliance with their own class rules regarding type, construction, equipment, maintenance and surveys of ships.
  2. Performance of statutory tasks on behalf of flag states – ensure compliance with IMO safety requirements and issue international ship safety certificates.

4.12 Government/Port Authories

- Deciding on policies to promote the development and growth of a port.
- Responsible for ensuring that national and international regulations are adhered to in port and surrounding jurisdiction areas.

  E.g., *International Safety Management Code*

MPA
SINGAPORE

4.13 The Singapore Chamber of Maritime Arbitration

- A specialist arbitration institution which provides a neutral, cost-effective and flexible framework for maritime and international trade arbitrations.



Review Questions

4. A _____ equips and exploits a ship to deliver cargoes at an agreed freight rate.

a) Ship builder

b) Ship owner

c) Terminal operator

d) Ship broker

Answer:

Review Questions

5. Which of the following is NOT among the top three ship builders in the world?

a) China

b) Japan

c) South Korea

d) Philippines

Answer:

Review Questions

6. Which of the following is NOT among the top 3 ship breakers in the world?

a) India

b) Pakistan

c) Sri Lanka

d) Bangladesh

Answer:

LO5

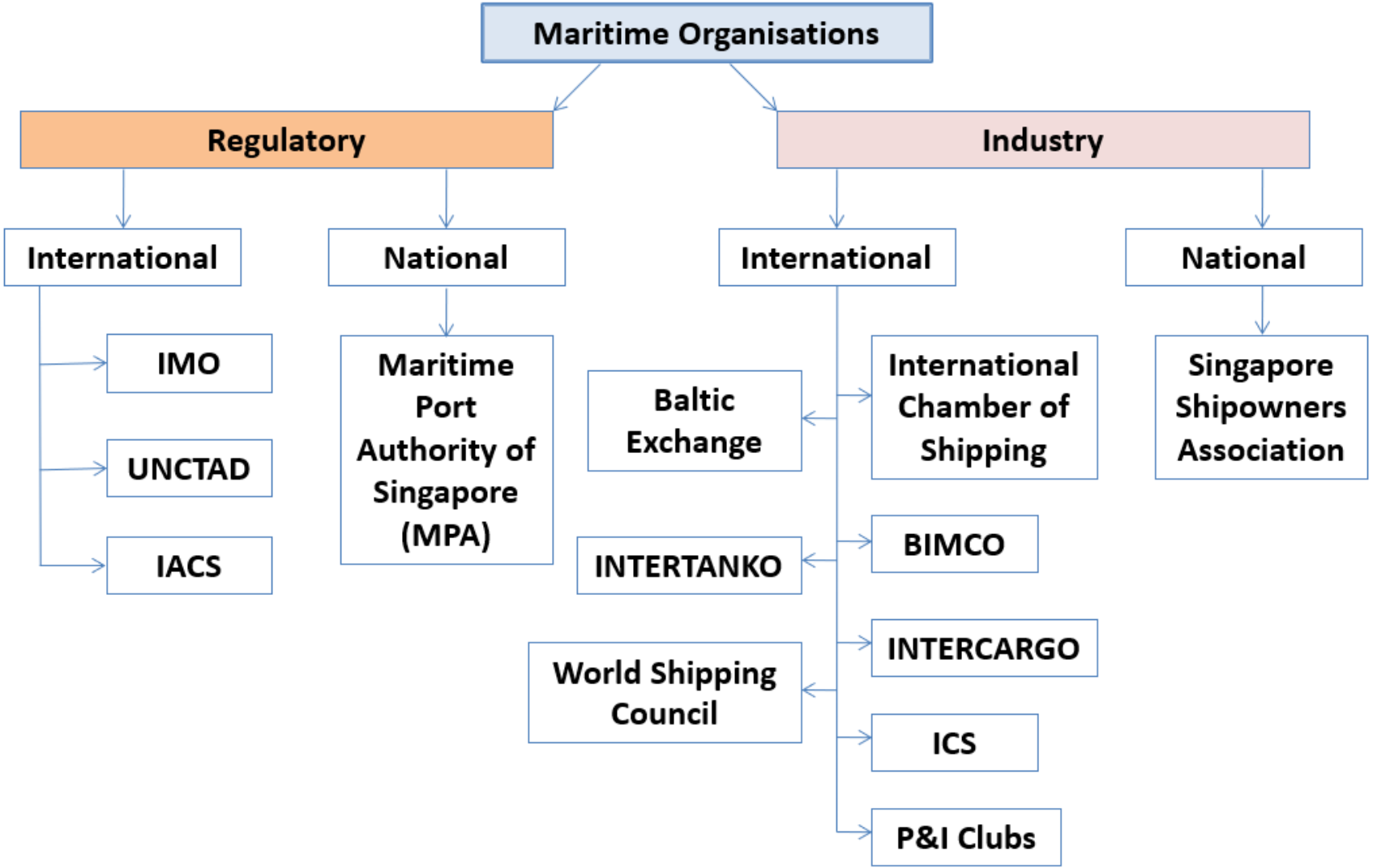
5. Maritime Organisations (Regulatory and International)

5. Maritime Organisations

- Shipping is regulated and influenced by a number of international, governmental (national level) and private organisations.

- Internationally, there never has been one single controlling organisation.

- Commercial and law reform agencies from the late 19th century have been instrumental in the coalescing of international opinion into treaties and conventions.


Maritime Organisations
Regulatory
Industry
International
National
International
National
IMO
UNCTAD
IACS
Maritime Port Authority of Singapore (MPA)
Baltic Exchange
International Chamber of Shipping
Singapore Shipowners Association
BIMCO
INTERTANKO
INTERCARGO
World Shipping Council
ICS
P&I Clubs

5.1 Maritime Organisations (Regulatory)



- **International Maritime Organisation (IMO)**

  - the United Nations specialised agency with responsibility for the safety and security of shipping and prevention of marine pollution by ships.

  - role is to create a level playing-field so that ship operators cannot address their financial issues by cutting corners and comprising on safety, security and environmental performance.

5.1 Maritime Organisations (Regulatory)

- **International Maritime Organisation (IMO)**
  - established to adopt legislation.
  - national governments are responsible for implementing it.

  - most important conventions:
  (a) International Convention for the Safety of Life at Sea (**SOLAS**);
  (b) International Convention for the Prevention of Pollution from Ships (**MARPOL**);
  (c) International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (**STCW**)

5.1 Maritime Organisations (Regulatory)

- **UN Conference on Trade and Development (UNCTAD)**

  - principal organ of the UN dealing with trade, investment and development issues.

  - primary objective is to formulate policies relating to all aspects of development include trade, aid, transport, finance and technology.

  - publishes "*Review of Maritime Transport*", annually.



5.1 Maritime Organisations (Industry)

- **International Chamber of Shipping (ICS)**

  - world's principal shipping organisation, representing around 80% of the world's merchant tonnage through membership of national shipowners' association.

  - concerns with all technical, legal, employment affairs and policy issues.

  - unique because it represents the global interests of all different trades in the industry.

5.1 Maritime Organisations (Industry)

- **Baltic and International Maritime Council (BIMCO)**

  - world's largest international shipping association with 2,300 members in 130 countries.

  - providing forum for exchange of information.

  - facilitate commercial operations by developing standard contracts and clauses.

  E.g. GENCON, GUARDCON, SALEFORM2012



The Baltic Exchange is the world's only independent source of **maritime market information** for the trading and settlement of physical and derivative contracts.

Baltic Exchange members are responsible for a **large proportion of all dry cargo and tanker fixtures as well as the sale and purchase of merchant vessels**.

The exchange provides **daily freight market prices** and **maritime shipping cost indices** which are used to settle freight futures (known as **Forward Freight Agreements** or FFAs)

Membership of the Baltic Exchange is not just limited to **shipbrokers**, **charterers** and **shipowners**, but also includes **financial institutions**, **maritime lawyers**, **educators**, **insurers** and **related associations**.

Baltic Exchange members undertake to commit to a code of conduct – **Our Word Our Bond**, overseen by the Baltic.

BALTIC DRY INDEX



The Baltic Dry Index (BDI) is a measure of what it costs to ship raw materials, including coal, iron ore and grain, around the world. The shipping or sea freight rates are issued daily by the Baltic Exchange. To compile the index, members of the Baltic Exchange call dry bulk shippers around the world to see what their prices are for more than 20 different shipping routes around the globe.

The data are then weighted together to create both the overall BDI and the size specific Supramax, Panamax and Capesize indices.

5.1 Maritime Organisations (Industry)

- **International Association of Independent Tanker Owners**

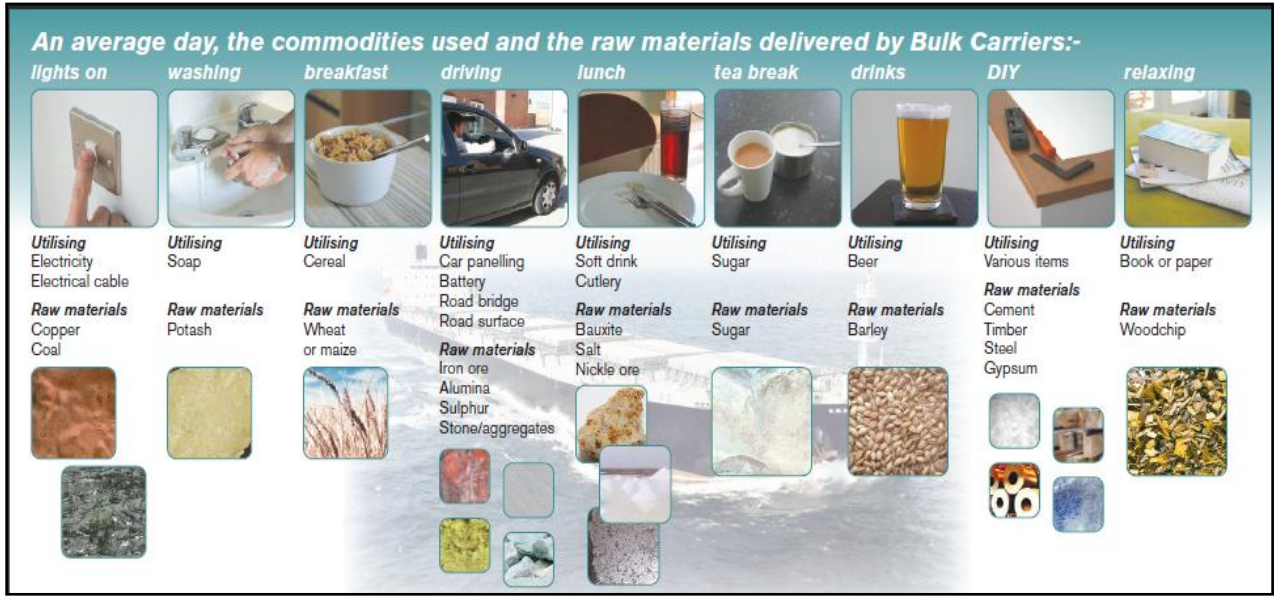
  - open to independent tanker owners and operators of oil and chemical tankers, i.e. non-oil companies and non-state-controlled tanker owners.

  - a forum where industry meets, policies discussed, and statements created.



International Association of Dry Cargo Shipowners

Main role is to work with members, regulators and other shipping associations to ensure that shipping operates safely, efficiently, environmentally and profitably.



INTERCARGO | Global Association of Dry Cargo Shipowners

5.1 Maritime Organisations (Industry)

- **World Shipping Council**





MAERSK

- goal is to provide a coordinated voice for the liner shipping industry.

- operate approximately **90 percent** of the global liner ship capacity, providing approximately 400 regularly scheduled services linking the continents of the world.

- includes world's ten largest container shipping companies.

5.1 Maritime Organisations (Industry)

- **Institute of Chartered Shipbrokers (ICS)**

  - founded in 1911 as the professional body for all members of the commercial shipping industry worldwide.

  - provides the shipping industry with highly qualified professionals being the only internationally recognized maritime professional body.

  - represents shipbrokers, ship managers and agents throughout the world.

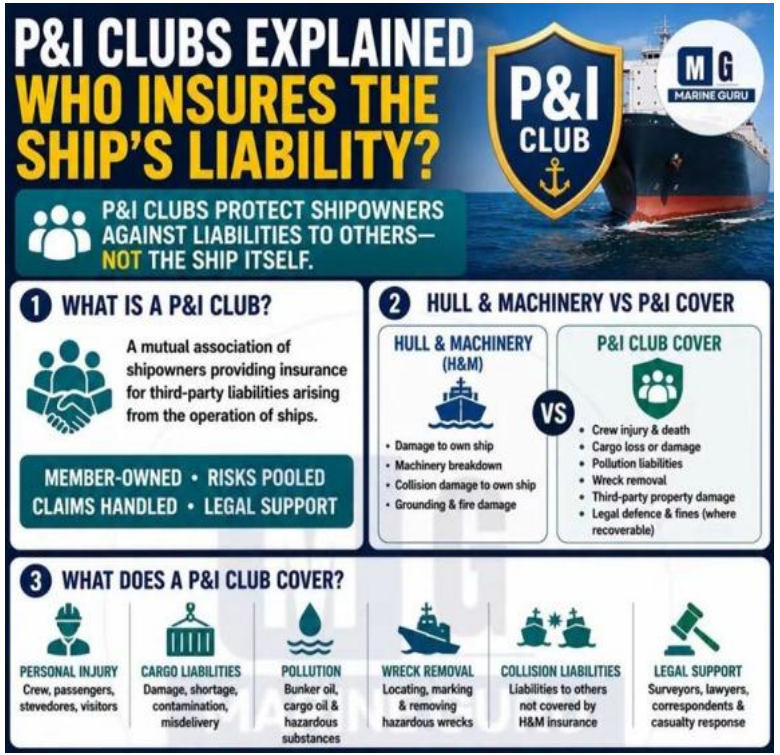
5.1 Maritime Organisations (Industry)

- **Protection and Indemnity (P&I) Clubs**

  - A marine insurance company only provides '*hull and machinery*' cover for shipowners, and '*cargo cover*' for cargo owners.

  - Provides insurance for third party risks such as damage to cargo, collision, war risk and environmentally pollution such as oil spills.

  - only ship owners, operators and charterers with solid reputations are allowed to join a P&I club.



UK P&I CLUB

5.1 Maritime Organisations (Industry)

- **Organization of the Petroleum Exporting Countries (OPEC)**

  - an international organization and economic cartel whose mission is to coordinate the policies of oil-producing countries.

  - goal: influence world oil prices through economic means.

  - consists of 11 countries: 5 from Middle East, 6 from Africa and 1 from South America.

OPEC+

Also known as the **Vienna Group**, was formed in late **2016** to regulate **global oil supply**.

PMF IAS

ORGANIZATION OF THE PETROLEUM EXPORTING COUNTRIES



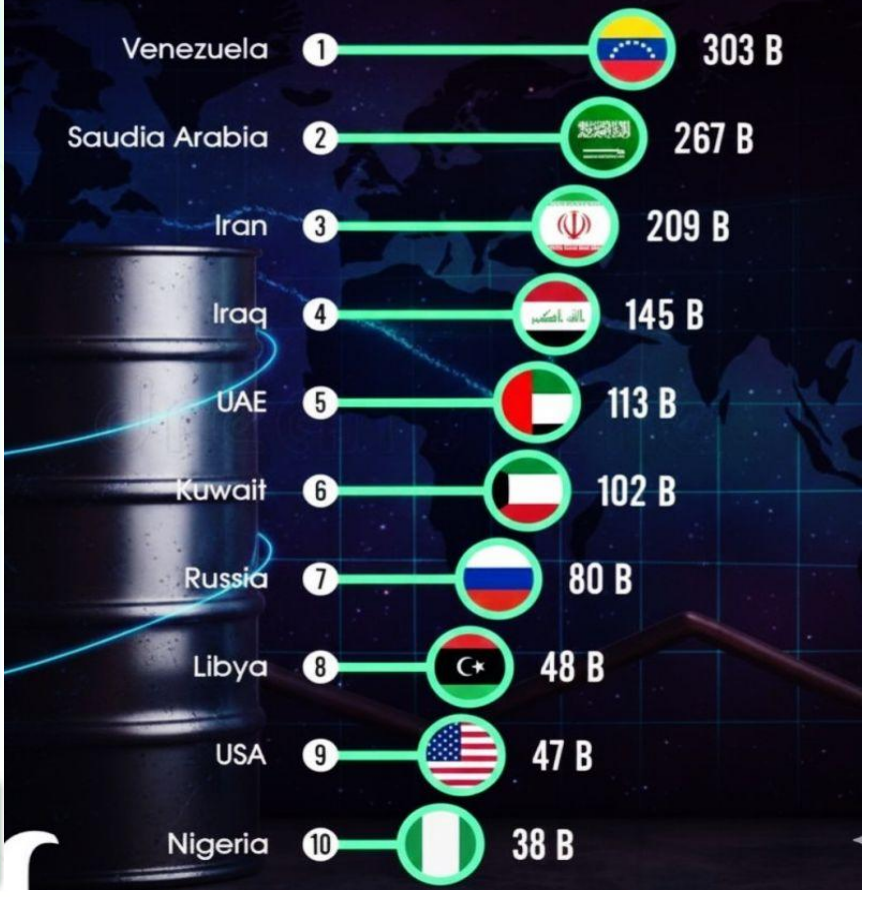


OPEC Members + Non OPEC Members → OPEC +

Saudi Arabia is the **largest producer** in OPEC+, followed by **Russia** and **Iraq**.

It currently controls **44% of global oil production** and **90%** of the world's proven **crude oil reserves**.

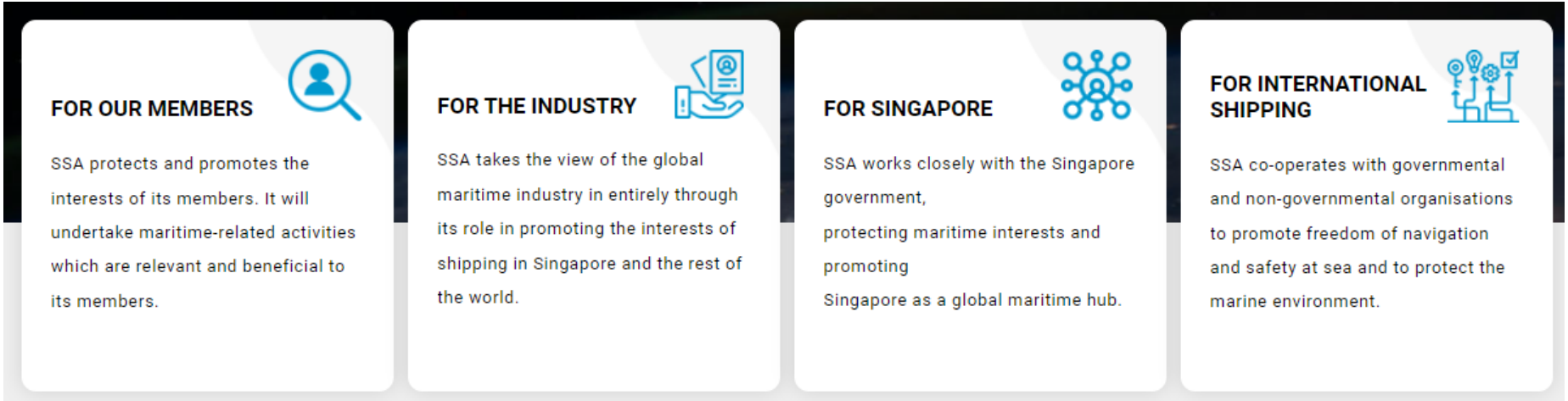
Countries With The Largest Oil Reserves



5.1 Maritime Organisations (Industry)

SSA SINGAPORE SHIPPING ASSOCIATION

- Formed in 1985.
- Represents over 485 member companies.



Review Questions

7. The Baltic Dry Index is a compilation of the freight rates of ___ different shipping routes around the world.

a. 17

b. 18

c. 19

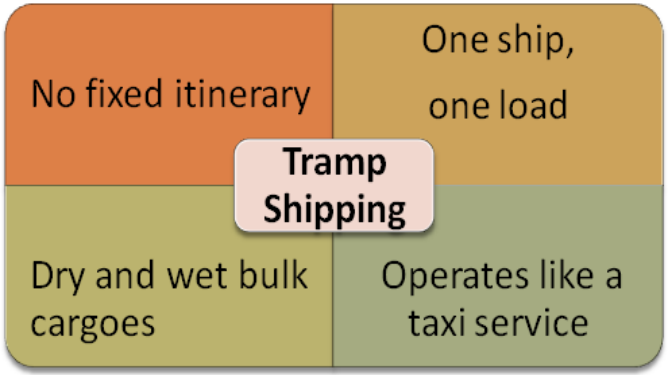
d. more than 20

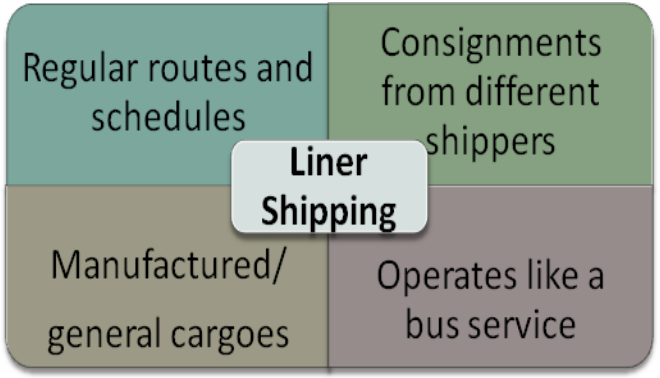
Answer:

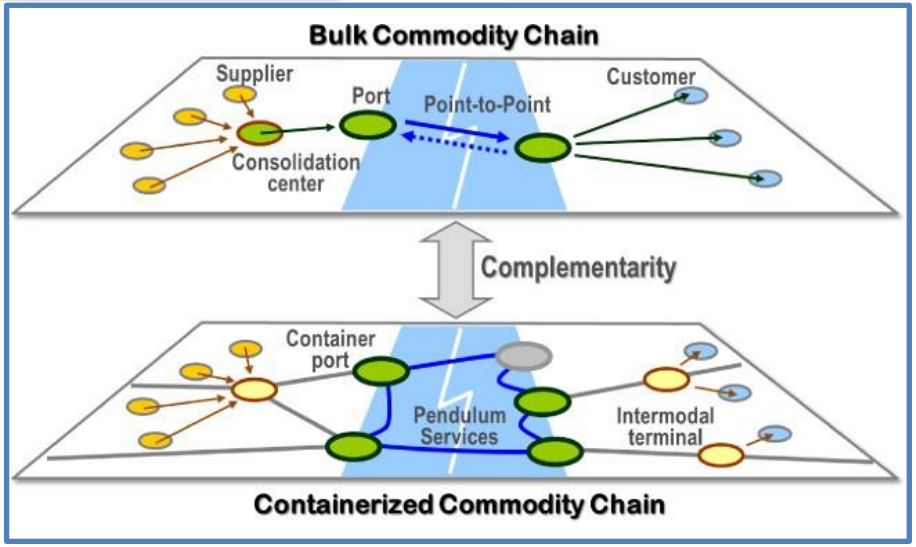
LO6

6. Two Main Types of Shipping Services

6. Two Main Types of Shipping Services



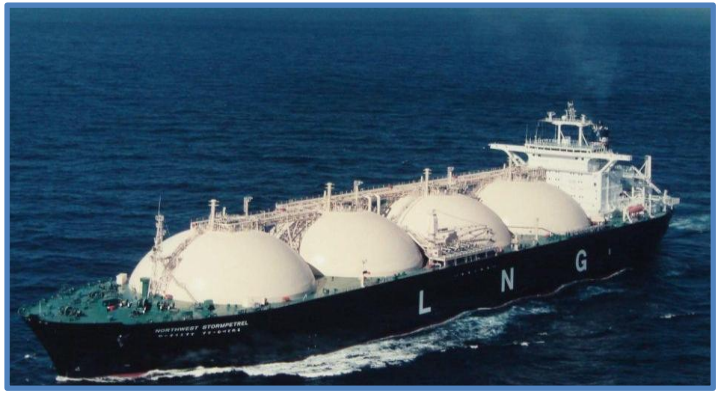




6.1 Tramp Shipping



Oil tanker



Gas Carrier



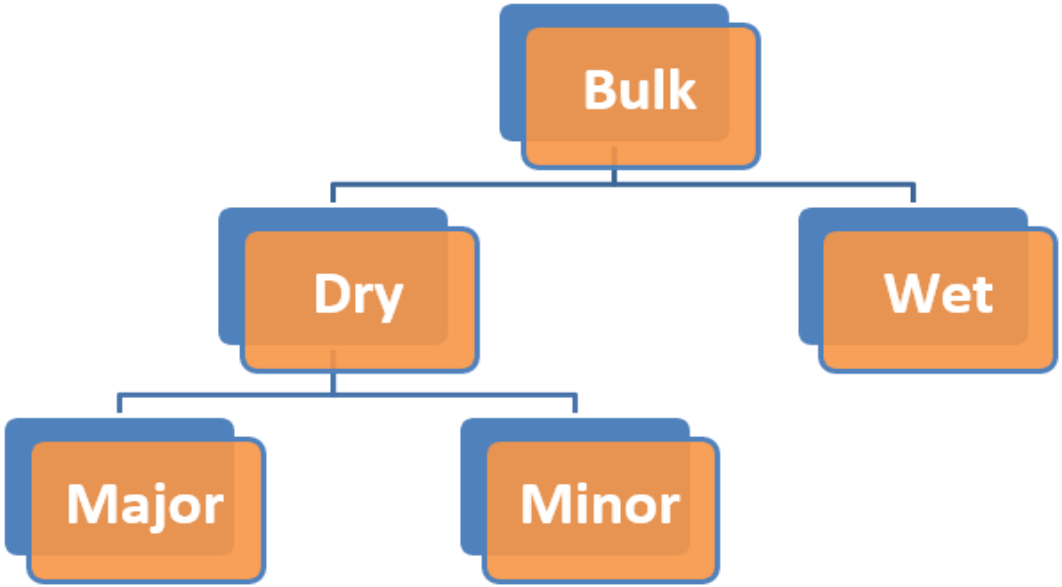
Dry Bulk Ship

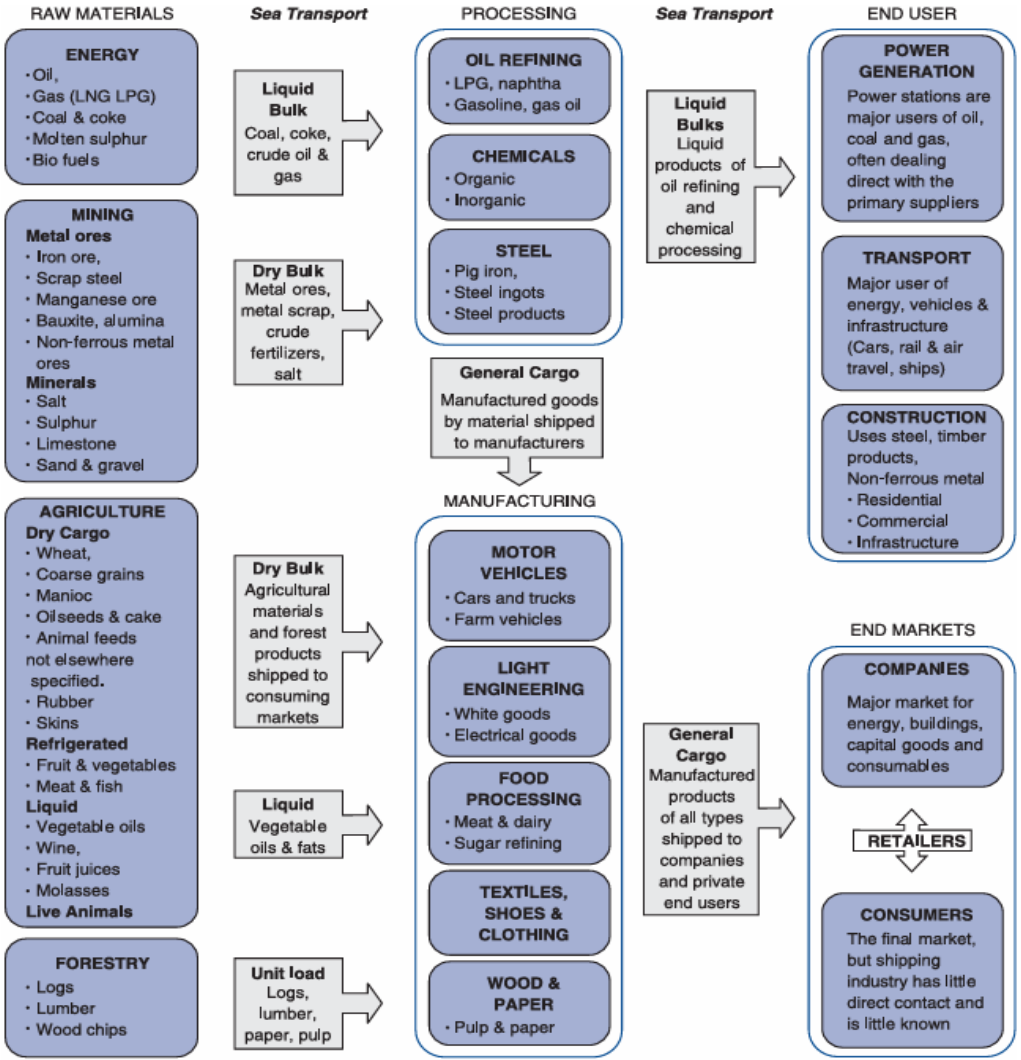


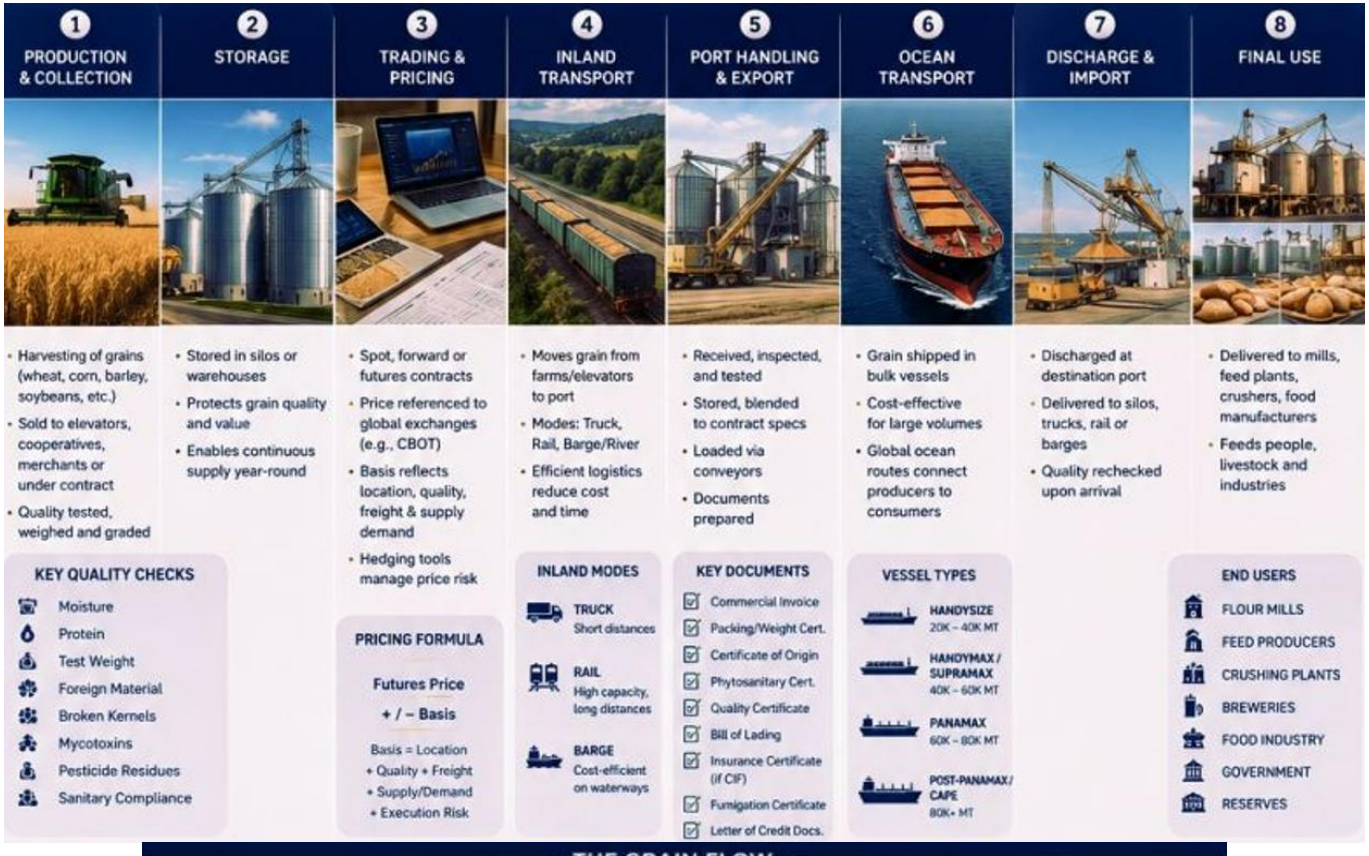
Dry Bulk Cargo

6.1 Tramp Shipping

- Tramp shipping is also known as **bulk** shipping.
- Divided into dry and wet (liquid) bulk cargoes.




RAW MATERIALS
Sea Transport
PROCESSING
Sea Transport
END USER
ENERGY
·Oil,
·Gas (LNG LPG)
·Coal & coke
·Molten sulphur
·Bio fuels
MINING
Metal ores
· Iron ore,
· Scrap steel
· Manganese ore
· Bauxite, alumina
· Non-ferrous metal ores
Minerals
· Salt
· Sulphur
· Limestone
· Sand & gravel
AGRICULTURE
Dry Cargo
· Wheat,
· Coarse grains
· Manioc
· Oilseeds & cake
· Animal feeds not elsewhere specified.
· Rubber
· Skins
Refrigerated
· Fruit & vegetables
· Meat & fish
Liquid
· Vegetable oils
· Wine,
· Fruit juices
· Molasses
Live Animals
FORESTRY
· Logs
· Lumber
· Wood chips
Liquid Bulk
Coal, coke, crude oil & gas
Dry Bulk
Metal ores, metal scrap, crude fertilizers, salt
Dry Bulk
Agricultural materials and forest products shipped to consuming markets
Liquid
Vegetable oils & fats
Unit load
Logs, lumber, paper, pulp
OIL REFINING
· LPG, naphtha
· Gasoline, gas oil
CHEMICALS
· Organic
· Inorganic
STEEL
· Pig iron,
· Steel ingots
· Steel products
General Cargo
Manufactured goods by material shipped to manufacturers
MANUFACTURING
MOTOR VEHICLES
· Cars and trucks
· Farm vehicles
LIGHT ENGINEERING
· White goods
· Electrical goods
FOOD PROCESSING
· Meat & dairy
· Sugar refining
TEXTILES, SHOES & CLOTHING
WOOD & PAPER
· Pulp & paper
Liquid Bulks
Liquid products of oil refining and chemical processing
General Cargo
Manufactured products of all types shipped to companies and private end users
POWER GENERATION
Power stations are major users of oil, coal and gas, often dealing direct with the primary suppliers
TRANSPORT
Major user of energy, vehicles & infrastructure (Cars, rail & air travel, ships)
CONSTRUCTION
Uses steel, timber products, Non-ferrous metal
· Residential
· Commercial
· Infrastructure
END MARKETS
COMPANIES
Major market for energy, buildings, capital goods and consumables
RETAILERS
CONSUMERS
The final market, but shipping industry has little direct contact and is little known


1
PRODUCTION & COLLECTION
2
STORAGE
3
TRADING & PRICING
4
INLAND TRANSPORT
5
PORT HANDLING & EXPORT
6
OCEAN TRANSPORT
7
DISCHARGE & IMPORT
8
FINAL USE
• Harvesting of grains (wheat, corn, barley, soybeans, etc.)
• Sold to elevators, cooperatives, merchants or under contract
• Quality tested, weighed and graded
• Stored in silos or warehouses
• Protects grain quality and value
• Enables continuous supply year-round
• Spot, forward or futures contracts
• Price referenced to global exchanges (e.g., CBOT)
• Basis reflects location, quality, freight & supply demand
• Hedging tools manage price risk
• Moves grain from farms/elevators to port
• Modes: Truck, Rail, Barge/River
• Efficient logistics reduce cost and time
• Received, inspected, and tested
• Stored, blended to contract specs
• Loaded via conveyors
• Documents prepared
• Grain shipped in bulk vessels
• Cost-effective for large volumes
• Global ocean routes connect producers to consumers
• Discharged at destination port
• Delivered to silos, trucks, rail or barges
• Quality rechecked upon arrival
• Delivered to mills, feed plants, crushers, food manufacturers
• Feeds people, livestock and industries
KEY QUALITY CHECKS
Moisture
Protein
Test Weight
Foreign Material
Broken Kernels
Mycotoxins
Pesticide Residues
Sanitary Compliance
PRICING FORMULA
Futures Price
+ / – Basis
Basis = Location + Quality + Freight + Supply/Demand + Execution Risk
INLAND MODES
TRUCK
Short distances
RAIL
High capacity, long distances
BARGE
Cost-efficient on waterways
KEY DOCUMENTS
Commercial Invoice
Packing/Weight Cert.
Certificate of Origin
Phytosanitary Cert.
Quality Certificate
Bill of Lading
Insurance Certificate (if CIF)
Fumigation Certificate
Letter of Credit Docs.
VESSEL TYPES
HANDYSIZE
20K – 40K MT
HANDYMAX / SUPRAMAX
40K – 60K MT
PANAMAX
60K – 80K MT
POST-PANAMAX / CAPE
80K+ MT
END USERS
FLOUR MILLS
FEED PRODUCERS
CRUSHING PLANTS
BREWERIES
FOOD INDUSTRY
GOVERNMENT
RESERVES


THE GRAIN FLOW
FARMER
ELEVATOR / COLLECTION
INLAND TRANSPORT
EXPORT TERMINAL
OCEAN TRANSPORT
IMPORT TERMINAL
MILL / FEED / INDUSTRY
CONSUMER

6.1 Tramp Shipping (Dry Bulk)

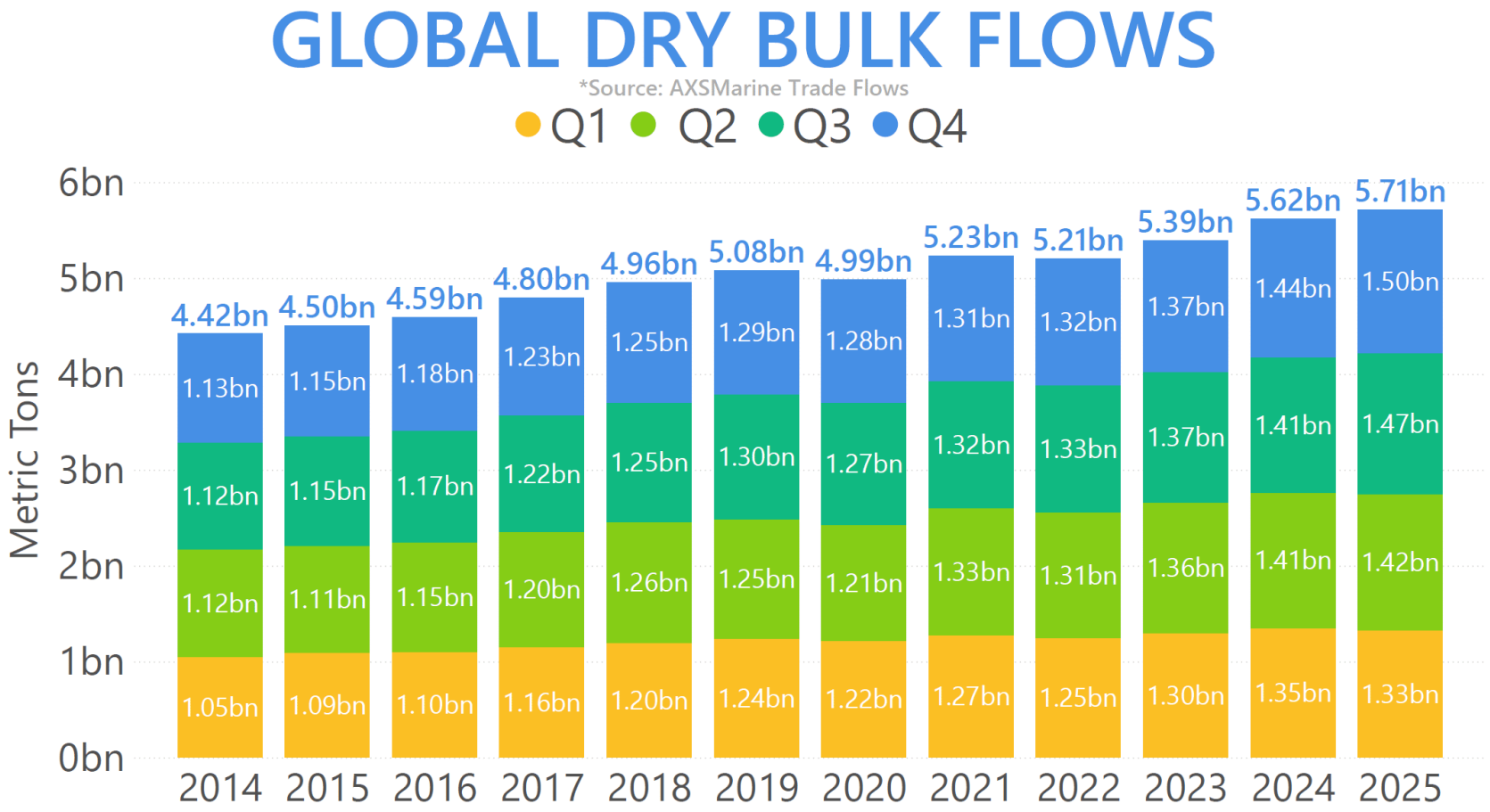


- Five major dry bulks – iron ore, grain, coal, phosphates and bauxite.
- Minor dry bulks – steel products, steel crap, cement, sugar, salt, forest products and wood chips.
- Wet bulk – crude oil, oil products, liquid chemicals such as caustic soda, vegetable oils and wine.





6.1 Tramp Shipping (Dry Bulk)

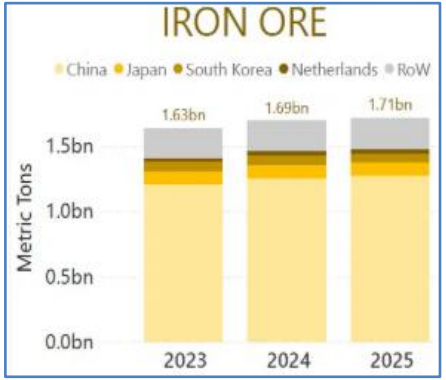


6.1 Tramp Shipping (Dry Bulk)

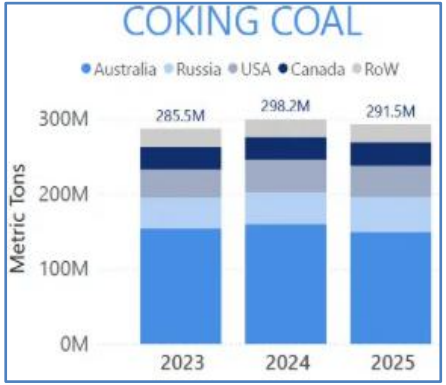
Exporter



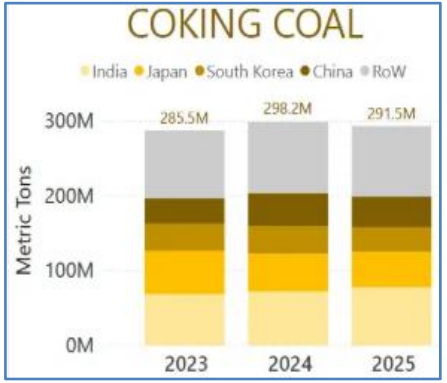
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Exporter



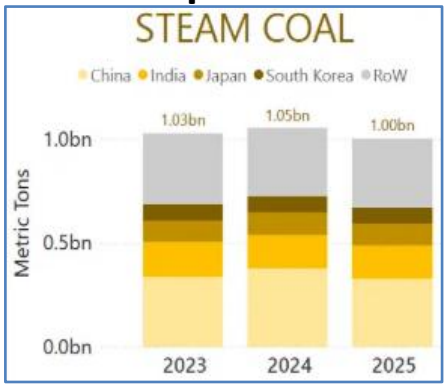
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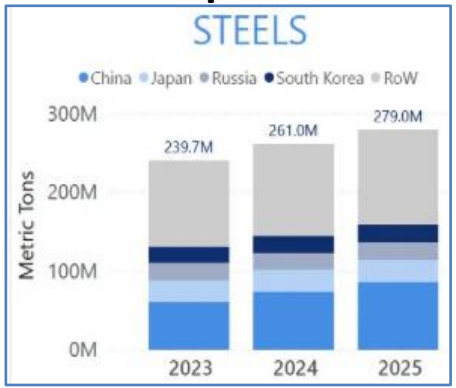
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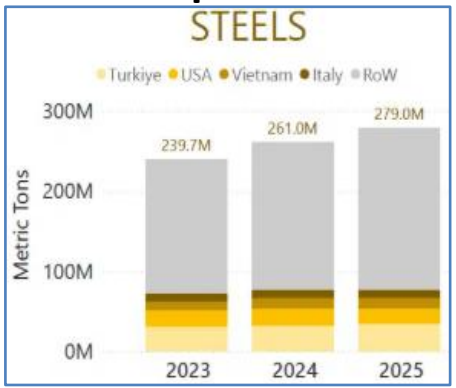
Importer



Exporter

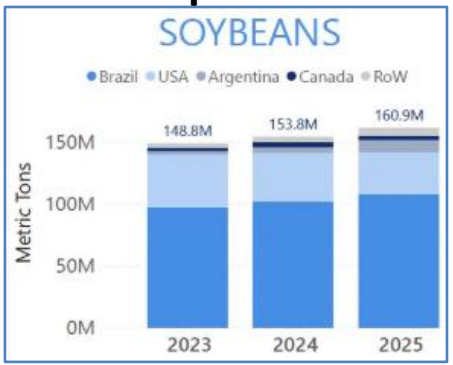


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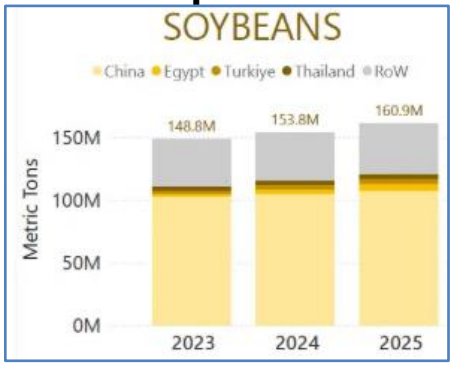


6.1 Tramp Shipping (Dry Bulk)

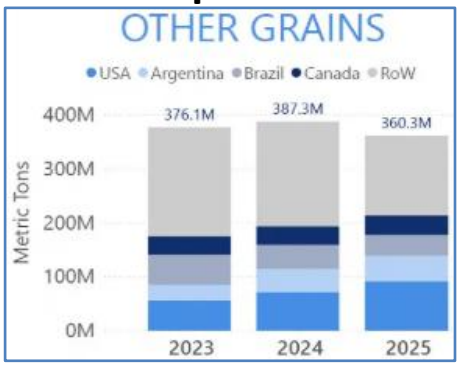
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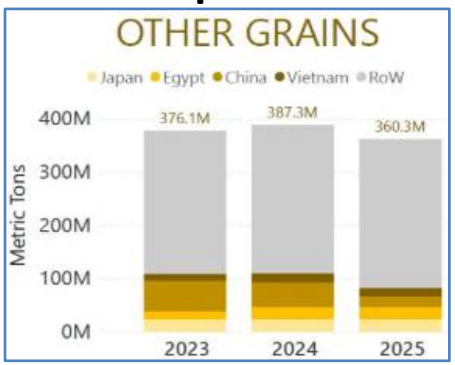
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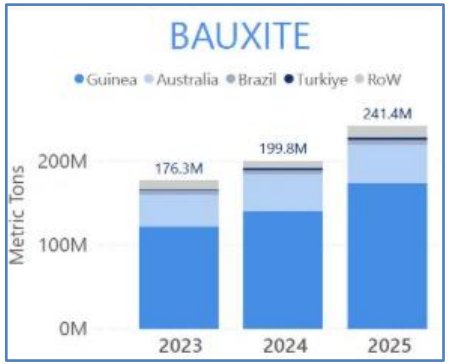
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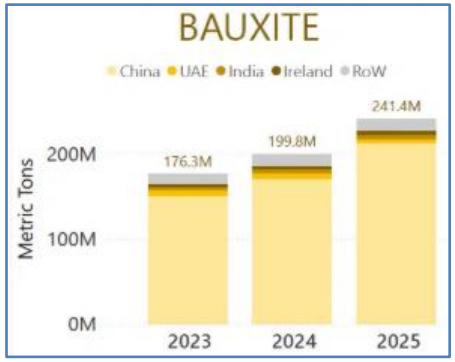
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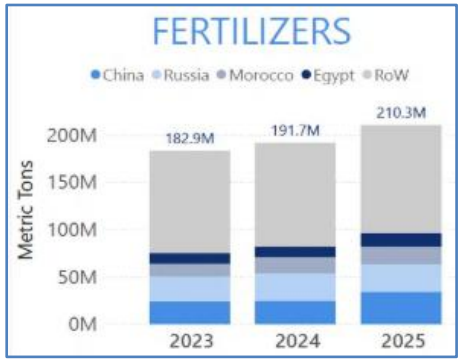
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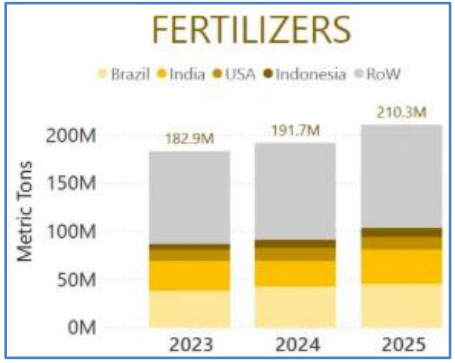
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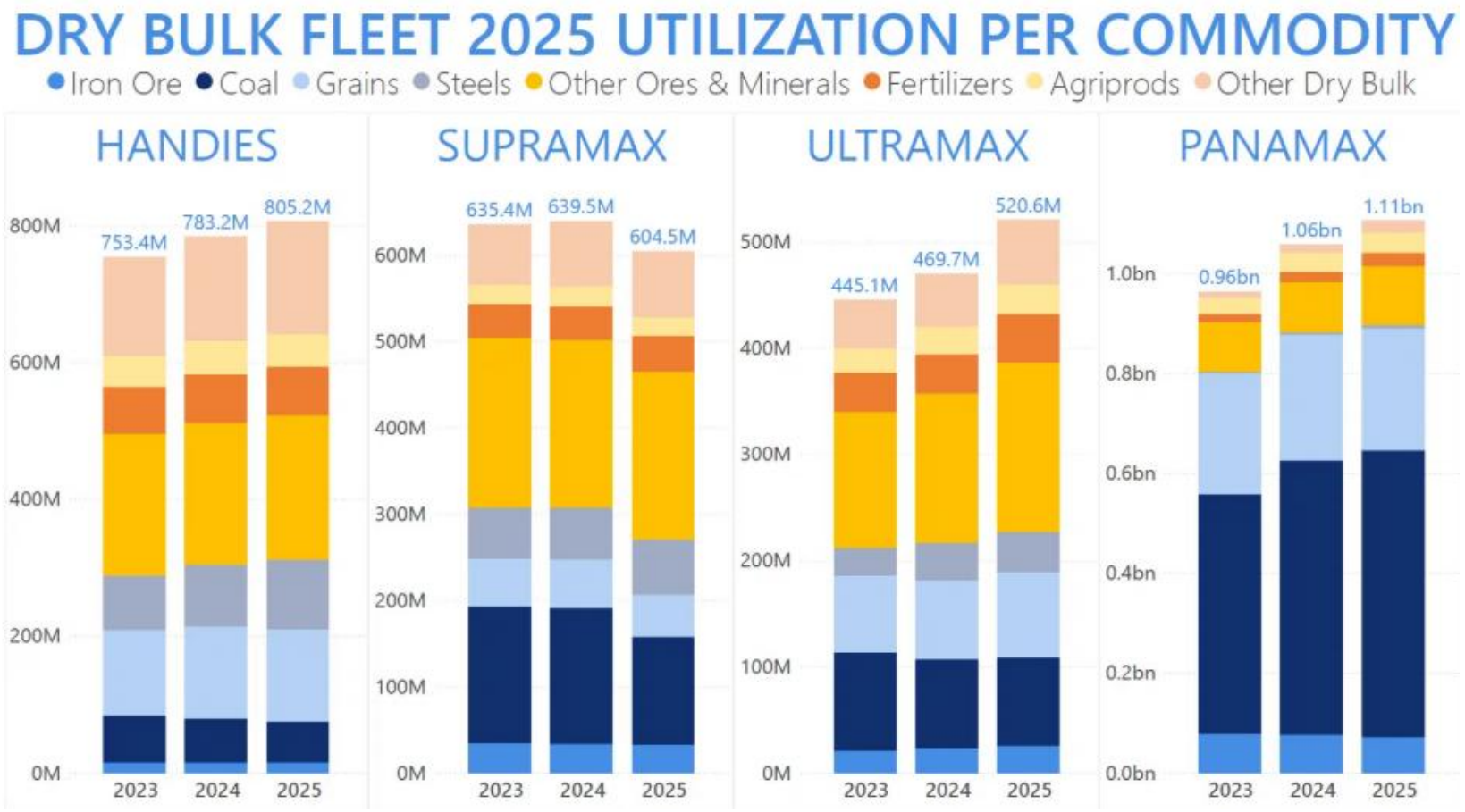


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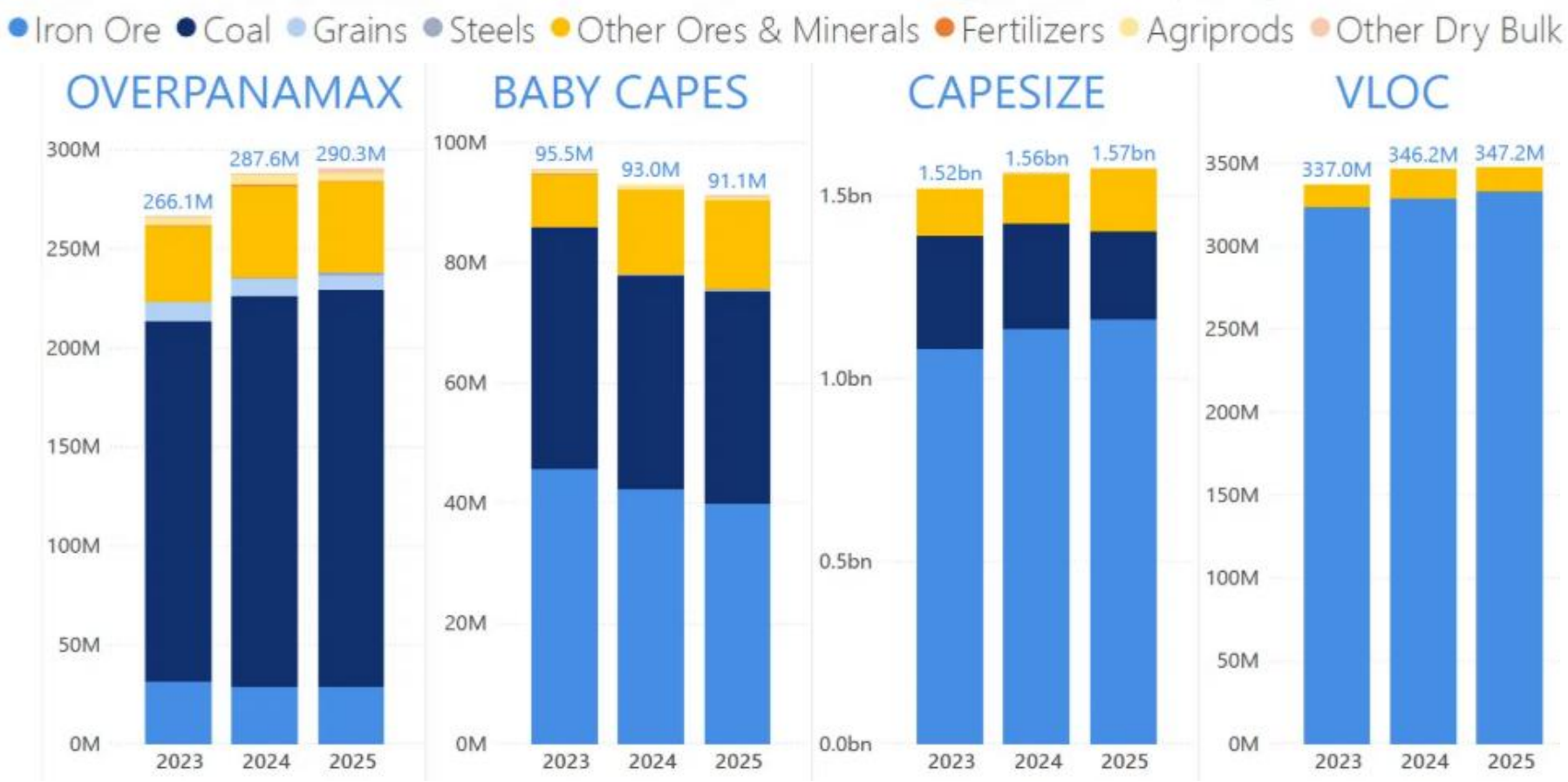


[Key Dry Bulk Data Trends: January 2026 - AXSMarine]

6.1 Tramp Shipping (Dry Bulk)



6.1 Tramp Shipping (Dry Bulk)

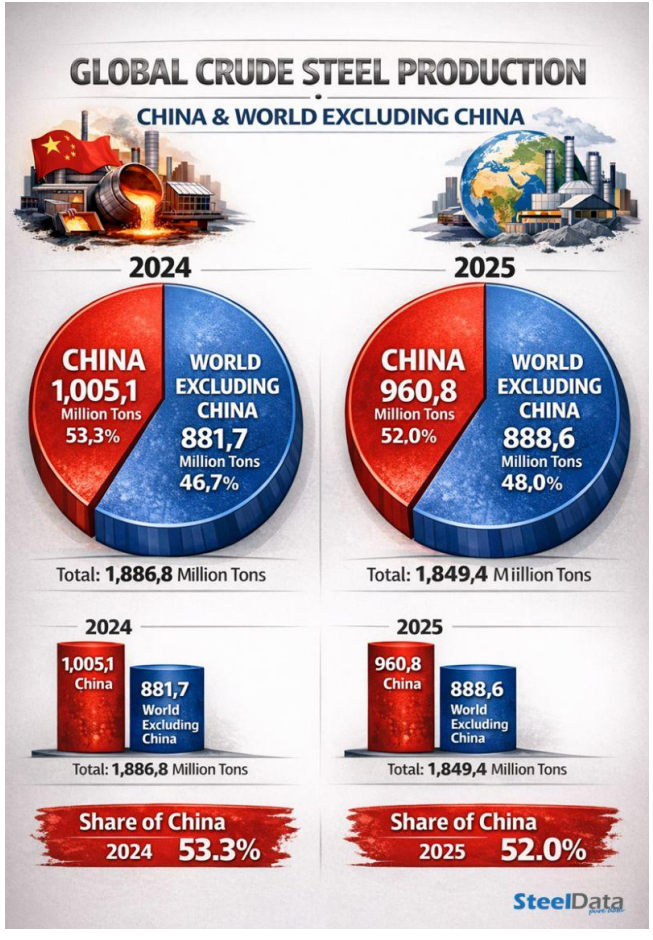


Global Iron Ore Trade

Rank	Country	2025 (mnt)	2024 (mnt)
1	Australia	↑ 924	902
2	Brazil	↑ 420	390
3	South Africa	↑ 65,7	61,8
4	Canada	↑ 61,9	60,8
5	Ukraine	↓ 31,9	35,7
6	India	↓ 26,4	37,7
7	Iran	↑ 24,8	22,6
8	Sweden	↑ 22,8	19,9
9	China	↓ 21,8	24,6
10	Peru	↓ 20,3	22,1
11	Netherlands	↑ 20,3	18,5
12	Chile	↓ 15,6	15,6
13	Russia	↑ 13,6	9,6
14	Sierra Leone	↑ 12,3	9,4
15	United States	↓ 5,2	10,2
Total		1760	1715



Global Steel Trade



22 July 2026

China's iron ore production fell by 7% y/y in 1H2026

Yuriy Grigorenko

At the same time, ore imports rose by 6.3% y/y over the same period

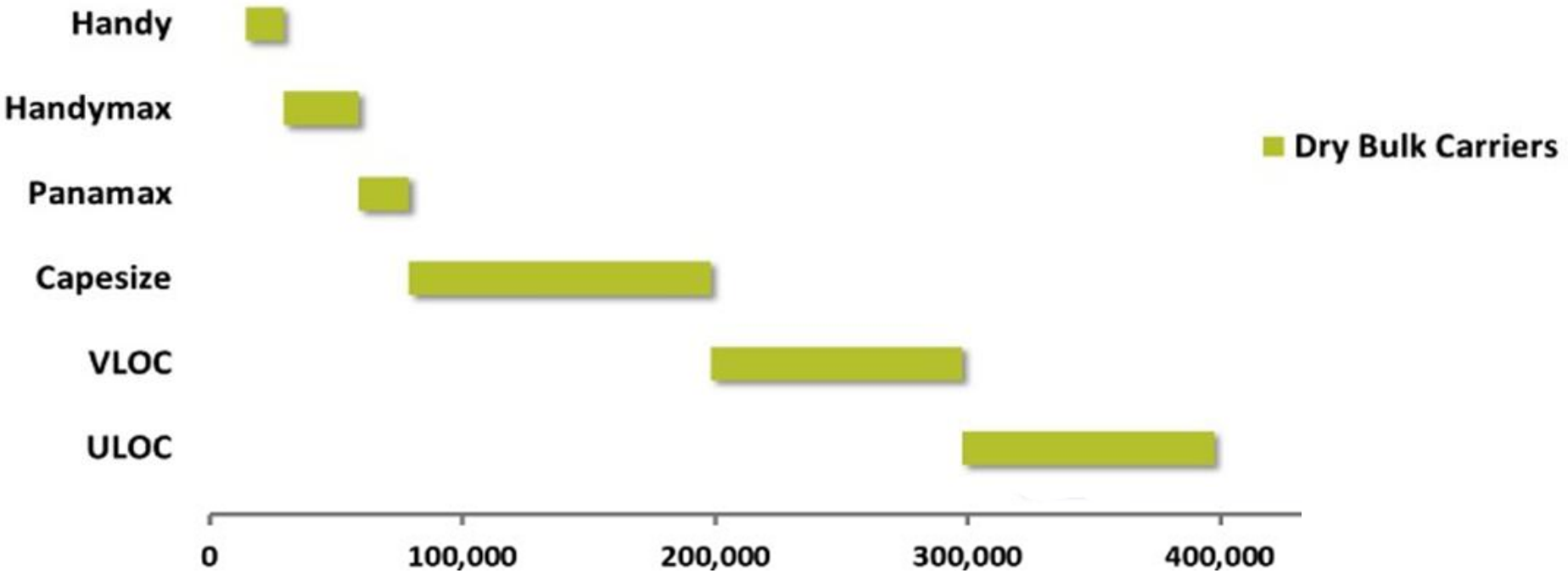
Iron ore production in China for the first six months of 2026 fell by 7% compared with the same period in 2025, totalling 466.9 million tonnes. The decline accelerated significantly in June, when output fell by 19.8% year-on-year – to 70.4 million tonnes, marking the lowest monthly level on record. This was reported by MySteel, citing data from the National Bureau of Statistics of China.

China's green steel push forces Australian iron ore giants to rethink strategies

China's efforts to strengthen its control over iron ore supply and decarburize its steel industry are ushering in a new era for the world's largest mining companies, BHP, Rio Tinto, and Fortescue.

01.08.2026 - 17:00

6.1 Tramp Shipping (Dry Bulk)



6.1 Tramp Shipping (Dry Bulk)

Dry bulk and ore carrier sizes:

i. _____ – 15,000-39,000DWT

ii. _____ – 40,000-50,000DWT

iii. _____ – 50,000-60,000DWT

iv. _____ – 62,000-65,000DWT (new design with eco main engine)

v. _____ – 70,000-80,000DWT

6.1 Tramp Shipping (Dry Bulk)

Dry bulk and ore carrier sizes:

vi. _____ – 80,000-85,000DWT (maximum permissible size to enter the port Kamsar in West Africa – one of the largest bauxite port in the world)

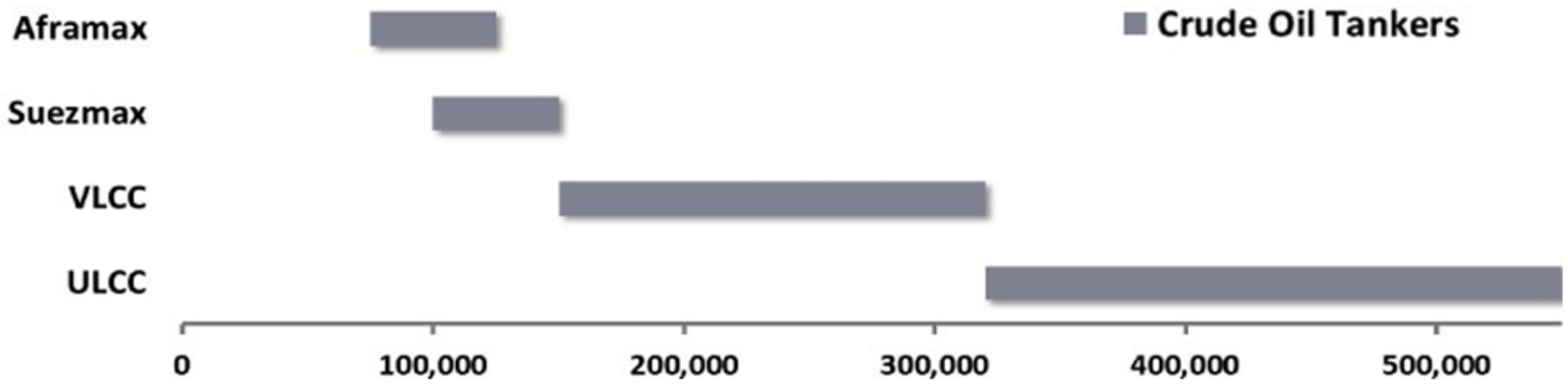
vii. _____ – 160,000-210,000DWT

viii. **Very Large Ore Carriers (VLOC)** – 210,000-300,000DWT

ix. **Ultra Large Ore Carriers (ULOC)** – 300,000-400,000DWT (new design with eco main engine)

6.1 Tramp Shipping (Wet Bulk)

- Wet bulk – crude oil, oil products, liquid chemicals such as caustic soda, vegetable oils and wine.



6.1 Tramp Shipping (Wet Bulk)

Crude oil tankers sizes:

i. _____ – 80,000-120,000DWT (ideally for short-medium haul oil trades)

ii. _____ – 120,000-160,000DWT (named after the Suez Canal)

iii. **Very Large Crude Carriers (VLCC)** – 160,000-320,000DWT (used in North Sea, Mediterranean and South Africa)

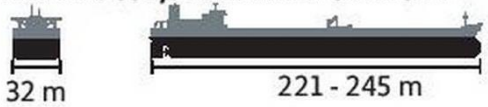
iv. **Ultra Large Crude Carriers (ULCC)** –320,000-350,000DWT (used for Middle East to Europe, Asia and North America)

6.1 Tramp Shipping (Wet Bulk)

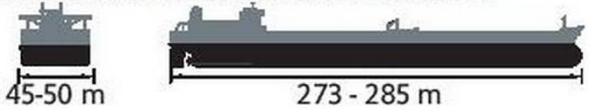
COASTAL TANKER



AFRAMAX: 650,000-BARREL CAPACITY



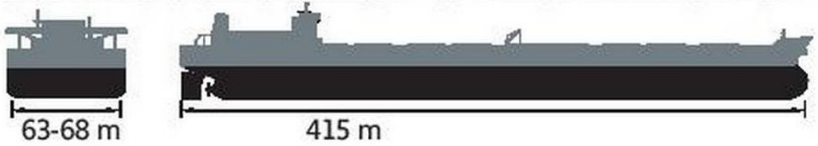
SUEZMAX: 1-MILLION BARREL CAPACITY

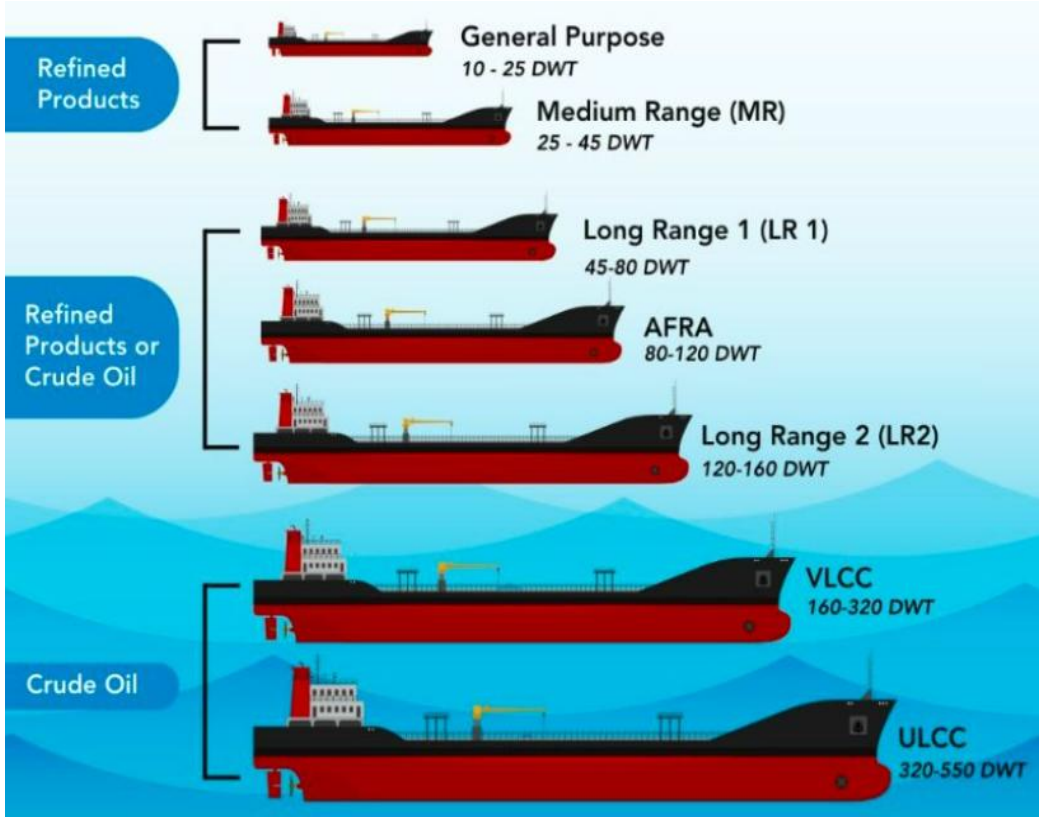


VERY LARGE CRUDE CARRIERS: 1.9 – 2.2-MILLION BARREL CAPACITY



ULTRA-LARGE CRUDE CARRIERS: 2 – 3.7-MILLION BARREL CAPACITY

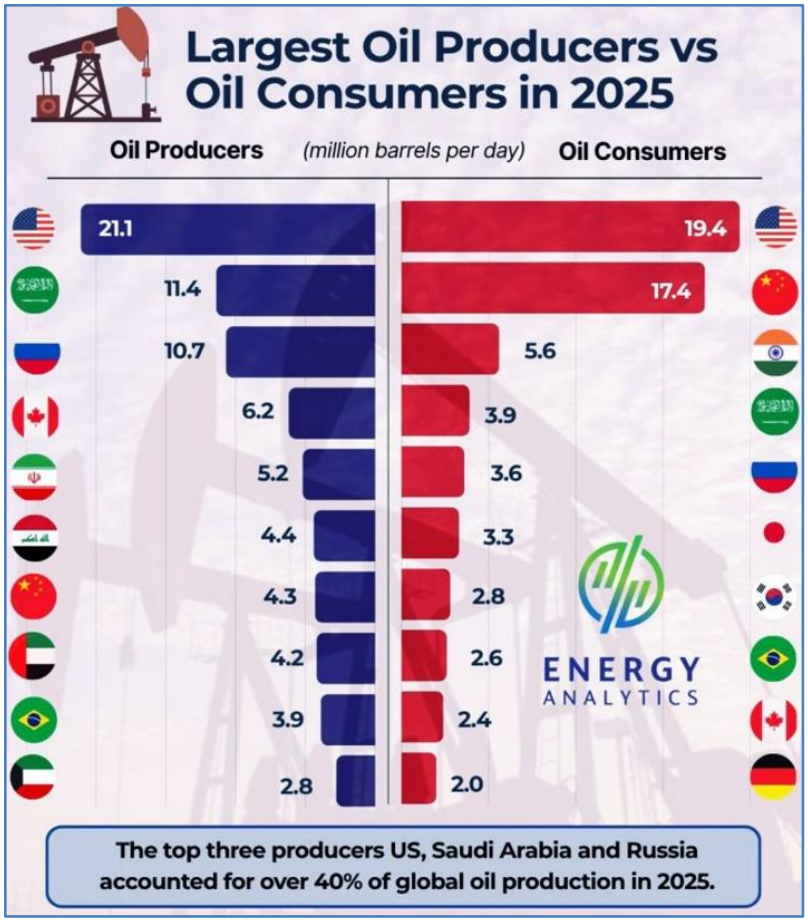


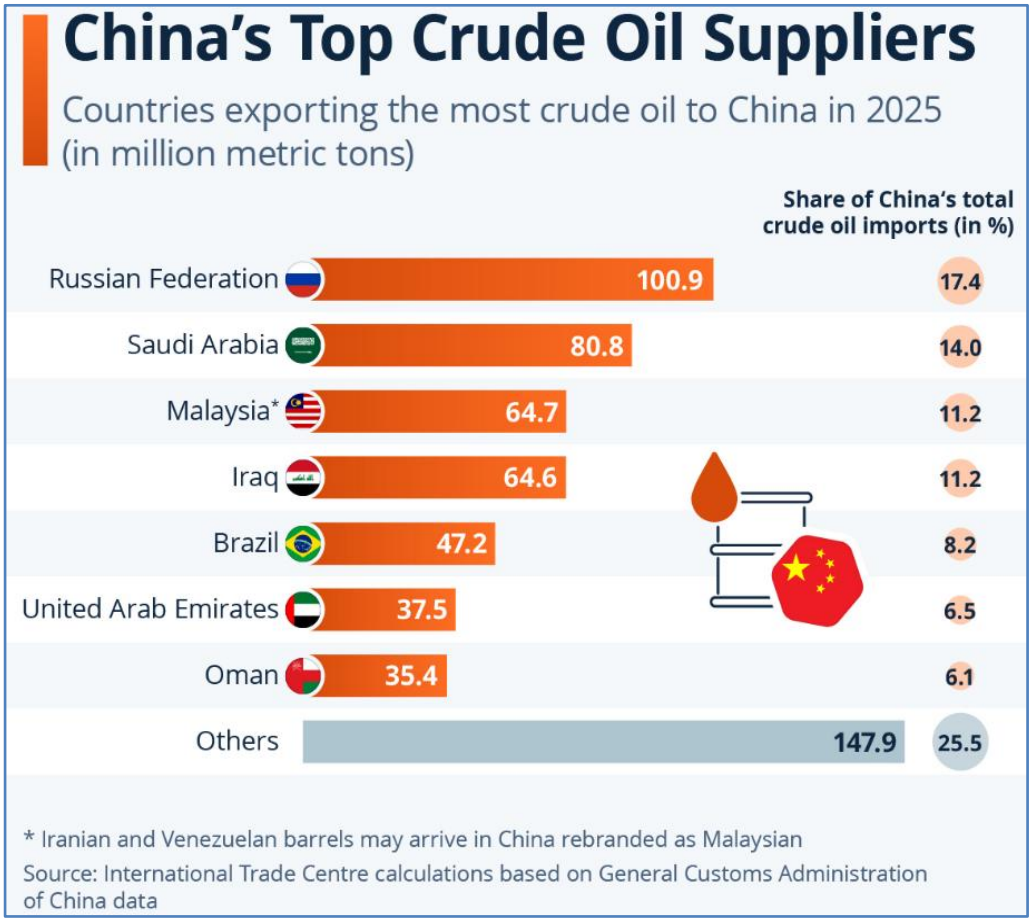


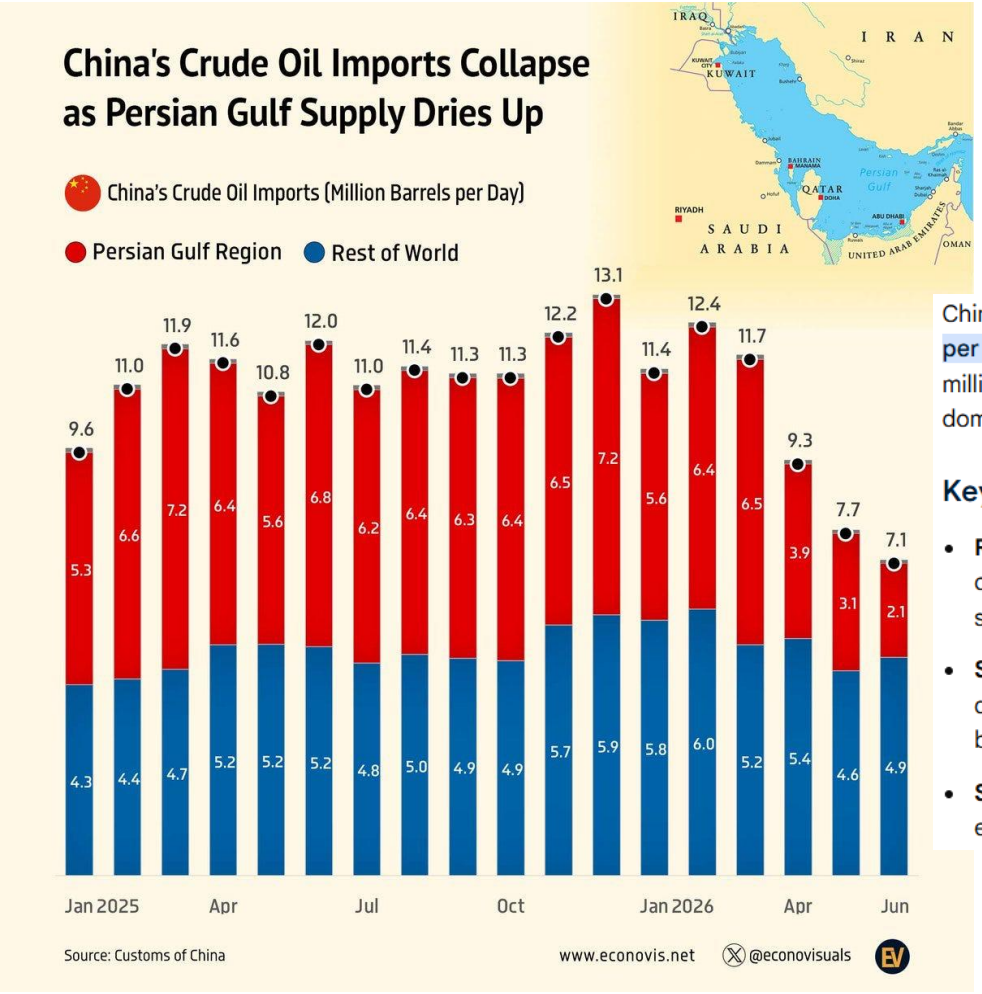
Futures & Indexes	Last	Change
WTI Crude	84.67	+1.08
Brent Crude	87.93	+1.05
Murban Crude	85.49	+1.06
Natural Gas	2.747	-0.011
Gasoline	3.114	-0.017
Heating Oil	4.122	-0.088
WTI Midland	85.96	+1.62
Mars	99.54	-3.59
Opec Basket	88.39	+2.53

OPEC Members (Daily Pricing)	Last
Arab Light	78.30
Kuwait Export Blend	85.00




Largest Oil Producers vs Oil Consumers in 2025
Oil Producers
(million barrels per day)
Oil Consumers
21.1
11.4
10.7
6.2
5.2
4.4
4.3
4.2
3.9
2.8
19.4
17.4
5.6
3.9
3.6
3.3
2.8
2.6
2.4
2.0
ENERGY ANALYTICS
The top three producers US, Saudi Arabia and Russia accounted for over 40% of global oil production in 2025.


China's Top Crude Oil Suppliers
Countries exporting the most crude oil to China in 2025 (in million metric tons)
Share of China's total crude oil imports (in %)
Russian Federation 100.9 17.4
Saudi Arabia 80.8 14.0
Malaysia* 64.7 11.2
Iraq 64.6 11.2
Brazil 47.2 8.2
United Arab Emirates 37.5 6.5
Oman 35.4 6.1
Others 147.9 25.5
* Iranian and Venezuelan barrels may arrive in China rebranded as Malaysian
Source: International Trade Centre calculations based on General Customs Administration of China data



China's crude oil imports fell sharply in mid-2026, dropping to roughly 7.12 million barrels per day (bpd) in June—a near 10-year low. This plunge from 2025's average of 11.6 million bpd was driven by Middle East conflict disruptions, high global prices, weak domestic fuel demand, and reliance on massive national stockpiles.

Key Trends & Drivers

- **Recent Slump:** Imports dropped over 41% year-on-year by June 2026, acting as a crucial global market stabilizer that prevented extreme energy price spikes during supply disruptions.
- **Stockpile Drawdown:** Instead of buying expensive overseas barrels, Beijing relied on drawing down its massive domestic reserves (estimated between 1.2 and 1.4 billion barrels).
- **Structural Shifts:** Rising adoption of electric vehicles (EVs) and tight refined product export quotas have permanently tempered baseline domestic oil demand.

Review Questions

8. Which of the following is not one of the characteristic of tramp shipping?

a. No fixed itinerary

b. Operates like a bus service

c. One ship, one load

d. Dry and wet bulk cargoes

Answer:

Review Questions

9. Which of the following is not one of the five major bulks of dry bulk shipping?

a. Phosphate

b. Bauxite

c. Iron ore

d. Steel

Answer:

Review Questions

10. Suezmax Tanker has _____ barrel capacity.

a. 650,000

b. 1 million

c. 1.9 million

d. 3.7 million

Answer:

6.2 Liner Shipping

- "Trade according to a *schedule of ports* of loading and discharge, usually adhering to a *published time-table* on set conditions of carriage and often charged at a *published rate of freight*".



Largest container ship in the world – MSC Irina

Review Questions

11. How many twenty-foot equivalent unit (TEUs) containers can MSC Irina hold?

a. 22,568 TEU

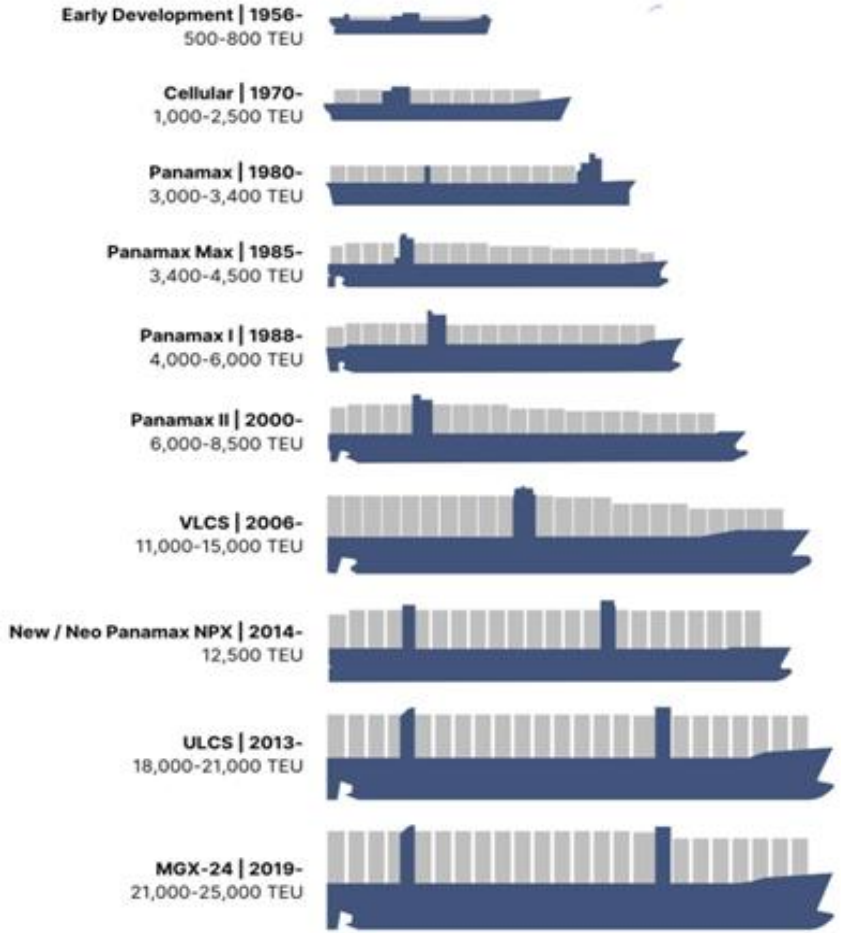
b. 23,456 TEU

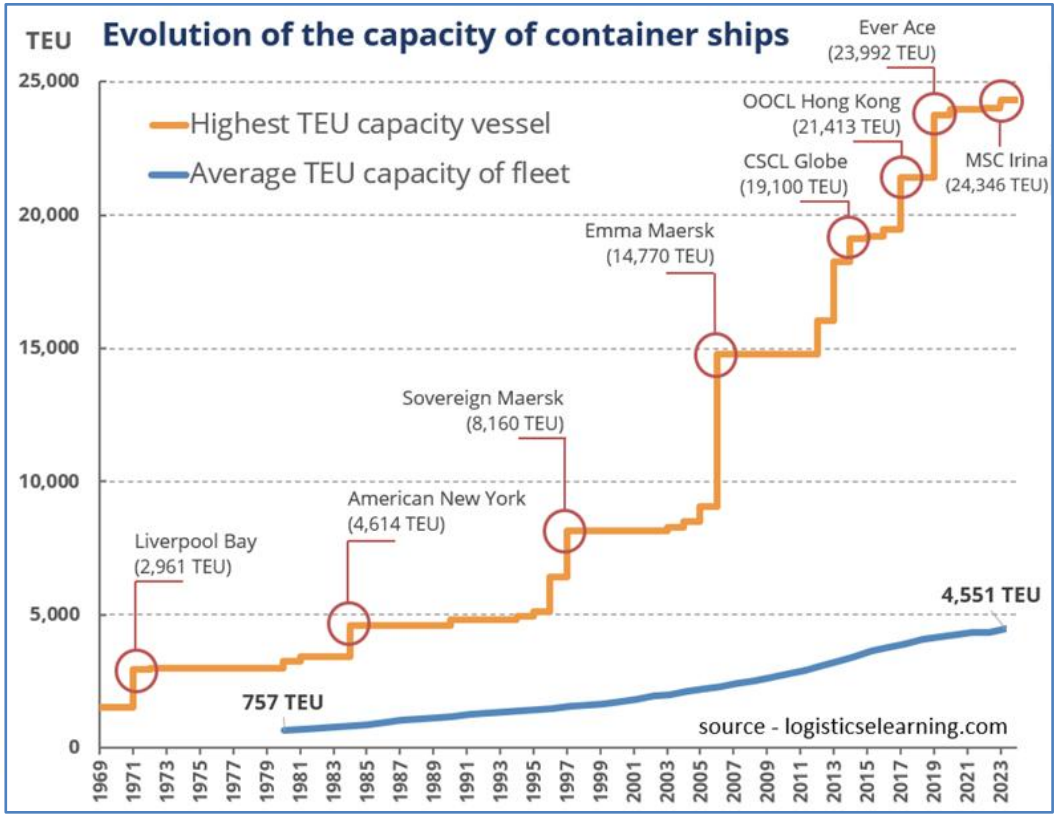
c. 24,346 TEU

d. 25,222 TEU

Answer:

The Evolution of Container Vessels & TEU Capacity





6.2 Liner Shipping (Trade Routes)

Frequency	1 sailing per week

Port Rotation

Loop A: Ho Chi Minh - Kaohsiung - Keelung - Hong Kong - Ho Chi Minh

Loop B: Ho Chi Minh – Kaohsiung – Taichung – Ho Chi Minh



Transit Time (Days)

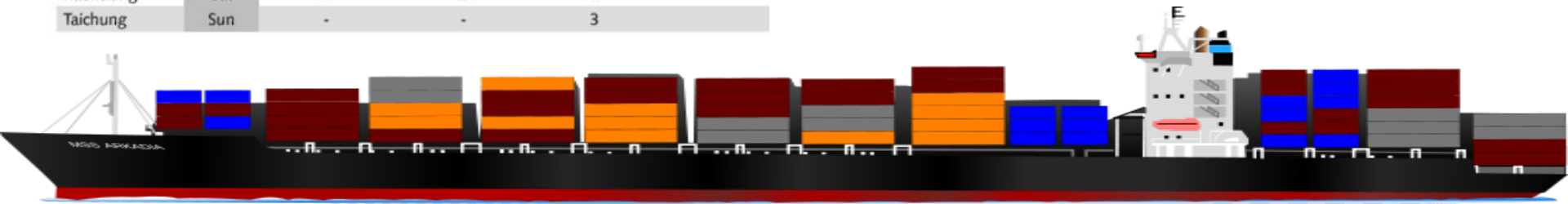
Loop A

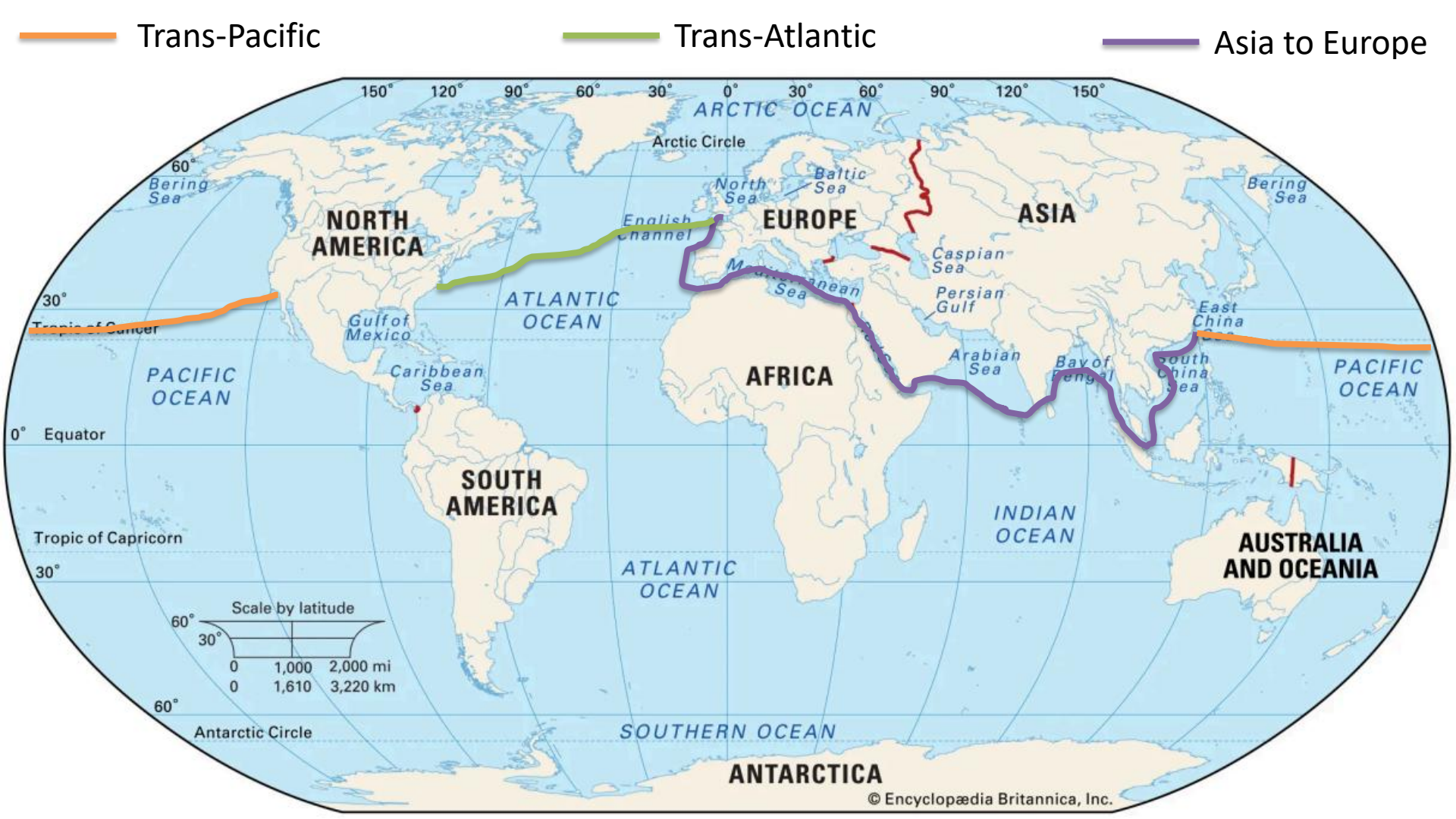
From \ To		Kaohsiung	Keelung	HongKong	Ho Chi Minh
		Mon	Tue	Thu	Mon
Ho Chi Minh	Thu	4	5	7	11
Kaohsiung	Mon	-	1	3	7
Keelung	Wed	-	-	1	5
HongKong	Fri	-	-	-	3

Transit Time (Days)

Loop B

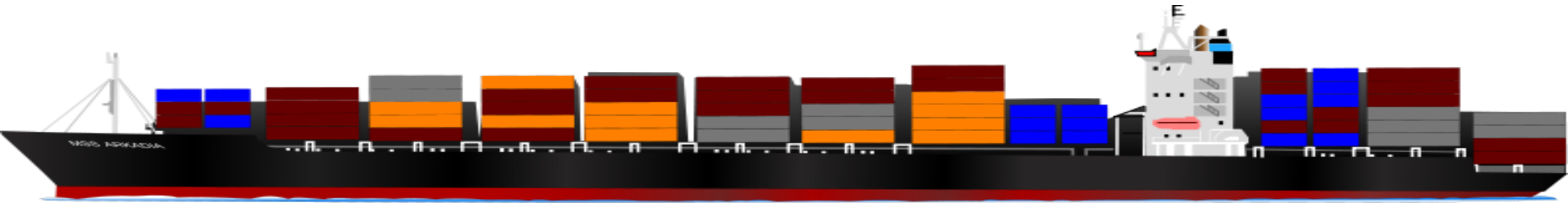
From \ To		Kaohsiung	Taichung	Ho Chi Minh
		Fri	Sat	Wed
Ho Chi Minh	Tue	3	4	8
Kaohsiung	Sat	-	0	4
Taichung	Sun	-	-	3




Trans-Pacific
Trans-Atlantic
Asia to Europe
ARCTIC OCEAN
Arctic Circle
NORTH AMERICA
EUROPE
ASIA
AFRICA
SOUTH AMERICA
AUSTRALIA AND OCEANIA
ANTARCTICA
PACIFIC OCEAN
ATLANTIC OCEAN
INDIAN OCEAN
SOUTHERN OCEAN
Bering Sea
North Sea
Baltic Sea
English Channel
Caspian Sea
Persian Gulf
Mediterranean Sea
Gulf of Mexico
Caribbean Sea
Arabian Sea
Bay of Bengal
East China Sea
South China Sea
0° Equator
Tropic of Capricorn
Antarctic Circle
Scale by latitude
0 1,000 2,000 mi
0 1,610 3,220 km
© Encyclopædia Britannica, Inc.

6.2 Liner Shipping (Shipping Companies)

Rank	Operator	Total		Owned		Chartered			Orderbook		
		Teu	Ships	TEU	Ship...	TEU	Ships	% Chart	TEU	Ship...	% existing
1	Mediterranean Shg Co	7,357,916	1005	4,723,159	757	2,634,757	248	35.8%	2,591,918	152	35.2%
2	Maersk	4,719,742	743	2,921,412	357	1,798,330	386	38.1%	1,201,971	90	25.5%
3	CMA CGM Group	4,407,216	731	2,895,513	376	1,511,703	355	34.3%	1,729,871	157	39.3%
4	COSCO Group	3,646,194	564	2,202,330	210	1,443,864	354	39.6%	1,646,116	138	45.1%
5	Hapag-Lloyd	2,394,122	287	1,455,918	134	938,204	153	39.2%	507,610	65	21.2%
6	ONE (Ocean Network Express)	2,179,800	272	1,029,008	110	1,150,792	162	52.8%	592,151	48	27.2%
7	Evergreen Line	2,022,899	242	1,459,367	168	563,532	74	27.9%	868,460	76	42.9%
8	HMM Co Ltd	1,033,593	97	850,162	73	183,431	24	17.7%	220,642	36	21.3%
9	Yang Ming Marine Transport Corp.	757,240	99	382,296	63	374,944	36	49.5%	268,360	21	35.4%
10	Zim	702,036	116	92,934	15	609,102	101	86.8%	181,836	23	25.9%
11	Wan Hai Lines	624,354	124	624,354	124				418,310	41	67%
12	PIL (Pacific Int. Line)	459,136	101	319,007	81	140,129	20	30.5%	318,862	30	69.4%



OCEAN FREIGHT MARKET

SHIPPING ALLIANCE LANDSCAPE CHANGE EFFECTIVE FEBRUARY 1ST 2025

2M Alliance	The Premier Alliance	The Ocean Alliance
10.7M TEU | 34.3%	5.9M TEU | 23.8%	9.8M TEU | 31.4%

Before 2025 | From 2025

MSC	Gemini Alliance	The Premier Alliance	The Ocean Alliance
7.1M TEU | 22.7%	6.7M TEU | 26.4%	3.6M TEU | 11.5%	9.8M TEU | 31.4%

Operating independently but collaborating with:

- Premier Alliance (Asia-Europe and Asia-Mediterranean services)
- ZIM (Transpacific services)

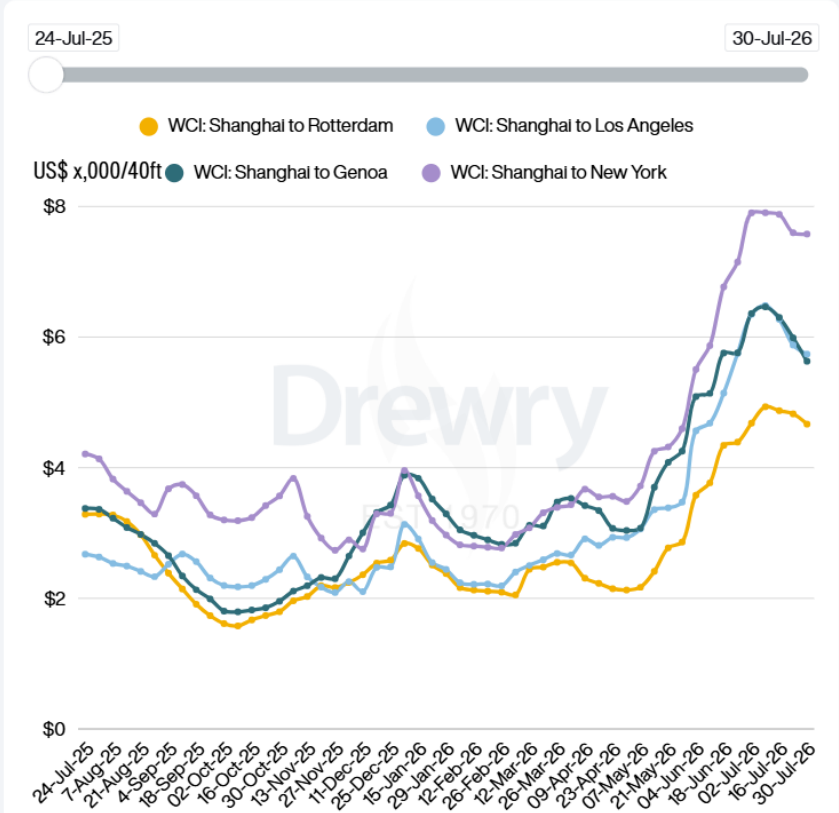
Operating independently but in partnership with MSC on select transpacific routes.

Rising container freight rates: Now more than half of Covid levels.



6.2 Liner Shipping (Freight Rates)

WCI Trade Routes from Shanghai (US$/40ft)



World Container Index

Routes from Shanghai
US$ / 40ft container - 30 Jul 2026

"Ongoing uncertainty surrounding global trade policies, geopolitical developments and port congestion is expected to influence market conditions and freight rate trends in the coming weeks."

Shanghai - Rotterdam	Shanghai - Genoa	Shanghai - Los Angeles	Shanghai - New York
$4,667	$5,630	$5,739	$7,578
WoW -3% ▼	WoW -6% ▼	WoW -2% ▼	WoW 0%
YoY 42% ▲	YoY 67% ▲	YoY 118% ▲	YoY 83% ▲

6.2 Liner Shipping (Container Terminal)





6.2 Liner Shipping (Container Sizes)

DRY VAN CONTAINER



The most common container for general cargo and everyday shipping needs.

REEFER CONTAINER



Temperature-controlled container for perishable and sensitive goods.

HIGH CUBE CONTAINER



Extra height for lightweight, bulky cargo and more loading capacity.

OPEN TOP CONTAINER



Topless container for overheight cargo that cannot be loaded from the doors.



Tank Container

Liquids, chemicals & food-grade fluids



Bulk Container

Grain, powder & loose materials



Ventilated Container

Ensures ventilation for coffee, cocoa & moisture-sensitive agricultural goods

6.2 Liner Shipping (Container Sizes)



Flat Rack Container

Heavy machinery & project cargo



Reefer Container

Temperature-control shipments



Open Side Container

Easy side loading for bulky goods

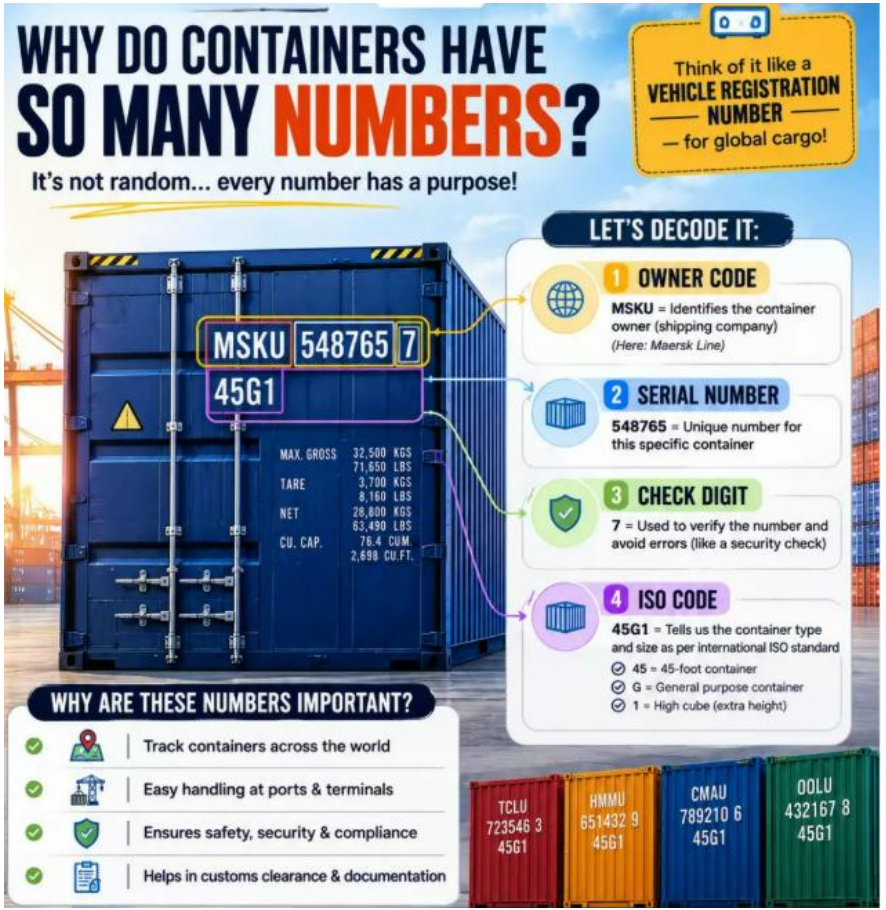


Platform Container

Extra-heavy & oversized cargo transport

6.2 Liner Shipping (Container Sizes)





6.3 Ton-Miles Concept

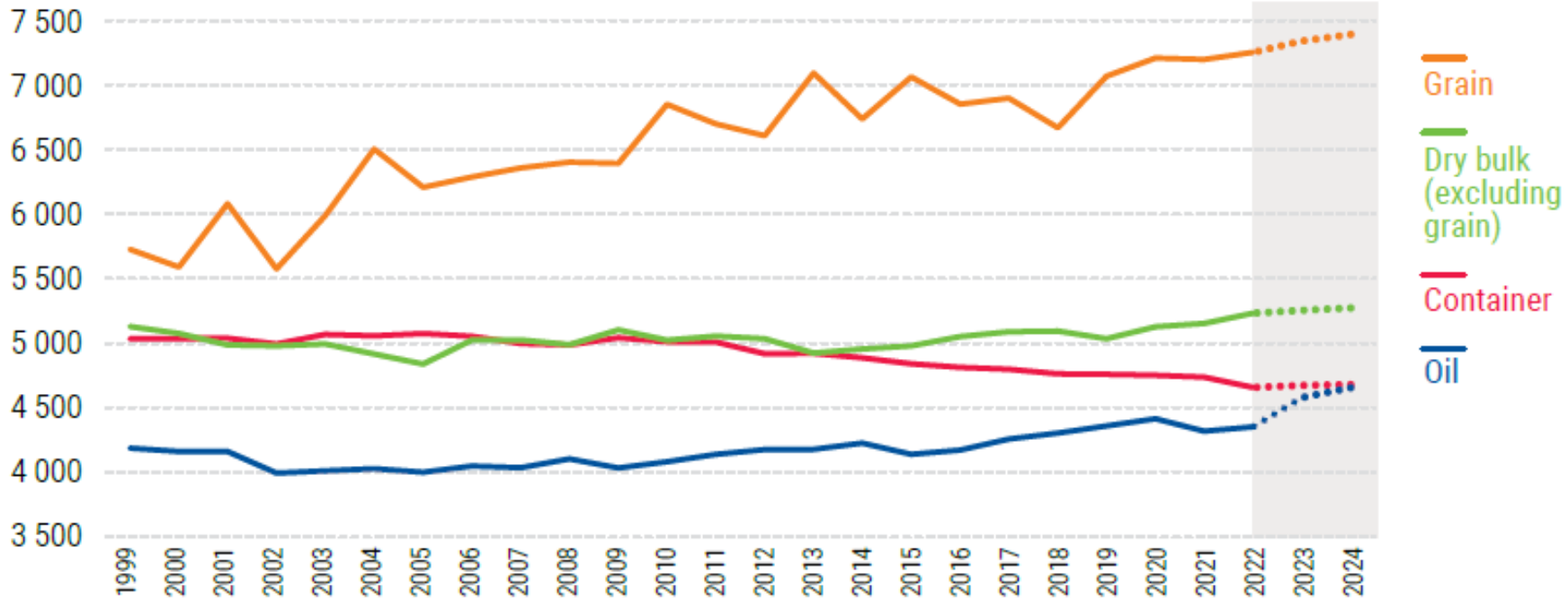
LO7

- Transport demand determined by a precise matrix of distances which determine the time it takes a ship to complete each voyage.

  E.g., *one ton of iron ore transported from Brazil to China (14772 miles) travels 3.6 times compared to one ton of iron ore shipped from Australia to China (4059 miles)*

6.3 Ton-Miles Concept

Figure 3 Average distance travelled, grain, other dry bulk, container and oil cargo, 1999–2024 (Nautical miles)



6.3 Ton-Miles Concept

- The distance effect is generally referred as the '*average haul (distance)*' of the trade.

E.g., *average distance travelled by grain trade has risen 32% between 2002 and 2023*.

ton-miles = tonnage of cargo shipped x average distance over which its transported.

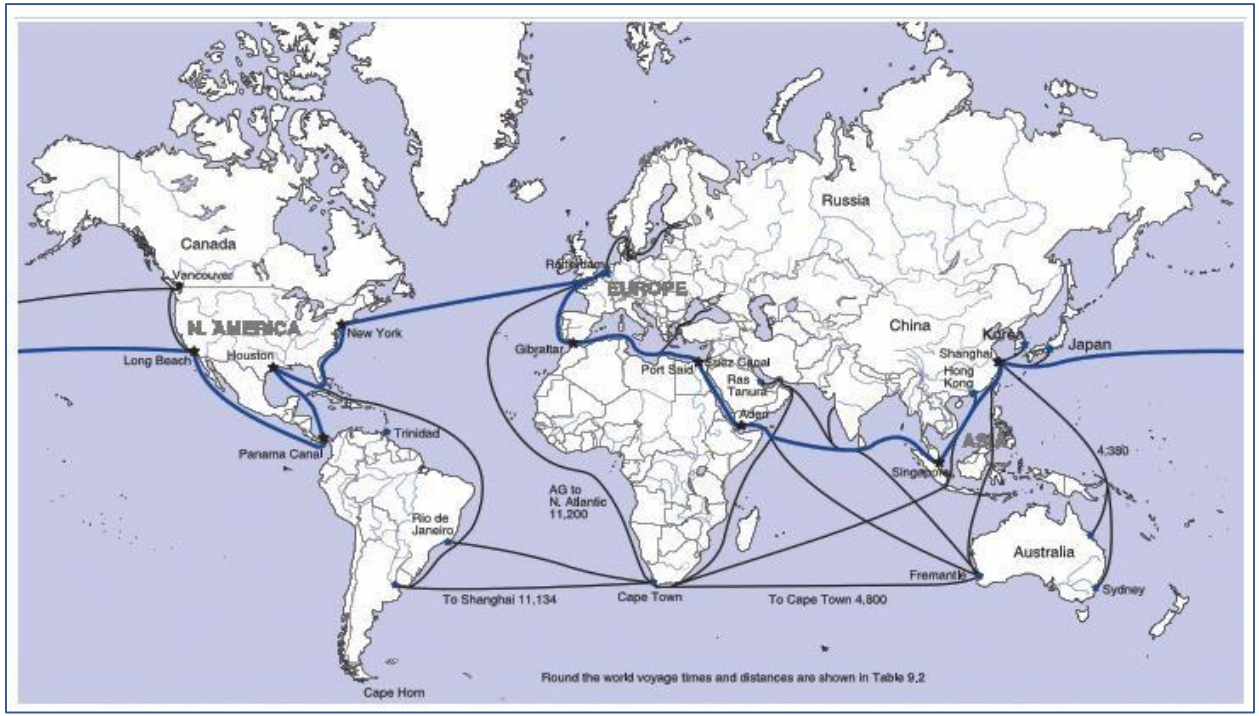
What is the implication of ton-miles on world trade?

LO8

7. Maritime Trading Networks

7. Maritime Trading Networks

- Maritime trade dominated by three economic centres – **North America, Europe and Asia**.



7.1 Major Economic Centres (Europe)



The major seas and ports of Europe

7.1 Major Economic Centres (North America)



The major seas and ports of North America

7.1 Major Economic Centres (Asia)

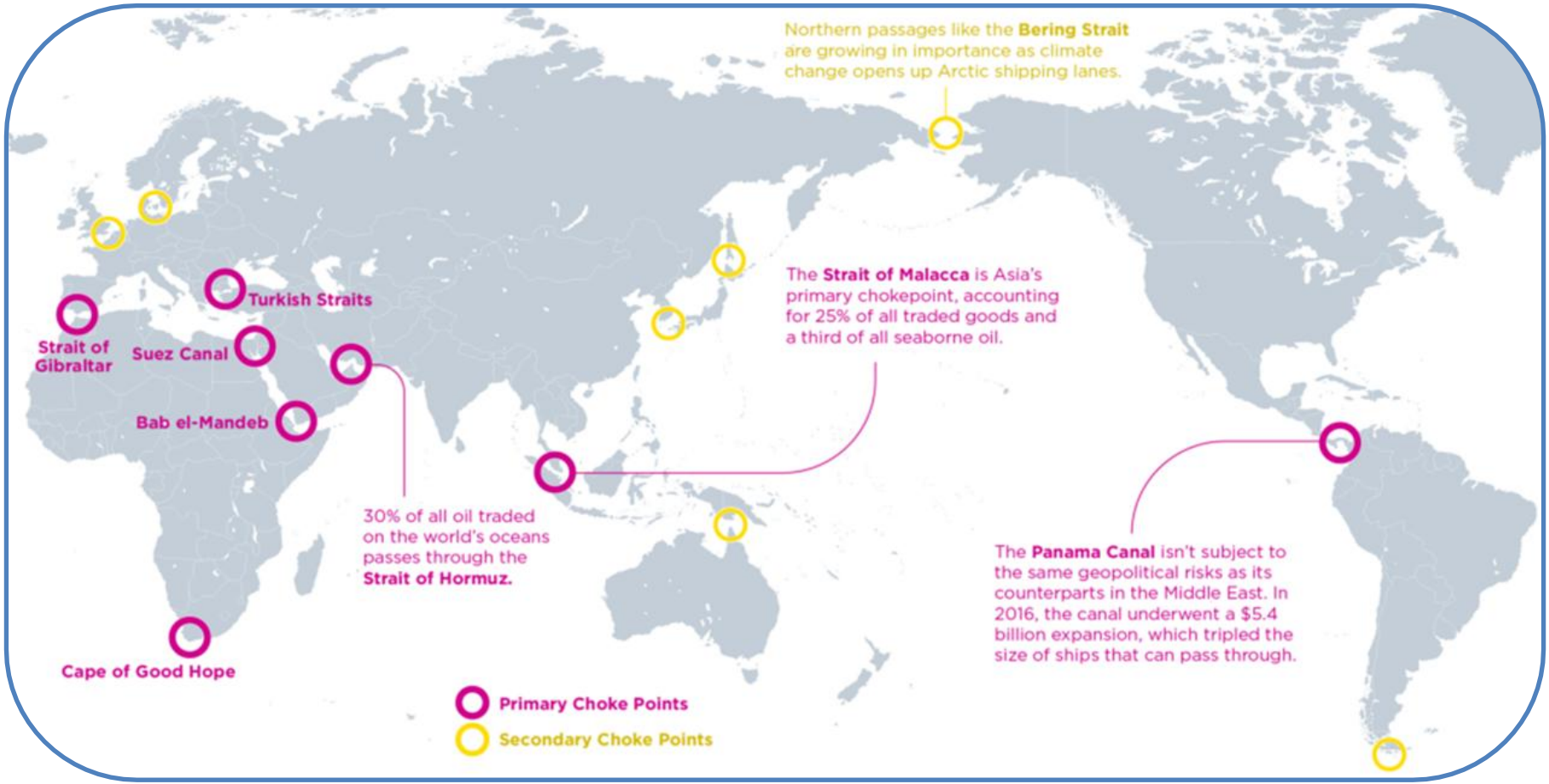


7.2 Major Maritime Chokepoints

- "***critical points*** along transport routes that facilitate the passage of substantial trade volumes, which serve as ***vital arteries for global commence***, connecting important regions worldwide".

"A key feature of maritime transport is its reliance on chokepoints: strategic, narrow maritime passages such as the Suez Canal and the Panama Canal. These critical waterways provide shortcuts on lengthy intercontinental maritime journeys and reduce time and costs. Yet the essential role of these chokepoints makes them particularly vulnerable to disruptions – whether climatic, economic, geopolitical or operational – leading to severe consequences for global shipping.", commented Rebeca Grynspan, Secretary-General of UNCTAD.

7.2 Major Maritime Chokepoints



7.2.1 Suez Canal

- Strategically located connecting the Mediterranean Sea at its northern end to the Rea Sea at its southern tip.
- Links Europe to Asian maritime routes leading to Indian and Pacific Oceans.



7.2.2 Bab el-Mandeb Strait (Red Sea)

- Connects the Red Sea to the Gulf of Aden and the Indian Ocean.
- Crucial for oil and natural gas from the Middle East.



7.2.3 Strait of Hormuz

- Connects Persian Gulf with the Gulf of Oman and the Arabian Sea.
- Significant portion of the world's petroleum passing through this chokepoint – approximately 20%.
- Geopolitical tensions can significantly impact oil prices and global energy security.



Strait of Hormuz

Connects oil and LNG production in the Middle East to global markets via the Arabian Sea and the Indian Ocean

Volume transported through the strait in 2024
(daily average)

Oil and petroleum products

20.3 MILLION barrels

LNG

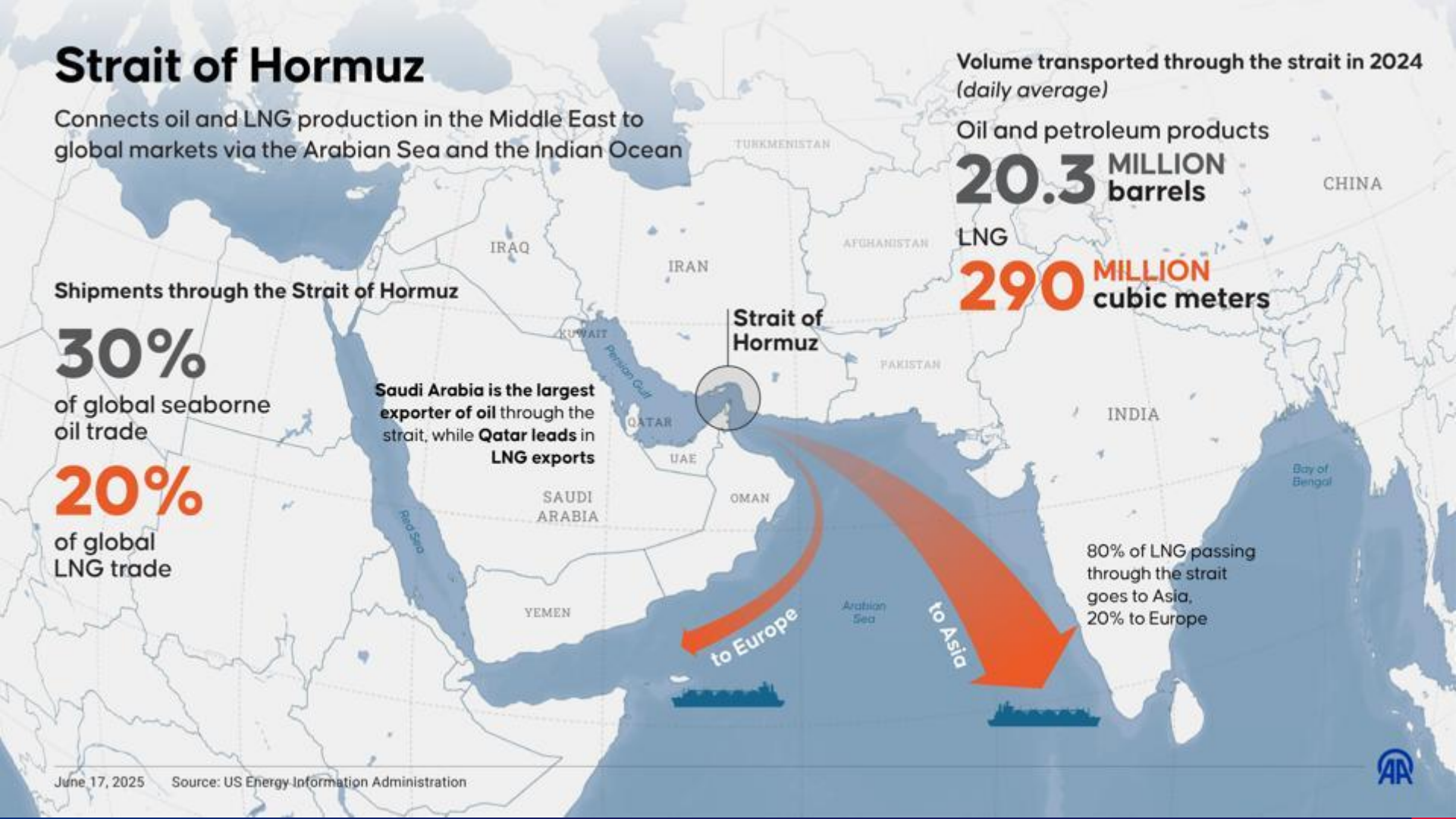
290 MILLION cubic meters

Shipments through the Strait of Hormuz

30%
of global seaborne oil trade

20%
of global LNG trade

Saudi Arabia is the largest exporter of oil through the strait, while **Qatar leads** in **LNG exports**



June 17, 2025 Source: US Energy Information Administration

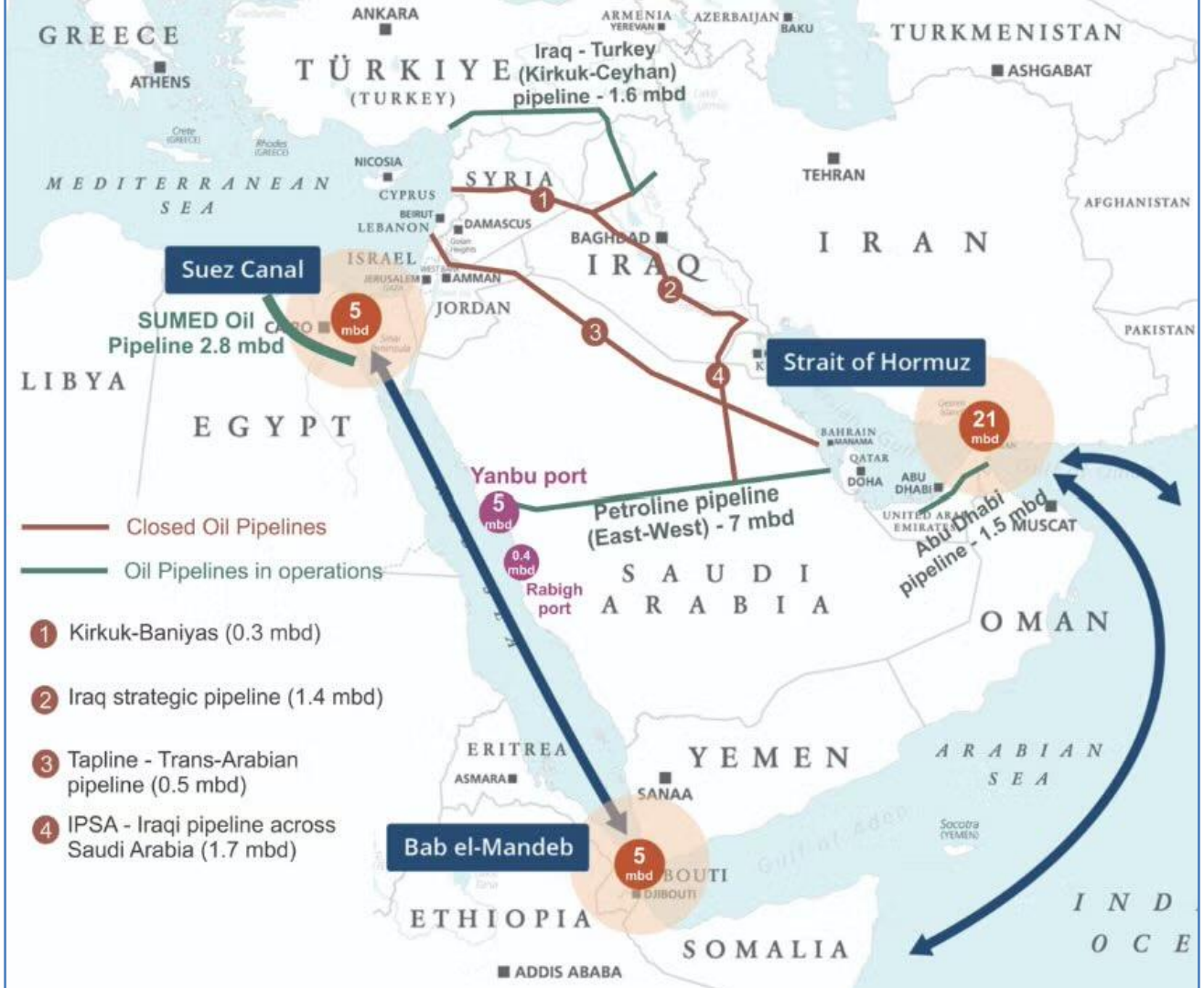
Hormuz Under Siege

How War Disrupted the World's Most Critical Oil Chokepoint



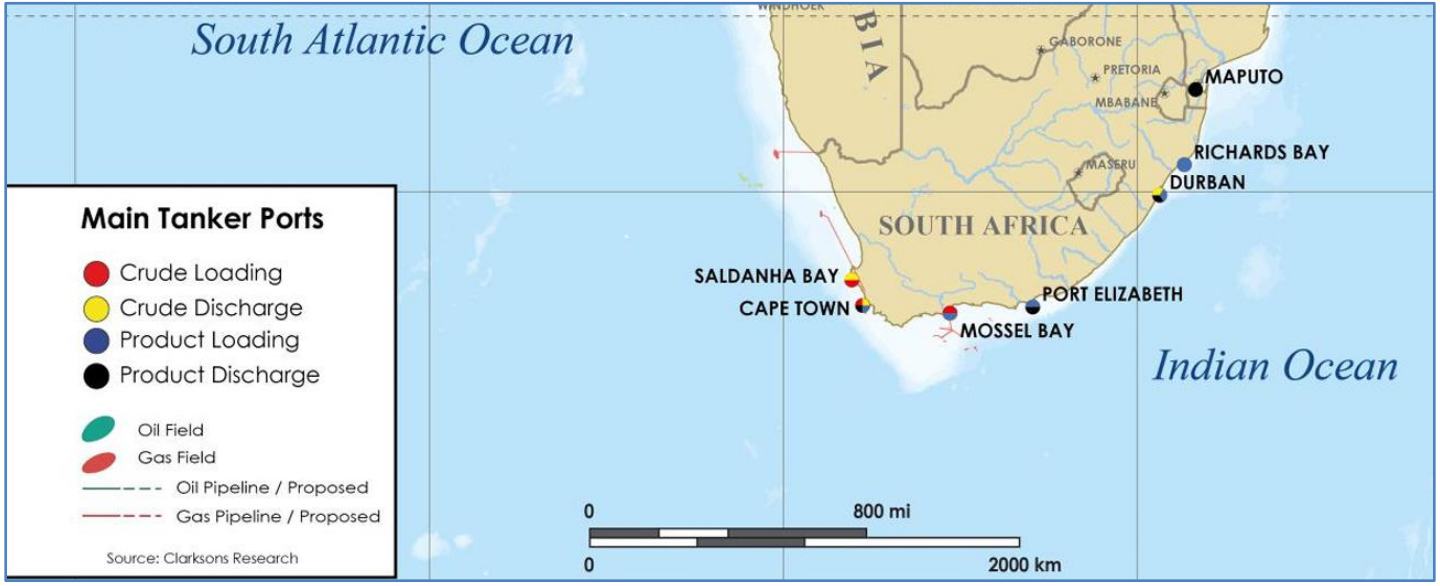
Tanker Transits (7-Day Moving Average)




GREECE
ATHENS
ANKARA
TÜRKIYE
(TURKEY)
ARMENIA
YEREVAN
AZERBAIJAN
BAKU
TURKMENISTAN
ASHGABAT
Iraq - Turkey
(Kirkuk-Ceyhan)
pipeline - 1.6 mbd
MEDITERRANEAN
SEA
NICOSIA
CYPRUS
SYRIA
BEIRUT
LEBANON
DAMASCUS
TEHRAN
AFGHANISTAN
ISRAEL
JERUSALEM
AMMAN
JORDAN
BAGHDAD
IRAQ
IRAN
Suez Canal
5
mbd
SUMED Oil
Pipeline 2.8 mbd
CAIRO
LIBYA
EGYPT
Strait of Hormuz
PAKISTAN
21
mbd
BAHRAIN
MANAMA
QATAR
DOHA
ABU
DHABI
UNITED ARAB
EMIRATES
Yanbu port
5
mbd
Petroline pipeline
(East-West) - 7 mbd
Abu Dhabi
pipeline - 1.5 mbd
MUSCAT
0.4
mbd
Rabigh
port
SAUDI
ARABIA
OMAN
Closed Oil Pipelines
Oil Pipelines in operations
1 Kirkuk-Baniyas (0.3 mbd)
2 Iraq strategic pipeline (1.4 mbd)
3 Tapline - Trans-Arabian
pipeline (0.5 mbd)
4 IPSA - Iraqi pipeline across
Saudi Arabia (1.7 mbd)
ERITREA
ASMARA
YEMEN
SANAA
ARABIAN
SEA
Socotra
(YEMEN)
Bab el-Mandeb
5
mbd
DJIBOUTI
ETHIOPIA
SOMALIA
ADDIS ABABA
IND
OCE

7.2.4 Cape of Good Hope

- Connects Indian Ocean with the Atlantic Ocean.
- Main commodities: containerized cargo, crude oil and dry bulks (iron ore and coal).

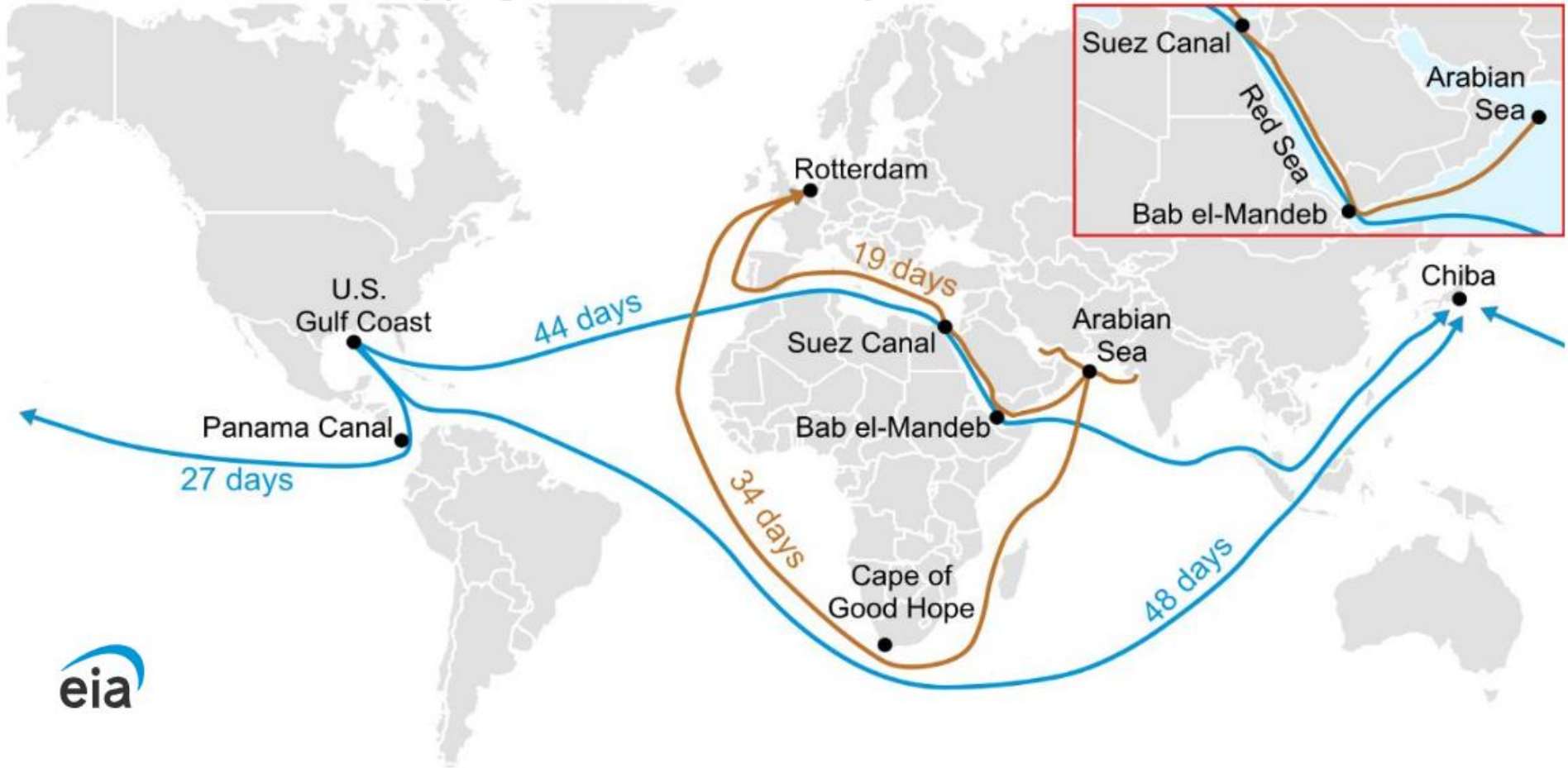


Alternative shipping route to avoid the Red Sea

Attacks by Yemen's Houthi rebels in the Red Sea have disrupted maritime trade along major shipping routes via the Suez Canal and Bab al-Mandeb Strait, leading to an alternative, longer route being taken.



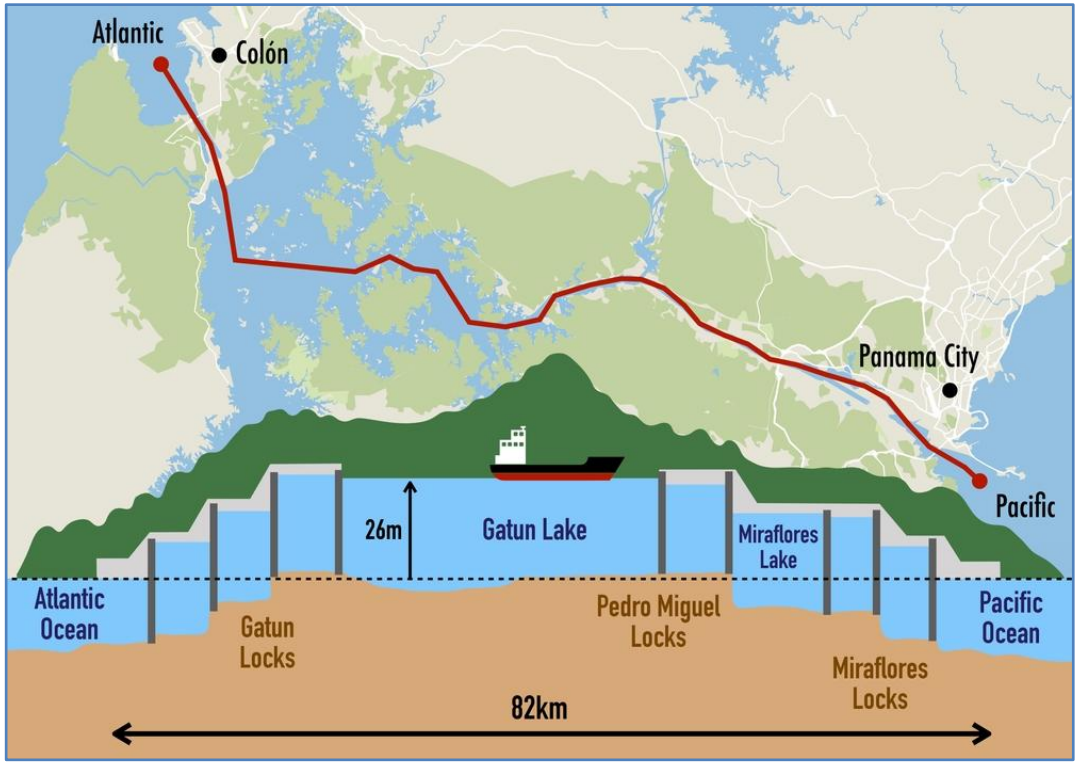
Selected commercial shipping routes, as of January 2024



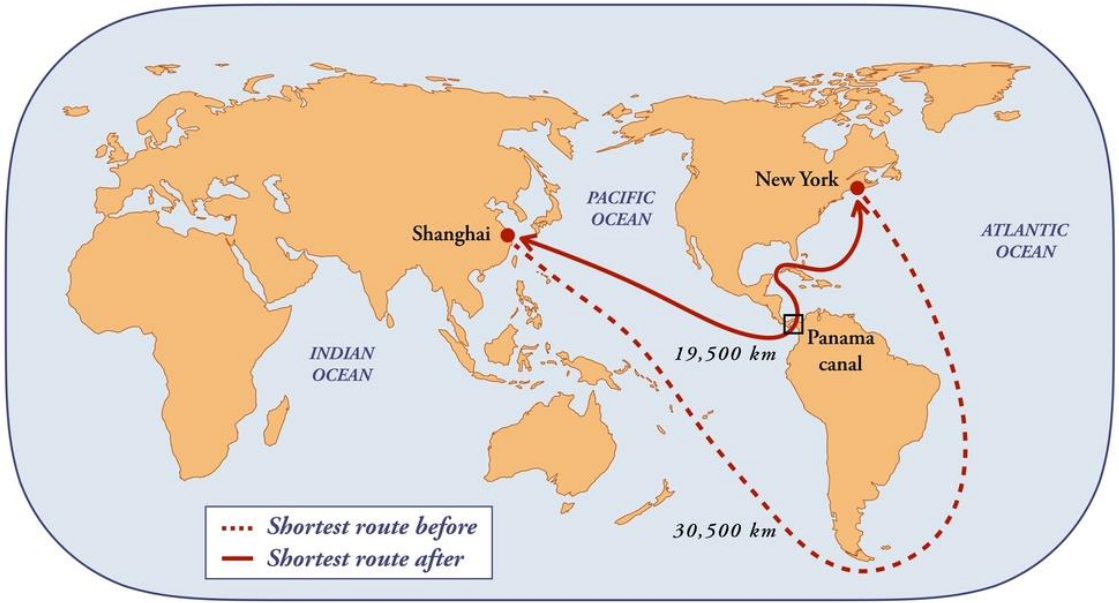
Data source: U.S. Energy Information Administration using calculations from Vortexa
Note: Voyage time is calculated for laden Suezmax tankers traveling at 14 knots without extended chokepoint delays.

7.2.5 Panama Canal

- Connects Pacific Ocean and the Atlantic Ocean.
- Average transit time for a vessel passing through is approximately **8 to 10 hours**, depending on vessel size, traffic congestion and weather conditions.



7.2.5 Panama Canal





7.2.6 Strait of Gibraltar

- Links the Mediterranean Sea with the Atlantic Ocean.
- Crucial for flow of crude oil and LNG, mainly to European markets.
- Approximately **300** ships cross every day.




GREENLAND
NORWEGIAN SEA
BARENTS SEA
KARA SEA
ICELAND
Faroe Islands
SWEDEN
FINLAND
NORWAY
RUSSIA
NORTH SEA
BALTIC SEA
ESTONIA
LATVIA
LITHUANIA
DENMARK
IRELAND
UNITED KINGDOM
BELARUS
POLAND
GERMANY
CELTIC SEA
CZECH REPUBLIC
SLOVAKIA
UKRAINE
KAZAKHSTAN
ATLANTIC OCEAN
FRANCE
AUSTRIA
HUNGARY
MOLDOVA
ROMANIA
ITALY
BULGARIA
BLACK SEA
GEORGIA
ARMENIA
AZERBAIJAN
CASPIAN SEA
UZBEKISTAN
TURKMENISTAN
PORTUGAL
SPAIN
GREECE
TURKEY (TÜRKIYE)
TUNISIA
CYPRUS
SYRIA
MEDITERRANEAN SEA
LEBANON
ISRAEL
IRAQ
IRAN
AFGHANISTAN
JORDAN
MOROCCO
KUWAIT
PAKISTAN
ALGERIA
LIBYA
EGYPT
Western Sahara
SAUDI ARABIA
U.A.E.
RED SEA
OMAN
MAURITANIA
MALI
NIGER
CHAD
SUDAN
ERITREA
YEMEN
ARABIAN SEA
SENEGAL
BURKINA FASO
GUINEA
NIGERIA
ETHIOPIA
SOMALIA

7.2.7 Strait of Malacca

- One of the world's busiest shipping lanes, connecting Indian Ocean to the South China Sea.
- Serves as a vital conduit for trade between Asia and Europe.
- Crucial passage for oil shipments from Middle East to Asia.

7.2.7 Strait of Malacca

Threats from land bridge/canal projects in Thailand



7.2.8 Turkish Straits

- Pair of vital waterways, the Bosphorus and Dardanelles, connecting the Black Sea to the Aegean and Mediterranean Sea.
- Transport of oil and grain from the Black Sea



7.2.8 Turkish Straits

Turkish Straits



Review Questions

12. _____ connects the Red Sea to the Gulf of Aden.

a. Bab el-Mandeb

b. Strait of Hormuz

c. Strait of Malacca

d. Strait of Gibraltar

Answer:

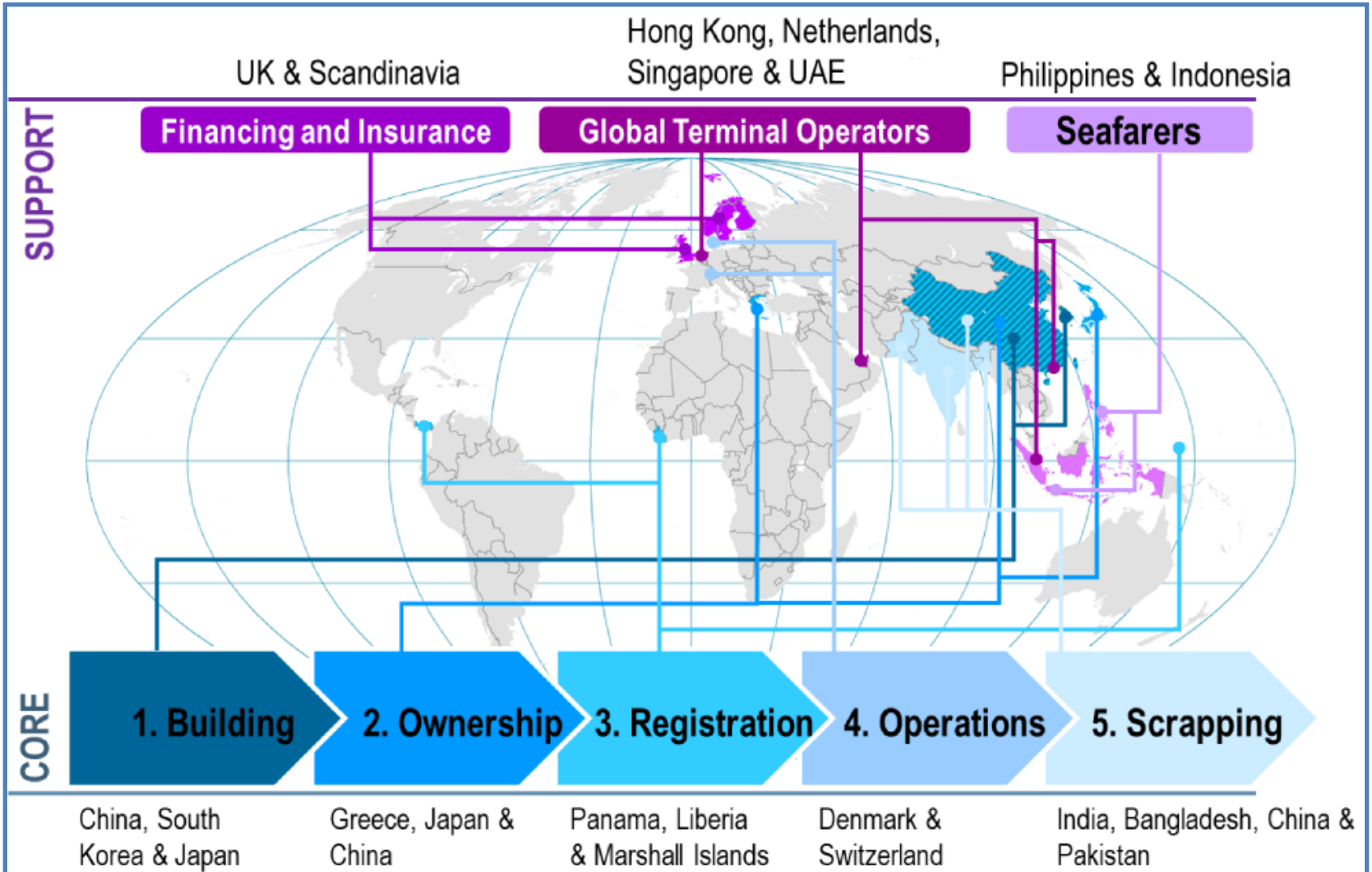
Summary

Summary

- Shipping is a derived demand of level of international trade.
- Shipping is a part of larger supply chains.
- Maritime eco-system is made up of multitude of actors.
- Shipping is regulated and influenced on international and national levels.

Summary

- There are two main types of shipping services – tramp and liner.
- Maritime trade dominated by three important trade centres – US, Europe and Asia and multiple choke points.


UK & Scandinavia
Hong Kong, Netherlands, Singapore & UAE
Philippines & Indonesia
SUPPORT
Financing and Insurance
Global Terminal Operators
Seafarers
CORE
1. Building
2. Ownership
3. Registration
4. Operations
5. Scrapping
China, South Korea & Japan
Greece, Japan & China
Panama, Liberia & Marshall Islands
Denmark & Switzerland
India, Bangladesh, China & Pakistan